# Atemoya:
# Quantitative Finance Framework

github.com/cb-g/atemoya

February 22, 2026

# Contents

**Table of Contents** ii

**1 Pricing** 1
1.1 Regime-Adaptive Downside Optimization . . . 1
1.1.1 Mathematical Framework . . . 1
1.1.1.1 Portfolio Returns . . . 1
1.1.2 Regime Detection . . . 1
1.1.2.1 Realized Volatility . . . 1
1.1.2.2 Historical Volatility Distribution . . . 1
1.1.2.3 Stress Weight Function . . . 1
1.1.3 Downside Risk Measures . . . 2
1.1.3.1 Lower Partial Moment of Order 1 . . . 2
1.1.3.2 Conditional Value-at-Risk . . . 2
1.1.3.3 Portfolio Beta . . . 2
1.1.4 Linear Program Formulation . . . 2
1.1.4.1 Decision Variables . . . 3
1.1.4.2 Objective Function . . . 3
1.1.4.3 Constraints . . . 3
1.1.4.4 Problem Structure . . . 4
1.1.5 Regime Adaptation Mechanism . . . 4
1.1.5.1 Stress-Conditional Beta Control . . . 4
1.1.5.2 Economic Rationale . . . 4
1.1.6 Solution and Rebalancing . . . 5
1.1.6.1 Optimization Frequency . . . 5
1.1.6.2 Rebalancing Decision Rule . . . 5
1.1.7 Performance Metrics . . . 5
1.1.7.1 Ex-Post Risk Evaluation . . . 5
1.1.7.2 Sharpe-Type Ratios . . . 5
1.1.8 Interpretation . . . 5
1.2 Market Regime Forecast . . . 6
1.2.1 GARCH(1,1) Volatility Model . . . 6
1.2.1.1 Model Specification . . . 6
1.2.1.2 Key Properties . . . 6
1.2.1.3 Maximum Likelihood Estimation . . . 6
1.2.1.4 Volatility Regime Classification . . . 6
1.2.2 Hidden Markov Model for Trend Detection . . . 6
1.2.2.1 Model Specification . . . 7
1.2.2.2 Baum-Welch Algorithm (EM) . . . 7
1.2.2.3 Viterbi Decoding . . . 7
1.2.2.4 State Relabeling . . . 7
1.2.3 Markov-Switching GARCH (MS-GARCH) . . . 7
1.2.3.1 Model Specification . . . 8
1.2.3.2 Interpretation . . . 8
1.2.3.3 Estimation . . . 8
1.2.3.4 Regime Classification . . . 8
1.2.4 Bayesian Online Changepoint Detection (BOCPD) . . . 8
1.2.4.1 Run Length Formulation . . . 8
1.2.4.2 Hazard Function . . . 8
1.2.4.3 Underlying Predictive Model . . . 9
1.2.4.4 Online Update Algorithm . . . 9
1.2.4.5 Changepoint Detection . . . 9
1.2.4.6 Regime Statistics . . . 9
1.2.5 Gaussian Process Regression . . . 9
1.2.5.1 Model Specification . . . 9
1.2.5.2 Kernel Functions . . . 10
1.2.5.3 GP Posterior . . . 10
1.2.5.4 Hyperparameter Optimization . . . 10

1.2.5.5 Regime Classification . . . 10
1.2.6 Model Comparison and Application . . . 11
1.2.6.1 Summary of Approaches . . . 11
1.2.6.2 Covered Call Suitability . . . 11
1.3 Options Hedging . . . 11
1.3.1 Motivation . . . 11
1.3.1.1 Hedging Strategies . . . 11
1.3.2 Black-Scholes-Merton Pricing . . . 11
1.3.2.1 European Call and Put . . . 11
1.3.2.2 Put-Call Parity . . . 12
1.3.2.3 Greeks . . . 12
1.3.3 SVI Volatility Surface . . . 12
1.3.3.1 Total Variance Formula . . . 12
1.3.3.2 Implied Volatility . . . 13
1.3.3.3 No-Arbitrage Conditions . . . 13
1.3.3.4 Calibration . . . 13
1.3.4 SABR Volatility Model . . . 13
1.3.4.1 Dynamics . . . 13
1.3.4.2 Hagan Approximation . . . 14
1.3.4.3 ATM Case . . . 14
1.3.4.4 Calibration . . . 14
1.3.4.5 SVI vs SABR . . . 14
1.3.5 Longstaff-Schwartz for American Options . . . 14
1.3.5.1 Algorithm . . . 14
1.3.6 Hedge Strategy Payoffs . . . 15
1.3.6.1 Protective Put . . . 15
1.3.6.2 Collar . . . 15
1.3.6.3 Vertical Put Spread . . . 15
1.3.6.4 Covered Call . . . 16
1.3.7 Pareto Frontier Optimization . . . 16
1.3.7.1 Multi-Objective Formulation . . . 16
1.3.7.2 Pareto Dominance . . . 16
1.3.7.3 Frontier Construction . . . 16
1.3.8 Interpretation . . . 16
1.4 Gamma Scalping . . . 17
1.4.1 Mathematical Framework . . . 17
1.4.1.1 Black-Scholes Model . . . 17
1.4.1.2 Greeks . . . 17
1.4.2 Delta-Hedging Mechanism . . . 18
1.4.2.1 Portfolio Construction . . . 18
1.4.2.2 Delta-Neutral Hedging . . . 18
1.4.2.3 Rehedging Strategies . . . 18
1.4.3 P&L Attribution . . . 19
1.4.3.1 Hedged Portfolio Dynamics . . . 19
1.4.3.2 Gamma P&L . . . 19
1.4.3.3 Theta Decay . . . 19
1.4.3.4 Vega P&L . . . 19
1.4.3.5 Total Attributed P&L . . . 19
1.4.4 Transaction Costs . . . 19
1.4.5 Profitability Condition . . . 20
1.4.5.1 Realized vs Implied Volatility . . . 20
1.4.5.2 Breakeven Realized Volatility . . . 20
1.4.6 Implementation . . . 20
1.4.6.1 Position Initialization . . . 20
1.4.6.2 Simulation Loop . . . 20
1.4.6.3 Expiry Handling . . . 21
1.4.7 Performance Metrics . . . 21
1.4.7.1 Total P&L . . . 21

1.4.7.2 Return on Capital . . . 21
1.4.7.3 Sharpe Ratio . . . 21
1.4.7.4 Hedge Effectiveness . . . 21
1.4.7.5 Realized Volatility . . . 21
1.4.8 Interpretation . . . 21
1.5 Volatility Arbitrage . . . 22
1.5.1 Motivation . . . 22
1.5.2 Realized Volatility Forecasting . . . 22
1.5.2.1 GARCH(1,1) Model . . . 22
1.5.2.2 Exponentially Weighted Moving Average (EWMA) . . . 22
1.5.2.3 Heterogeneous Autoregressive (HAR) Model . . . 22
1.5.3 IV–RV Mispricing Detection . . . 22
1.5.3.1 Arbitrage Signal . . . 22
1.5.3.2 Threshold Strategy . . . 23
1.5.4 No-Arbitrage Constraint Violations . . . 23
1.5.4.1 Butterfly Arbitrage . . . 23
1.5.4.2 Calendar Arbitrage . . . 23
1.5.4.3 Put-Call Parity Violation . . . 23
1.5.5 Delta-Neutral Volatility Trading . . . 23
1.5.5.1 Straddle with Delta Hedging . . . 23
1.5.5.2 P&L Attribution . . . 23
1.5.6 Risk Management . . . 23
1.5.6.1 Greeks Exposure . . . 23
1.5.6.2 VaR and Stress Testing . . . 24
1.5.7 Interpretation . . . 24
1.6 Variance Swaps . . . 24
1.6.1 Motivation . . . 24
1.6.2 Variance Swap Payoff . . . 24
1.6.3 Realized Variance Estimators . . . 25
1.6.3.1 Parkinson (5× efficiency) . . . 25
1.6.3.2 Garman-Klass (8× efficiency) . . . 25
1.6.3.3 Rogers-Satchell (drift-independent) . . . 25
1.6.3.4 Yang-Zhang (14× efficiency) . . . 25
1.6.4 Variance Forecasting . . . 25
1.6.4.1 EWMA . . . 25
1.6.4.2 GARCH(1,1) . . . 25
1.6.5 Carr-Madan Replication . . . 26
1.6.5.1 Model-Free Fair Variance . . . 26
1.6.5.2 Static Replication . . . 26
1.6.5.3 Portfolio Greeks . . . 26
1.6.6 Variance Risk Premium (VRP) . . . 26
1.6.6.1 Definition . . . 26
1.6.6.2 Signal Generation . . . 26
1.6.6.3 VRP Significance Testing . . . 27
1.6.6.4 Backtesting . . . 27
1.6.7 Jump Risk and Tail Events . . . 27
1.6.8 Interpretation . . . 27
1.7 Skew Trading . . . 27
1.7.1 Motivation . . . 27
1.7.2 Skew Metrics . . . 28
1.7.2.1 Risk Reversal (RR25) . . . 28
1.7.2.2 Butterfly (BF25) . . . 28
1.7.2.3 Historical Context . . . 28
1.7.3 Risk Reversal Trade . . . 28
1.7.3.1 Structure . . . 28
1.7.3.2 Greeks . . . 28
1.7.3.3 P&L Drivers . . . 28
1.7.4 Butterfly Spread Trade . . . 29

1.7.4.1 Structure . . . 29
1.7.4.2 Greeks . . . 29
1.7.4.3 P&L Attribution . . . 29
1.7.5 Mean Reversion Signal Construction . . . 29
1.7.5.1 Z-Score Trigger . . . 29
1.7.5.2 Bollinger Band Strategy . . . 29
1.7.5.3 Exit Rules . . . 29
1.7.6 Risk Management . . . 29
1.7.6.1 Delta Hedging . . . 29
1.7.6.2 Scenario Analysis . . . 30
1.7.6.3 Correlation Risk . . . 30
1.7.7 Interpretation . . . 30
1.8 FX Hedging & Futures Options . . . 30
1.8.1 Motivation . . . 30
1.8.2 Forward and Futures Pricing . . . 30
1.8.2.1 Covered Interest Rate Parity . . . 30
1.8.2.2 Forward Points . . . 31
1.8.2.3 Futures Price . . . 31
1.8.2.4 Basis . . . 31
1.8.3 Black-76 Model for Futures Options . . . 31
1.8.3.1 Model Setup . . . 31
1.8.3.2 Option Pricing Formulas . . . 31
1.8.3.3 Key Differences from Black-Scholes . . . 32
1.8.3.4 Greeks for Futures Options . . . 32
1.8.4 Margin and Capital Efficiency . . . 32
1.8.4.1 Initial Margin . . . 32
1.8.4.2 Maintenance Margin . . . 32
1.8.4.3 Variation Margin . . . 33
1.8.4.4 SPAN Methodology . . . 33
1.8.5 Hedge Ratio Optimization . . . 33
1.8.5.1 Exposure Calculation . . . 33
1.8.5.2 Futures Contracts Needed . . . 33
1.8.5.3 Minimum Variance Hedge Ratio . . . 33
1.8.5.4 Hedge Effectiveness . . . 33
1.8.6 Roll Management . . . 34
1.8.6.1 Roll Decision . . . 34
1.8.6.2 Roll P&L . . . 34
1.8.7 Hedge Strategies . . . 34
1.8.7.1 Static Hedge . . . 34
1.8.7.2 Dynamic Hedge (Time-Based) . . . 34
1.8.7.3 Minimum Variance Hedge . . . 34
1.8.7.4 Optimal Cost Hedge . . . 34
1.8.8 Performance Metrics . . . 35
1.8.8.1 Unhedged P&L . . . 35
1.8.8.2 Hedged P&L . . . 35
1.8.8.3 Sharpe Ratio . . . 35
1.8.8.4 Maximum Drawdown . . . 35
1.8.9 Interpretation . . . 35
1.9 Dispersion Trading . . . 36
1.9.1 Motivation . . . 36
1.9.2 Implied Correlation from Volatility . . . 36
1.9.2.1 Volatility Decomposition . . . 36
1.9.2.2 Implied Correlation . . . 36
1.9.3 Dispersion Metrics . . . 36
1.9.3.1 Dispersion Level . . . 36
1.9.3.2 Dispersion Z-Score . . . 37
1.9.4 Trade Construction . . . 37
1.9.4.1 Long Dispersion . . . 37

1.9.4.2 Short Dispersion . . . 37
1.9.4.3 Delta Neutrality . . . 37
1.9.5 P&L Attribution . . . 37
1.9.6 Interpretation . . . 38
1.10 Earnings Volatility Scanner . . . 38
1.10.1 Motivation . . . 38
1.10.2 Term Structure Analysis . . . 38
1.10.2.1 Term Structure Slope . . . 38
1.10.2.2 Term Structure Ratio . . . 38
1.10.2.3 IV/RV Comparison . . . 39
1.10.3 Expected Move Calculation . . . 39
1.10.4 Kelly Criterion Sizing . . . 39
1.10.4.1 Kelly Fraction . . . 39
1.10.4.2 Fractional Kelly . . . 39
1.10.5 Trade Structures . . . 39
1.10.5.1 Calendar Spread . . . 39
1.10.5.2 Short Straddle . . . 39
1.10.6 Filter Criteria . . . 40
1.10.7 Interpretation . . . 40
1.11 Forward Factor Scanner . . . 40
1.11.1 Motivation . . . 40
1.11.2 Forward Volatility Calculation . . . 40
1.11.2.1 Variance Decomposition . . . 40
1.11.2.2 Forward Variance . . . 40
1.11.2.3 Forward Volatility . . . 41
1.11.3 Forward Factor . . . 41
1.11.3.1 Definition . . . 41
1.11.3.2 Interpretation . . . 41
1.11.4 Calendar Spread Structures . . . 41
1.11.4.1 ATM Calendar . . . 41
1.11.4.2 Double Calendar . . . 41
1.11.5 Position Sizing . . . 41
1.11.5.1 Kelly Fraction by Forward Factor . . . 41
1.11.6 Interpretation . . . 42
1.12 Pairs Trading . . . 42
1.12.1 Motivation . . . 42
1.12.2 Cointegration Testing . . . 42
1.12.2.1 Engle-Granger Two-Step Method . . . 42
1.12.2.2 Augmented Dickey-Fuller Test . . . 42
1.12.2.3 Lag Selection . . . 43
1.12.2.4 Johansen Cointegration Test . . . 43
1.12.3 Spread Construction . . . 43
1.12.3.1 OLS Hedge Ratio . . . 43
1.12.3.2 Total Least Squares Regression . . . 43
1.12.3.3 Spread Z-Score . . . 44
1.12.4 Mean Reversion Half-Life . . . 44
1.12.4.1 AR(1) Model . . . 44
1.12.4.2 Half-Life Calculation . . . 44
1.12.4.3 Dynamic Half-Life Monitoring . . . 44
1.12.5 Trading Signals . . . 45
1.12.5.1 Entry Rules . . . 45
1.12.5.2 Exit Rules . . . 45
1.12.6 Position Sizing . . . 45
1.12.6.1 Dollar Neutrality . . . 45
1.12.6.2 Beta Neutrality . . . 45
1.12.7 Performance Metrics . . . 45
1.12.8 Interpretation . . . 45
1.13 Pre-Earnings Straddle Scanner . . . 46

1.13.1 Motivation . . . 46
1.13.2 Signal Construction . . . 46
1.13.2.1 Four Predictive Signals . . . 46
1.13.3 Linear Regression Model . . . 46
1.13.3.1 Prediction Equation . . . 46
1.13.3.2 Typical Coefficients . . . 46
1.13.4 Implied Move Calculation . . . 47
1.13.5 Kelly Sizing . . . 47
1.13.5.1 Historical Statistics . . . 47
1.13.5.2 Kelly Fraction . . . 47
1.13.5.3 Conditional Sizing . . . 47
1.13.6 Recommendation Thresholds . . . 47
1.13.7 Interpretation . . . 47
1.14 Skew Verticals Scanner . . . 48
1.14.1 Motivation . . . 48
1.14.2 Skew Metrics . . . 48
1.14.2.1 Normalized Skew . . . 48
1.14.2.2 Skew Z-Score . . . 48
1.14.2.3 Variance Risk Premium . . . 48
1.14.3 Vertical Spread Construction . . . 48
1.14.3.1 Bull Call Spread . . . 48
1.14.3.2 Bear Put Spread . . . 49
1.14.3.3 Strike Selection . . . 49
1.14.4 Probability of Profit . . . 49
1.14.4.1 Lognormal Approximation . . . 49
1.14.4.2 Expected Value . . . 49
1.14.5 Multi-Factor Edge Score . . . 49
1.14.6 Filter Criteria . . . 50
1.14.7 Recommendation Thresholds . . . 50
1.14.8 Interpretation . . . 50
1.15 Perpetual Futures Pricing . . . 50
1.15.1 Contract Types . . . 50
1.15.1.1 Linear Perpetual Futures . . . 50
1.15.1.2 Inverse Perpetual Futures . . . 50
1.15.1.3 Quanto Perpetual Futures . . . 50
1.15.2 Funding Mechanism . . . 51
1.15.3 No-Arbitrage Pricing . . . 51
1.15.3.1 Linear Perpetual Futures Price . . . 51
1.15.3.2 Inverse Perpetual Futures Price . . . 51
1.15.3.3 Quanto Perpetual Futures Price . . . 51
1.15.4 Perfect Anchoring . . . 51
1.15.5 Random Maturity Representation . . . 52
1.15.6 Basis Analysis . . . 52
1.15.7 Everlasting Options . . . 52
1.15.7.1 Black-Scholes Everlasting Options . . . 52
1.15.7.2 Greeks . . . 53
1.15.8 Implementation . . . 53
1.15.9 Practical Considerations . . . 53
1.15.9.1 Funding Rate Annualization . . . 53
1.15.9.2 Interest Rate Proxy . . . 53
1.15.9.3 Arbitrage Opportunities . . . 53
1.16 Liquidity Analysis . . . 54
1.16.1 Structural Liquidity Metrics . . . 54
1.16.1.1 Amihud Illiquidity Ratio . . . 54
1.16.1.2 Turnover Ratio . . . 54
1.16.1.3 Spread Proxy . . . 54
1.16.1.4 Volume Volatility . . . 54
1.16.1.5 Relative Volume . . . 54

1.16.2 Composite Liquidity Score . . . 55
1.16.3 Volume-Based Predictive Signals . . . 55
1.16.3.1 On-Balance Volume (OBV) . . . 55
1.16.3.2 Volume Surge Detection . . . 55
1.16.3.3 Smart Money Flow . . . 55
1.16.4 Composite Signal Score . . . 56
1.16.5 Interpretation . . . 56

**2 Valuation** **57**
2.1 DCF (Deterministic) . . . 57
2.1.1 Discounted Cash Flow Foundation . . . 57
2.1.2 Cash Flow Mechanics . . . 57
2.1.2.1 Free Cash Flow to Equity . . . 57
2.1.2.2 Free Cash Flow to Firm . . . 57
2.1.3 Cost of Capital . . . 58
2.1.3.1 Levered Beta . . . 58
2.1.3.2 Cost of Equity (CAPM) . . . 58
2.1.3.3 Currency-Adjusted Equity Risk Premium . . . 58
2.1.3.4 Weighted Average Cost of Capital . . . 58
2.1.4 Growth Rate Estimation . . . 58
2.1.4.1 FCFE Growth Rate . . . 58
2.1.4.2 FCFF Growth Rate . . . 59
2.1.5 Present Value Calculation . . . 59
2.1.5.1 FCFE-Based Valuation . . . 59
2.1.5.2 FCFF-Based Valuation . . . 60
2.1.5.3 Reconciliation . . . 60
2.1.5.4 Intrinsic Value Per Share . . . 60
2.1.6 Interpretation . . . 61
2.1.7 Sector-Specific Models . . . 61
2.1.7.1 Bank Valuation (Excess Return Model) . . . 61
2.1.7.2 Insurance Valuation (Float-Based Model) . . . 62
2.1.7.3 Oil & Gas E&P Valuation (NAV Model) . . . 62
2.1.8 Sensitivity Analysis . . . 63
2.2 DCF (Probabilistic) . . . 64
2.2.1 Motivation . . . 64
2.2.2 Input Sampling . . . 64
2.2.2.1 Historical Distributions . . . 64
2.2.2.2 Sampling Distributions . . . 64
2.2.2.3 Outlier Squashing . . . 64
2.2.3 Growth Rate Sampling . . . 65
2.2.3.1 Fundamental Growth Identity . . . 65
2.2.3.2 Bayesian Smoothing with Sector Priors . . . 65
2.2.3.3 Growth Rate Distribution . . . 65
2.2.4 Stochastic Discount Rates . . . 65
2.2.4.1 Discount Rate Component Sampling . . . 66
2.2.4.2 Cost of Capital Recomputation . . . 66
2.2.5 Cash Flow Projection . . . 66
2.2.5.1 Explicit Forecast Period . . . 66
2.2.5.2 Time-Varying Growth Rates . . . 66
2.2.5.3 Cash Flow Computation . . . 67
2.2.5.4 Terminal Value . . . 67
2.2.5.5 Present Value . . . 67
2.2.6 Statistical Analysis . . . 67
2.2.6.1 Simulation Matrix . . . 67
2.2.6.2 Kernel Density Estimation . . . 68
2.2.6.3 Value Surplus View . . . 68
2.2.7 Multi-Asset Portfolio Frontiers . . . 68
2.2.7.1 Simulation Matrix . . . 68

2.2.7.2 Random Portfolio Generation . . . 68
2.2.7.3 Portfolio Return Distribution . . . 69
2.2.7.4 Portfolio Statistics . . . 69
2.2.7.5 Efficient Frontier Plots . . . 70
2.2.8 Interpretation . . . 70
2.3 GARP / PEG Analysis . . . 71
2.3.1 PEG Ratio . . . 71
2.3.1.1 Growth Rate Selection . . . 71
2.3.1.2 PEGY Ratio . . . 71
2.3.1.3 Fair Value Estimation . . . 71
2.3.2 GARP Scoring . . . 71
2.3.3 PEG Assessment . . . 71
2.4 Growth Analysis . . . 72
2.4.1 Rule of 40 . . . 72
2.4.2 Operating Leverage . . . 72
2.4.3 EV/Revenue Efficiency . . . 72
2.4.4 Implied Growth Rate . . . 72
2.4.5 Growth Classification . . . 72
2.4.6 Growth Scoring . . . 73
2.5 Relative Valuation . . . 74
2.5.1 Peer Similarity Scoring . . . 74
2.5.2 Multiple Analysis . . . 74
2.5.2.1 Premium/Discount . . . 74
2.5.2.2 Implied Valuation . . . 74
2.5.3 Fair Premium Adjustment . . . 74
2.5.4 Relative Score . . . 74
2.5.5 Multiples Analyzed . . . 74
2.6 Normalized Multiples . . . 75
2.6.1 Multiple Extraction . . . 75
2.6.1.1 Price Multiples . . . 75
2.6.1.2 Enterprise Value Multiples . . . 75
2.6.2 Percentile Ranking . . . 75
2.6.3 Premium/Discount and Implied Price . . . 75
2.6.3.1 Premium/Discount . . . 75
2.6.3.2 Implied Price . . . 75
2.6.4 Quality Adjustment . . . 76
2.6.4.1 Growth Premium . . . 76
2.6.4.2 Margin Premium . . . 76
2.6.4.3 Return Premium . . . 76
2.6.4.4 Total Fair Premium . . . 76
2.6.5 Composite Scoring . . . 76
2.6.5.1 Raw Composite Percentile . . . 76
2.6.5.2 Quality-Adjusted Percentile . . . 76
2.6.5.3 Confidence Score . . . 76
2.6.5.4 Value Score . . . 77
2.6.6 Interpretation . . . 77
2.7 Dividend Income Analysis . . . 78
2.7.1 Dividend Discount Models . . . 78
2.7.1.1 Gordon Growth Model (Constant Growth DDM) . . . 78
2.7.1.2 Two-Stage DDM . . . 78
2.7.1.3 H-Model (Gradual Growth Decline) . . . 78
2.7.2 Dividend Coverage . . . 78
2.7.3 Chowder Number . . . 78
2.7.4 Dividend Aristocrat Classification . . . 78
2.7.5 Payout Ratio Assessment . . . 79
2.7.6 Dividend Safety Scoring . . . 79
2.8 REIT Valuation . . . 80
2.8.1 Funds From Operations (FFO) . . . 80

2.8.2 Adjusted FFO (AFFO) . . . 80
2.8.3 Net Asset Value (NAV) . . . 80
2.8.3.1 Sector Cap Rates . . . 80
2.8.4 Valuation Multiples . . . 80
2.8.5 REIT Payout Analysis . . . 80
2.8.6 Mortgage REIT Metrics . . . 81
2.9 ETF Analysis . . . 82
2.9.1 Premium/Discount to NAV . . . 82
2.9.2 Tracking Error . . . 82
2.9.3 Total Cost of Ownership . . . 82
2.9.4 ETF Quality Scoring . . . 82
2.9.5 Derivatives ETF Analysis . . . 82
2.9.5.1 Capture Ratios . . . 82
2.9.5.2 Covered Call ETFs . . . 83
2.9.5.3 Volatility ETF Decay . . . 83
2.9.5.4 Leveraged ETF Volatility Drag . . . 83
2.9.5.5 Buffer ETFs . . . 83
2.10 Crypto Treasury Valuation . . . 84
2.10.1 Net Asset Value . . . 84
2.10.2 Multiple of NAV (mNAV) . . . 84
2.10.3 Premium/Discount . . . 84
2.10.4 Per-Share Exposure . . . 84
2.10.5 Implied Crypto Prices . . . 84
2.10.6 Unrealized Gains . . . 84
2.11 NLP Sentiment Analysis . . . 85
2.11.1 Document Sources . . . 85
2.11.2 Embedding-Based Change Detection . . . 85
2.11.3 Historical Centroid . . . 85
2.11.4 Hedging Language Detection . . . 85
2.11.5 Commitment Strength Scoring . . . 85
2.11.6 Signal Ontology . . . 85
2.11.7 Composite Scoring . . . 86
2.11.8 Interpretation . . . 86
2.12 Tail Risk Forecast . . . 87
2.12.1 Realized Variance . . . 87
2.12.2 HAR-RV Model . . . 87
2.12.3 Jump Detection . . . 87
2.12.4 Value-at-Risk . . . 87
2.12.5 Expected Shortfall . . . 87
2.12.6 Interpretation . . . 88

**3 Monitoring** **89**
3.1 Systematic Risk Early-Warning Signals . . . 89
3.1.1 Covariance Decomposition . . . 89
3.1.1.1 Sample Covariance Matrix . . . 89
3.1.1.2 Eigenvalue Decomposition . . . 89
3.1.1.3 Signal 1: Variance Explained by First Eigenvalue . . . 89
3.1.1.4 Signal 2: Variance Explained by Eigenvalues 2–5 . . . 89
3.1.2 Distance and Graph Construction . . . 90
3.1.2.1 Correlation-Based Distance . . . 90
3.1.2.2 Minimum Spanning Tree . . . 90
3.1.3 Eigenvector Centrality . . . 90
3.1.3.1 MST Adjacency Matrix . . . 90
3.1.3.2 Centrality Computation . . . 90
3.1.3.3 Signal 3: Mean Eigenvector Centrality . . . 90
3.1.3.4 Signal 4: Standard Deviation of Centrality . . . 91
3.1.4 Regime Classification . . . 91
3.1.4.1 Composite Scoring . . . 91

3.1.5 Transition Probability . . . . . 91
3.1.5.1 Signal Normalization . . . . . 91
3.1.5.2 Logistic Combination . . . . . 91

# 1 Pricing

Pricing tools enable the investor to construct optimal portfolios that systematically adapt to market conditions. Rather than assuming time-invariant risk, we recognize that markets exhibit distinct volatility regimes and that tail risk becomes paramount during periods of stress.

## 1.1 Regime-Adaptive Downside Optimization

### 1.1.1 Mathematical Framework

Let $\mathcal{A} = \{1, \ldots, N\}$ denote the universe of $N$ risky assets with returns $R_t = (R_{t,1}, \ldots, R_{t,N})^\top$ observed over periods $t = 1, \ldots, T$. Let $b_t$ denote the benchmark return (e.g., market index) and $r_t^c$ the risk-free cash return at time $t$.

#### 1.1.1.1 Portfolio Returns

A portfolio is specified by weight vector $\mathbf{w} = (w_1, \ldots, w_N)^\top$ with $w_i \geq 0$ for all $i \in \mathcal{A}$ and cash weight $w_c \geq 0$, satisfying the full-investment constraint:

$$\sum_{i=1}^{N} w_i + w_c = 1$$

The portfolio return in scenario $t$ is:

$$p_t = \sum_{i=1}^{N} w_i R_{t,i} + w_c r_t^c = \mathbf{w}^\top \mathbf{R}_t + w_c r_t^c$$

The active return (excess over benchmark) is:

$$a_t = p_t - b_t$$

### 1.1.2 Regime Detection

#### 1.1.2.1 Realized Volatility

Given a return series $\{r_t\}_{t=1}^T$ and rolling window size $h$ (typically 20 trading days), the annualized realized volatility at time $T$ is:

$$\sigma_{\text{realized}}(T) = \sqrt{252} \cdot \sqrt{\frac{1}{h} \sum_{t=T-h+1}^{T} (r_t - \bar{r})^2}$$

where $\bar{r} = \frac{1}{h} \sum_{t=T-h+1}^{T} r_t$ and 252 is the approximate number of trading days per year.

#### 1.1.2.2 Historical Volatility Distribution

To establish regime thresholds, we compute rolling volatilities over a lookback period of $L$ years (typically 3–5 years, i.e., $L \times 252$ trading days):

$$\mathcal{V} = \{\sigma_{\text{realized}}(t) : t = h, h+1, \ldots, L \times 252\}$$

Let $Q_p(\mathcal{V})$ denote the $p$-th quantile of the volatility distribution $\mathcal{V}$.

#### 1.1.2.3 Stress Weight Function

Define lower and upper volatility thresholds:

$$\sigma_L = Q_{0.25}(\mathcal{V})$$
$$\sigma_U = Q_{0.75}(\mathcal{V})$$

The stress weight at current volatility $\sigma$ is:

$$s(\sigma) = \begin{cases} 0 & \text{if } \sigma \leq \sigma_L \quad \text{(calm regime)} \\ \frac{\sigma - \sigma_L}{\sigma_U - \sigma_L} & \text{if } \sigma_L < \sigma < \sigma_U \quad \text{(transition)} \\ 1 & \text{if } \sigma \geq \sigma_U \quad \text{(stress regime)} \end{cases}$$

This smooth transition function avoids abrupt regime switches and provides a continuous measure $s \in [0, 1]$ of market stress intensity.

### 1.1.3 Downside Risk Measures

#### 1.1.3.1 Lower Partial Moment of Order 1

The First Lower Partial Moment (LPM1) measures the expected shortfall below a threshold $\tau < 0$:

$$\text{LPM1}(\mathbf{w}) = \frac{1}{T} \sum_{t=1}^{T} \max(0, \tau - a_t)$$

where $a_t$ is the active return in scenario $t$. This captures the average magnitude of underperformance relative to the threshold, providing a convex risk measure suitable for optimization.

#### 1.1.3.2 Conditional Value-at-Risk

Conditional Value-at-Risk (CVaR) at confidence level $1 - \alpha$ (typically $\alpha = 0.05$ for 95% CVaR) is defined as the expected loss conditional on exceeding the Value-at-Risk (VaR) threshold.

Define the loss at scenario $t$ as:

$$\ell_t = \max(0, -a_t)$$

The VaR at level $1 - \alpha$ is the $1 - \alpha$ quantile of the loss distribution:

$$\text{VaR}_{1-\alpha} = \inf\{\eta : \mathbb{P}[\ell \leq \eta] \geq 1 - \alpha\}$$

The CVaR is then:

$$\text{CVaR}_{1-\alpha} = \mathbb{E}[\ell \mid \ell \geq \text{VaR}_{1-\alpha}]$$

We use the dual representation (Rockafellar–Uryasev):

$$\text{CVaR}_{1-\alpha}(\mathbf{w}) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{\alpha T} \sum_{t=1}^{T} \max(0, \ell_t - \eta) \right\}$$

This formulation allows CVaR to be incorporated directly into a convex optimization problem.

#### 1.1.3.3 Portfolio Beta

The portfolio beta relative to the benchmark is the weighted average of individual asset betas:

$$\beta(\mathbf{w}) = \sum_{i=1}^{N} w_i \beta_i$$

where $\beta_i$ is the beta of asset $i$, estimated via exponentially-weighted covariance with the benchmark.

### 1.1.4 Linear Program Formulation

The portfolio optimization problem is formulated as a convex linear program with auxiliary slack variables.

#### 1.1.4.1 Decision Variables

- $\mathbf{w} \in \mathbb{R}_+^N$: asset weights
- $w_c \in \mathbb{R}_+$: cash weight
- $\mathbf{s}_{\text{LPM}} \in \mathbb{R}_+^T$: LPM1 slack variables
- $\eta \in \mathbb{R}$: CVaR threshold
- $\mathbf{u} \in \mathbb{R}_+^T$: CVaR excess loss slack variables
- $\mathbf{z} \in \mathbb{R}_+^N$: turnover slack variables
- $v \in \mathbb{R}_+$: beta deviation slack variable

#### 1.1.4.2 Objective Function

$$\min_{\mathbf{w}, w_c, \mathbf{s}_{\text{LPM}}, \eta, \mathbf{u}, \mathbf{z}, v} \underbrace{\frac{\lambda_{\text{LPM}}}{T} \sum_{t=1}^{T} s_{\text{LPM},t}}_{\text{LPM1 penalty}} + \underbrace{\lambda_{\text{CVaR}} \left( \eta + \frac{1}{(1-\alpha)T} \sum_{t=1}^{T} u_t \right)}_{\text{CVaR penalty}} + \underbrace{\kappa \sum_{i=1}^{N} z_i}_{\text{turnover cost}} + \underbrace{\lambda_\beta \cdot s(\sigma) \cdot v}_{\text{stress beta penalty}}$$

where:

$\lambda_{\text{LPM}} > 0$: LPM1 risk aversion parameter

$\lambda_{\text{CVaR}} > 0$: CVaR risk aversion parameter

$\lambda_\beta > 0$: beta deviation penalty (only active in stress)

$\kappa = c + \gamma$: combined transaction cost and turnover penalty

$s(\sigma) \in [0, 1]$: current stress weight

#### 1.1.4.3 Constraints

Let $\mathbf{R} \in \mathbb{R}^{T \times N}$ be the matrix of asset scenario returns, $\mathbf{b} \in \mathbb{R}^T$ the benchmark returns, $\mathbf{r}^c \in \mathbb{R}^T$ the cash returns, and $\mathbf{w}_{\text{prev}} \in \mathbb{R}^N$ the previous portfolio weights.

**(1) Full investment:**

$$\sum_{i=1}^{N} w_i + w_c = 1$$

**(2) Active returns:**

$$\mathbf{a} = \mathbf{R}\mathbf{w} + \mathbf{r}^c w_c - \mathbf{b}$$

**(3) LPM1 slack constraints:**

$$s_{\text{LPM},t} \geq \tau - a_t, \quad s_{\text{LPM},t} \geq 0, \quad \forall t = 1, \ldots, T$$

**(4) CVaR slack constraints:**

$$u_t \geq -a_t - \eta, \quad u_t \geq 0, \quad \forall t = 1, \ldots, T$$

**(5) Turnover slack constraints (L1 norm):**

$$z_i \geq w_i - w_{\text{prev},i}, \quad z_i \geq -(w_i - w_{\text{prev},i}), \quad z_i \geq 0, \quad \forall i = 1, \ldots, N$$

**(6) Beta deviation slack constraints (L1 norm):**

$$v \geq \beta(\mathbf{w}) - \beta_{\text{target}}, \quad v \geq -(\beta(\mathbf{w}) - \beta_{\text{target}}), \quad v \geq 0$$

**(7) Non-negativity:**

$$\mathbf{w} \geq \mathbf{0}, \quad w_c \geq 0$$

#### 1.1.4.4 Problem Structure

The resulting optimization problem is a convex linear program of the form:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} \geq \mathbf{b} \\ & \mathbf{A}_{\text{eq}}\mathbf{x} = \mathbf{b}_{\text{eq}} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{x} = (\mathbf{w}, w_c, \mathbf{s}_{\text{LPM}}, \eta, \mathbf{u}, \mathbf{z}, v)^\top$ is the concatenated decision vector. This can be solved efficiently using interior-point methods (e.g., CLARABEL, SCS, OSQP solvers).

### 1.1.5 Regime Adaptation Mechanism

#### 1.1.5.1 Stress-Conditional Beta Control

The key innovation is that the beta penalty term is modulated by the stress weight $s(\sigma)$:

$$\text{Beta Penalty} = \lambda_\beta \cdot s(\sigma) \cdot |\beta(\mathbf{w}) - \beta_{\text{target}}|$$

This design has three regimes:

1. **Calm regime** ($s \approx 0$): Beta penalty is negligible. The optimizer focuses purely on downside risk (LPM1, CVaR) and turnover minimization. Market exposure is unconstrained.
2. **Transition regime** ($0 < s < 1$): Beta penalty gradually increases. The portfolio begins defensive positioning while maintaining some market exposure.
3. **Stress regime** ($s \approx 1$): Beta penalty is fully active. The portfolio is strongly penalized for deviating from the target beta (typically $\beta_{\text{target}} \in [0.5, 0.7]$), forcing defensive positioning with reduced market sensitivity.

#### 1.1.5.2 Economic Rationale

During calm markets, investors can tolerate higher market beta to capture upside. During stress, systematic market exposure becomes the dominant risk factor, overwhelming idiosyncratic considerations. By dynamically adjusting beta exposure based on volatility regime, the strategy:

- Maintains participation during bull markets
- Reduces drawdowns during bear markets
- Avoids whipsaw from binary regime classification
- Respects transaction costs through turnover penalties

### 1.1.6 Solution and Rebalancing

#### 1.1.6.1 Optimization Frequency

The LP is solved at each rebalancing period (typically weekly or monthly). The solution provides:

- Optimal weights $\mathbf{w}^*$ and cash $w_c^*$
- Realized downside risk: LPM1$^*$, CVaR$^*$
- Turnover incurred: $\|\mathbf{w}^* - \mathbf{w}_{\text{prev}}\|_1$
- Portfolio beta: $\beta(\mathbf{w}^*)$

#### 1.1.6.2 Rebalancing Decision Rule

To avoid excessive trading, rebalancing is triggered only when:

$$\|\mathbf{w}_{\text{current}} - \mathbf{w}^*\|_1 > \epsilon$$

where $\epsilon$ is a drift tolerance threshold (typically 0.05–0.10).

### 1.1.7 Performance Metrics

#### 1.1.7.1 Ex-Post Risk Evaluation

After observing realized returns $\{a_t\}$, we evaluate:

- **LPM1**: $\text{LPM1} = \frac{1}{T}\sum_{t=1}^{T} \max(0, \tau - a_t)$
- **CVaR**: $\text{CVaR}_{95} = \eta^* + \frac{1}{0.05 \cdot T}\sum_{t=1}^{T} \max(0, \ell_t - \eta^*)$
- **Realized volatility**: $\sigma_p = \sqrt{\text{Var}(a_t)}$
- **Turnover**: $\sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}|$

#### 1.1.7.2 Sharpe-Type Ratios

While the objective minimizes downside risk rather than variance, we can still compute:

- **Sharpe ratio**: $\frac{\mathbb{E}[a_t]}{\sigma_p}$
- **Sortino ratio**: $\frac{\mathbb{E}[a_t]}{\sqrt{\text{LPM2}}}$ where LPM2 uses squared shortfalls
- **CVaR ratio**: $\frac{\mathbb{E}[a_t]}{\text{CVaR}_{95}}$

### 1.1.8 Interpretation

The regime-adaptive downside optimization framework provides:

1. **Tail risk protection**: CVaR directly targets the expected severity of worst-case scenarios, unlike variance-based methods that treat upside and downside equally.
2. **Systematic risk management**: Beta control during stress prevents catastrophic drawdowns from market beta exposure, the primary driver of portfolio losses during crises.
3. **Continuous adaptation**: The smooth stress weight $s(\sigma) \in [0, 1]$ avoids the instability of binary regime switching while maintaining responsiveness to changing market conditions.
4. **Transaction cost awareness**: Turnover penalties and drift thresholds prevent excessive rebalancing, ensuring net-of-cost performance.
5. **Convex formulation**: The LP structure guarantees global optimality and computational efficiency, enabling real-time portfolio construction even for large asset universes.

The framework rejects the mean-variance paradigm in favor of downside-focused objectives that better align with investor utility and observed market behavior during stress episodes.

## 1.2 Market Regime Forecast

Market regime detection is fundamental to adaptive portfolio management. Different market environments—bull trends, bear trends, sideways consolidation, high volatility, low volatility—call for different strategies. This section presents four complementary approaches to regime classification, ranging from classical econometric models to Bayesian online methods.

### 1.2.1 GARCH(1,1) Volatility Model

The Generalized Autoregressive Conditional Heteroskedasticity model captures volatility clustering—the empirical observation that large returns tend to be followed by large returns (of either sign).

#### 1.2.1.1 Model Specification

Let $r_t$ denote the log return at time $t$. The GARCH(1,1) model specifies:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1) \tag{1}$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \tag{2}$$

where:

- $\omega > 0$: baseline variance (intercept)
- $\alpha \geq 0$: ARCH coefficient (shock reaction)
- $\beta \geq 0$: GARCH coefficient (persistence)
- $\alpha + \beta < 1$: stationarity condition

#### 1.2.1.2 Key Properties

The **persistence** $\rho = \alpha + \beta$ measures how long volatility shocks persist. Values near 1 indicate highly persistent volatility clustering. The **unconditional (long-run) volatility** is:

$$\bar{\sigma} = \sqrt{\frac{\omega}{1 - \alpha - \beta}} \tag{3}$$

The model implies an exponential decay of volatility shocks:

$$\mathbb{E}[\sigma_{t+h}^2 | \mathcal{F}_t] = \bar{\sigma}^2 + \rho^h (\sigma_t^2 - \bar{\sigma}^2) \tag{4}$$

#### 1.2.1.3 Maximum Likelihood Estimation

Parameters $\theta = (\omega, \alpha, \beta)$ are estimated by maximizing the log-likelihood:

$$\mathcal{L}(\theta) = -\frac{1}{2} \sum_{t=1}^{T} \left[ \log(2\pi) + \log(\sigma_t^2) + \frac{(r_t - \mu)^2}{\sigma_t^2} \right] \tag{5}$$

subject to $\omega > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta < 1$. Optimization uses Nelder-Mead or BFGS with penalty terms for constraint violations.

#### 1.2.1.4 Volatility Regime Classification

Current volatility is classified relative to historical percentiles:

$$\text{Vol Regime} = \begin{cases} \text{High Vol} & \text{if } \sigma_t > P_{80}(\sigma_{1:T}) \\ \text{Low Vol} & \text{if } \sigma_t < P_{20}(\sigma_{1:T}) \\ \text{Normal Vol} & \text{otherwise} \end{cases} \tag{6}$$

### 1.2.2 Hidden Markov Model for Trend Detection

While GARCH captures volatility dynamics, trend classification requires modeling the latent market state. A Hidden Markov Model (HMM) with Gaussian emissions provides a principled approach.

#### 1.2.2.1 Model Specification

Let $S_t \in \{1, 2, 3\}$ denote the hidden state (Bull, Bear, Sideways) at time $t$. The HMM specifies:

$$P(S_t = j | S_{t-1} = i) = A_{ij} \quad \text{(transition probabilities)} \tag{7}$$

$$r_t | S_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2) \quad \text{(emission distributions)} \tag{8}$$

The transition matrix $\mathbf{A}$ is row-stochastic: $\sum_j A_{ij} = 1$ for all $i$.

#### 1.2.2.2 Baum-Welch Algorithm (EM)

Parameters $\theta = (\mathbf{A}, \boldsymbol{\mu}, \boldsymbol{\sigma}^2, \boldsymbol{\pi})$ are estimated via Expectation-Maximization.

**E-step (Forward-Backward):**

Define forward probabilities $\alpha_t(i) = P(r_{1:t}, S_t = i)$ and backward probabilities $\beta_t(i) = P(r_{t+1:T} | S_t = i)$:

$$\alpha_t(j) = \left[ \sum_{i=1}^{K} \alpha_{t-1}(i) A_{ij} \right] b_j(r_t) \tag{9}$$

$$\beta_t(i) = \sum_{j=1}^{K} A_{ij} b_j(r_{t+1}) \beta_{t+1}(j) \tag{10}$$

where $b_j(r) = \mathcal{N}(r; \mu_j, \sigma_j^2)$ is the emission density.

The posterior state probabilities are:

$$\gamma_t(i) = P(S_t = i | r_{1:T}) = \frac{\alpha_t(i)\beta_t(i)}{\sum_{j=1}^{K} \alpha_t(j)\beta_t(j)} \tag{11}$$

**M-step (Parameter Update):**

$$\hat{\mu}_k = \frac{\sum_{t=1}^{T} \gamma_t(k) r_t}{\sum_{t=1}^{T} \gamma_t(k)} \tag{12}$$

$$\hat{\sigma}_k^2 = \frac{\sum_{t=1}^{T} \gamma_t(k)(r_t - \hat{\mu}_k)^2}{\sum_{t=1}^{T} \gamma_t(k)} \tag{13}$$

$$\hat{A}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \tag{14}$$

where $\xi_t(i,j) = P(S_t = i, S_{t+1} = j | r_{1:T})$.

#### 1.2.2.3 Viterbi Decoding

The most likely state sequence is found via dynamic programming:

$$\delta_t(j) = \max_i \left[ \delta_{t-1}(i) A_{ij} \right] b_j(r_t) \tag{15}$$

with backtracking to recover $\hat{S}_{1:T}$.

#### 1.2.2.4 State Relabeling

HMM states are arbitrary—state 1 is not inherently "Bull." Post-fitting, states are relabeled by sorting emission means: $\mu_{\text{Bull}} > \mu_{\text{Sideways}} > \mu_{\text{Bear}}$.

### 1.2.3 Markov-Switching GARCH (MS-GARCH)

MS-GARCH integrates regime switching directly into volatility dynamics, allowing GARCH parameters to vary across regimes.

#### 1.2.3.1 Model Specification

Let $S_t \in \{1, \ldots, K\}$ be the latent regime. The MS-GARCH model specifies:

$$r_t = \mu_{S_t} + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t \tag{16}$$

$$\sigma_t^2 = \omega_{S_t} + \alpha_{S_t}\varepsilon_{t-1}^2 + \beta_{S_t}\sigma_{t-1}^2 \tag{17}$$

$$P(S_t = j | S_{t-1} = i) = P_{ij} \tag{18}$$

Each regime $k$ has its own parameter vector $(\mu_k, \omega_k, \alpha_k, \beta_k)$, capturing regime-specific volatility dynamics.

#### 1.2.3.2 Interpretation

For a 3-regime model (Low/Normal/High volatility):

- **Low Vol Regime**: Small $\omega_k$, high $\beta_k$ (calm, persistent)
- **Normal Vol Regime**: Moderate parameters
- **High Vol Regime**: Large $\omega_k$, high $\alpha_k$ (reactive to shocks)

#### 1.2.3.3 Estimation

Estimation uses EM with the forward algorithm adapted for GARCH dynamics:

$$\alpha_t(j) \propto \left[\sum_{i=1}^{K} \alpha_{t-1}(i) P_{ij}\right] \cdot \phi\left(\frac{r_t - \mu_j}{\sigma_t^{(j)}}\right) / \sigma_t^{(j)} \tag{19}$$

where $\sigma_t^{(j)}$ is the conditional volatility under regime $j$, computed using regime-weighted lagged values.

#### 1.2.3.4 Regime Classification

Regimes are relabeled by unconditional volatility:

$$\bar{\sigma}_k = \sqrt{\frac{\omega_k}{1 - \alpha_k - \beta_k}} \tag{20}$$

The regime with smallest $\bar{\sigma}_k$ is labeled "Low Vol," etc.

### 1.2.4 Bayesian Online Changepoint Detection (BOCPD)

BOCPD provides real-time regime change detection by maintaining a posterior distribution over *run lengths*—the time since the last changepoint.

#### 1.2.4.1 Run Length Formulation

Let $r_t$ denote the run length at time $t$ (time since last regime change). The key recursion is:

$$P(r_t | x_{1:t}) \propto \sum_{r_{t-1}} P(r_t | r_{t-1}) \cdot P(x_t | r_{t-1}, x_{1:t-1}) \cdot P(r_{t-1} | x_{1:t-1}) \tag{21}$$

#### 1.2.4.2 Hazard Function

The prior probability of a changepoint is given by the hazard function $H(r)$. A constant hazard $H(r) = 1/\lambda$ implies geometric run lengths with expected duration $\lambda$:

$$P(r_t | r_{t-1}) = \begin{cases} H(r_{t-1}) & \text{if } r_t = 0 \text{ (changepoint)} \\ 1 - H(r_{t-1}) & \text{if } r_t = r_{t-1} + 1 \text{ (growth)} \end{cases} \tag{22}$$

#### 1.2.4.3 Underlying Predictive Model

Within each regime segment, observations are modeled using a conjugate Normal-Inverse-Gamma prior:

$$\mu|\sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2/\kappa_0) \tag{23}$$

$$\sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0) \tag{24}$$

The predictive distribution for a new observation given sufficient statistics $(n, \bar{x}, S)$ is Student-$t$:

$$x_{\text{new}}|x_{1:n} \sim t_{2\alpha_n}\left(\mu_n, \frac{\beta_n(\kappa_n + 1)}{\alpha_n \kappa_n}\right) \tag{25}$$

where the posterior parameters update via:

$$\kappa_n = \kappa_0 + n, \quad \mu_n = \frac{\kappa_0 \mu_0 + n\bar{x}}{\kappa_n} \tag{26}$$

$$\alpha_n = \alpha_0 + \frac{n}{2}, \quad \beta_n = \beta_0 + \frac{S}{2} + \frac{\kappa_0 n(\bar{x} - \mu_0)^2}{2\kappa_n} \tag{27}$$

#### 1.2.4.4 Online Update Algorithm

For each new observation $x_t$:

1. Compute predictive probabilities $P(x_t|r_{t-1})$ for each run length
2. Compute growth probabilities: $P(r_t = r_{t-1} + 1) \propto (1 - H) \cdot P(x_t|r_{t-1}) \cdot P(r_{t-1})$
3. Compute changepoint probability: $P(r_t = 0) \propto \sum_{r_{t-1}} H \cdot P(x_t|r_{t-1}) \cdot P(r_{t-1})$
4. Normalize and update sufficient statistics

#### 1.2.4.5 Changepoint Detection

A changepoint is detected when:

$$P(r_t \leq \tau | x_{1:t}) > \theta \tag{28}$$

for threshold $\tau$ (recent changepoint) and detection threshold $\theta$ (e.g., 0.5).

#### 1.2.4.6 Regime Statistics

Current regime characteristics are computed as expectations over the run length distribution:

$$\hat{\mu}_{\text{regime}} = \sum_r P(r_t = r|x_{1:t}) \cdot \bar{x}^{(r)} \tag{29}$$

$$\hat{\sigma}^2_{\text{regime}} = \sum_r P(r_t = r|x_{1:t}) \cdot s^{2(r)} \tag{30}$$

where $\bar{x}^{(r)}$ and $s^{2(r)}$ are the sample mean and variance for run length $r$.

### 1.2.5 Gaussian Process Regression

Gaussian Processes provide a non-parametric approach to regime detection with full uncertainty quantification. The GP posterior mean captures the underlying trend, while posterior variance indicates regime stability.

#### 1.2.5.1 Model Specification

A Gaussian Process defines a distribution over functions:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \tag{31}$$

where $m(\mathbf{x})$ is the mean function (typically zero) and $k(\mathbf{x}, \mathbf{x}')$ is the covariance (kernel) function.

For return modeling with time index $t$, observations are:

$$r_t = f(t) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_n^2) \tag{32}$$

#### 1.2.5.2 Kernel Functions

The kernel encodes assumptions about function smoothness:

**Squared Exponential (RBF):**

$$k_{\mathrm{SE}}(t,t') = \sigma_f^2 \exp\left(-\frac{(t-t')^2}{2\ell^2}\right) \tag{33}$$

**Matérn 5/2** (moderate smoothness, recommended):

$$k_{\mathrm{M52}}(t,t') = \sigma_f^2 \left(1 + \frac{\sqrt{5}|t-t'|}{\ell} + \frac{5(t-t')^2}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}|t-t'|}{\ell}\right) \tag{34}$$

**Rational Quadratic:**

$$k_{\mathrm{RQ}}(t,t') = \sigma_f^2 \left(1 + \frac{(t-t')^2}{2\alpha\ell^2}\right)^{-\alpha} \tag{35}$$

Parameters: $\ell$ (length scale), $\sigma_f^2$ (signal variance), $\sigma_n^2$ (noise variance).

#### 1.2.5.3 GP Posterior

Given training data $\mathbf{X}, \mathbf{y}$ and test points $\mathbf{X}_*$, the posterior is:

$$\mathbf{f}_* | \mathbf{X}, \mathbf{y}, \mathbf{X}_* \sim \mathcal{N}(\bar{\mathbf{f}}_*, \mathrm{cov}(\mathbf{f}_*)) \tag{36}$$

$$\bar{\mathbf{f}}_* = \mathbf{K}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \tag{37}$$

$$\mathrm{cov}(\mathbf{f}_*) = \mathbf{K}_{**} - \mathbf{K}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{K}_* \tag{38}$$

where $\mathbf{K} = k(\mathbf{X}, \mathbf{X})$, $\mathbf{K}_* = k(\mathbf{X}, \mathbf{X}_*)$, $\mathbf{K}_{**} = k(\mathbf{X}_*, \mathbf{X}_*)$.

#### 1.2.5.4 Hyperparameter Optimization

Hyperparameters $\theta = (\ell, \sigma_f, \sigma_n)$ are optimized by maximizing the log marginal likelihood:

$$\log p(\mathbf{y}|\mathbf{X}, \theta) = -\frac{1}{2}\mathbf{y}^T \mathbf{K}_y^{-1} \mathbf{y} - \frac{1}{2}\log|\mathbf{K}_y| - \frac{n}{2}\log(2\pi) \tag{39}$$

where $\mathbf{K}_y = \mathbf{K} + \sigma_n^2 \mathbf{I}$.

#### 1.2.5.5 Regime Classification

**Trend** is classified from the posterior mean at the current time:

$$\text{Trend} = \begin{cases} \text{Bull} & \text{if } \bar{f}_T \times 252 > 10\% \\ \text{Bear} & \text{if } \bar{f}_T \times 252 < -5\% \\ \text{Sideways} & \text{otherwise} \end{cases} \tag{40}$$

**Volatility regime** is classified by comparing GP uncertainty to historical volatility:

$$\text{Vol Regime} = \begin{cases} \text{High Vol} & \text{if } \sqrt{\mathrm{var}(f_T)} > 1.5 \times \sigma_{\mathrm{hist}} \\ \text{Low Vol} & \text{if } \sqrt{\mathrm{var}(f_T)} < 0.7 \times \sigma_{\mathrm{hist}} \\ \text{Normal Vol} & \text{otherwise} \end{cases} \tag{41}$$

**Anomaly score** measures how far the current observation deviates from the GP prediction:

$$z = \frac{|r_T - \bar{f}_T|}{\sqrt{\mathrm{var}(f_T)}} \tag{42}$$

High anomaly scores suggest potential regime change or unusual market behavior.

### 1.2.6 Model Comparison and Application

#### 1.2.6.1 Summary of Approaches

| Model | Primary Focus | Estimation | Online? | Uncertainty? |
|---|---|---|---|---|
| GARCH(1,1) | Volatility dynamics | MLE | No | No |
| HMM | Trend classification | Baum-Welch | No | Probabilities |
| MS-GARCH | Regime-dependent vol | EM | No | Probabilities |
| BOCPD | Changepoint detection | Bayesian | Yes | Run length dist. |
| GP | Trend + uncertainty | MLE | No | Full posterior |

#### 1.2.6.2 Covered Call Suitability

For income-generating covered call ETFs, regime classification determines strategy suitability. The optimal environment is *sideways trend* with *high volatility*—premium income is maximized while directional risk (opportunity cost from capped upside, or loss from declining underlying) is minimized.

| Trend × Vol Regime | Suitability | Rationale |
|---|---|---|
| Sideways + High Vol | 5/5 | Optimal: high premium, low directional risk |
| Sideways + Normal Vol | 4/5 | Good: decent premium, range-bound |
| Bear + High Vol | 4/5 | Premium cushions losses |
| Bull + High Vol | 3/5 | High premium but opportunity cost |
| Sideways + Low Vol | 3/5 | Compressed premiums |
| Bear + Normal Vol | 3/5 | Moderate cushion |
| Bull + Normal Vol | 2/5 | Opportunity cost dominates |
| Bear + Low Vol | 2/5 | Low premium, downside risk |
| Bull + Low Vol | 1/5 | Worst: low premium + capped upside |

## 1.3 Options Hedging

### 1.3.1 Motivation

Portfolio protection requires hedging downside risk while managing cost. Traditional portfolio insurance involves buying protective puts, but this can be expensive. This framework explores the **Pareto frontier** of cost vs protection across four hedging strategies, enabling investors to select optimal trade-offs based on their risk tolerance and budget constraints.

#### 1.3.1.1 Hedging Strategies

1. **Protective Put**: Maximum downside protection (buy put options at strike $K_{\text{put}}$)
2. **Collar**: Bounded risk/reward (buy put + sell call for lower net cost)
3. **Vertical Spread**: Limited protection at reduced cost (buy/sell puts at different strikes)
4. **Covered Call**: Income generation with slight protection (sell call against holdings)

### 1.3.2 Black-Scholes-Merton Pricing

#### 1.3.2.1 European Call and Put

Let $S$ denote the current stock price, $K$ the strike price, $T$ the time to expiry, $r$ the risk-free rate, $q$ the dividend yield, and $\sigma$ the volatility.

**Call price:**

$$C(S, K, T, r, q, \sigma) = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2)$$

**Put price:**

$$P(S, K, T, r, q, \sigma) = Ke^{-rT}N(-d_2) - Se^{-qT}N(-d_1)$$

where:

$$d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

and $N(\cdot)$ is the cumulative standard normal distribution.

#### 1.3.2.2 Put-Call Parity

The no-arbitrage relationship between European calls and puts:

$$C - P = Se^{-qT} - Ke^{-rT}$$

This ensures consistent pricing across the option chain and enables arbitrage detection.

#### 1.3.2.3 Greeks

**Delta** ($\Delta$): Sensitivity to underlying price

$$\Delta_{\text{call}} = e^{-qT}N(d_1)$$
$$\Delta_{\text{put}} = e^{-qT}(N(d_1) - 1)$$

**Gamma** ($\Gamma$): Convexity (rate of change of delta)

$$\Gamma = \frac{e^{-qT}n(d_1)}{S\sigma\sqrt{T}}$$

where $n(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ is the standard normal PDF.

**Vega** ($\mathcal{V}$): Sensitivity to volatility (per 1% change)

$$\mathcal{V} = Se^{-qT}n(d_1)\sqrt{T}$$

**Theta** ($\Theta$): Time decay (per day)

$$\Theta_{\text{call}} = -\frac{Sn(d_1)\sigma e^{-qT}}{2\sqrt{T}} - rKe^{-rT}N(d_2) + qSe^{-qT}N(d_1)$$
$$\Theta_{\text{put}} = -\frac{Sn(d_1)\sigma e^{-qT}}{2\sqrt{T}} + rKe^{-rT}N(-d_2) - qSe^{-qT}N(-d_1)$$

**Rho** ($\rho$): Sensitivity to interest rate (per 1% change)

$$\rho_{\text{call}} = KTe^{-rT}N(d_2)$$
$$\rho_{\text{put}} = -KTe^{-rT}N(-d_2)$$

### 1.3.3 SVI Volatility Surface

The Stochastic Volatility Inspired (SVI) model parametrizes total implied variance as a function of log-moneyness.

#### 1.3.3.1 Total Variance Formula

$$w(k;\theta) = a + b\left\{\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2}\right\}$$

where:

- $k = \ln(K/F)$ is log-moneyness ($F$ is the forward price)

- $\theta = (a, b, \rho, m, \sigma)$ are SVI parameters
- $a \geq 0$: vertical translation
- $b \geq 0$: slope of the wings
- $\rho \in [-1, 1]$: rotation (skew direction)
- $m$: horizontal translation (ATM shift)
- $\sigma > 0$: curvature at the ATM

#### 1.3.3.2 Implied Volatility

$$\mathrm{IV}(k, T) = \sqrt{\frac{w(k; \theta)}{T}}$$

#### 1.3.3.3 No-Arbitrage Conditions

To prevent arbitrage opportunities:

**Butterfly arbitrage:**

$$\frac{\partial^2 C}{\partial K^2} \geq 0 \quad \Rightarrow \quad \frac{b}{\sigma} \geq |\rho|$$

**Calendar arbitrage:**

$$\frac{\partial w}{\partial T} \geq 0 \quad \Rightarrow \quad a \geq 0$$

#### 1.3.3.4 Calibration

Minimize sum of squared errors between model and market total variance:

$$\min_{\theta} \sum_{i=1}^{N} \left[ w(k_i; \theta) - w_{\mathrm{market}}(k_i) \right]^2$$

subject to the no-arbitrage constraints. Solved via differential evolution (global optimizer) with fallback to L-BFGS-B.

### 1.3.4 SABR Volatility Model

The SABR (Stochastic Alpha Beta Rho) model captures the dynamics of forward prices with stochastic volatility. Unlike SVI, which is a static parametrization, SABR describes a stochastic process.

#### 1.3.4.1 Dynamics

Under the risk-neutral measure:

$$\begin{aligned} dF_t &= \sigma_t F_t^{\beta} \, dW_t^1 \\ d\sigma_t &= \nu \sigma_t \, dW_t^2 \\ dW_t^1 \cdot dW_t^2 &= \rho \, dt \end{aligned}$$

where:

- $F_t$ is the forward price
- $\sigma_t$ is the stochastic volatility with initial value $\alpha = \sigma_0$
- $\beta \in [0, 1]$: CEV exponent ($0$ = normal, $0.5$ = typical equity, $1$ = lognormal)
- $\nu > 0$: volatility of volatility
- $\rho \in (-1, 1)$: correlation between price and vol processes

#### 1.3.4.2 Hagan Approximation

Hagan et al. (2002) derive a closed-form approximation for implied volatility. For the general case ($F \neq K$):

$$\sigma_{\text{impl}}(K,T) = \frac{\alpha}{(FK)^{(1-\beta)/2}} \cdot \frac{z}{\chi(z)} \cdot \frac{1}{1 + \frac{(1-\beta)^2}{24}\ln^2\frac{F}{K} + \frac{(1-\beta)^4}{1920}\ln^4\frac{F}{K}} \cdot (1+\epsilon T)$$

where:

$$\begin{aligned} z &= \frac{\nu}{\alpha}(FK)^{(1-\beta)/2}\ln\frac{F}{K} \\ \chi(z) &= \ln\left(\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right) \\ \epsilon &= \frac{(1-\beta)^2\alpha^2}{24\bar{F}^{2(1-\beta)}} + \frac{\rho\beta\nu\alpha}{4\bar{F}^{1-\beta}} + \frac{(2-3\rho^2)\nu^2}{24} \end{aligned}$$

with $\bar{F} = (F+K)/2$.

#### 1.3.4.3 ATM Case

When $F = K$:

$$\sigma_{\text{ATM}} = \frac{\alpha}{F^{1-\beta}} \cdot (1+\epsilon T)$$

#### 1.3.4.4 Calibration

For each expiry, calibrate $(\alpha, \beta, \rho, \nu)$ by minimizing:

$$\min_{\alpha,\rho,\nu} \sum_{i=1}^{N} \left[\sigma_{\text{impl}}(K_i) - \sigma_{\text{market}}(K_i)\right]^2$$

with $\beta$ typically fixed (e.g., $\beta = 0.5$). Solved via differential evolution.

#### 1.3.4.5 SVI vs SABR

SVI is the default model for this framework:

- SVI: 5 free parameters per expiry, more flexible static fit, built-in no-arbitrage checks
- SABR: 3 free parameters (with $\beta$ fixed), Hagan approximation can become unstable for deep OTM or very short-dated options

SABR is available as an alternative when the user requires a stochastic volatility model or needs to capture vol-of-vol dynamics.

### 1.3.5 Longstaff-Schwartz for American Options

American options allow early exercise, requiring Monte Carlo simulation with backward induction.

#### 1.3.5.1 Algorithm

**Step 1:** Simulate $M$ price paths forward to expiry using geometric Brownian motion:

$$S_{i,t+1} = S_{i,t}\exp\left[\left(r - q - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_{i,t}\right]$$

where $Z_{i,t} \sim \mathcal{N}(0,1)$.

**Step 2:** At expiry $T$, compute intrinsic value:

$$V_{i,T} = \max(0, h(S_{i,T}))$$

where $h(S) = K - S$ for puts, $h(S) = S - K$ for calls.

**Step 3:** Backward induction for $t = T-1, T-2, \ldots, 1$:

1. Filter in-the-money paths: $\mathcal{I}_t = \{i : h(S_{i,t}) > 0\}$
2. Regress continuation value on basis functions:

$$C(S_{i,t}) \approx \sum_{j=0}^{J} \beta_j L_j(S_{i,t})$$

   where $L_j$ are Laguerre polynomials and the target is $e^{-r\Delta t} V_{i,t+1}$

3. Exercise decision:

$$V_{i,t} = \begin{cases} h(S_{i,t}) & \text{if } h(S_{i,t}) > C(S_{i,t}) \quad \text{(exercise)} \\ e^{-r\Delta t} V_{i,t+1} & \text{otherwise} \quad \text{(continue)} \end{cases}$$

**Step 4:** Average discounted payoffs:

$$\text{Option Price} = \frac{1}{M} \sum_{i=1}^{M} e^{-r\tau_i} V_{i,\tau_i}$$

where $\tau_i$ is the optimal exercise time for path $i$.

### 1.3.6 Hedge Strategy Payoffs

#### 1.3.6.1 Protective Put

Portfolio value at expiry:

$$V_T = n \cdot \max(S_T, K_{\text{put}})$$

where $n$ is the number of shares hedged.

**Cost:** Put premium $P(S, K_{\text{put}}, T, r, q, \sigma)$

**Protection:** Downside floored at $K_{\text{put}}$. No upside cap.

#### 1.3.6.2 Collar

Portfolio value at expiry:

$$V_T = n \cdot \min\left(\max(S_T, K_{\text{put}}), K_{\text{call}}\right)$$

where $K_{\text{put}} < K_{\text{call}}$.

**Cost:** $P(K_{\text{put}}) - C(K_{\text{call}})$ (can be $\leq 0$ for zero-cost collar)

**Protection:** Bounded between $K_{\text{put}}$ and $K_{\text{call}}$.

#### 1.3.6.3 Vertical Put Spread

Buy put at $K_{\text{long}}$, sell put at $K_{\text{short}}$ where $K_{\text{long}} > K_{\text{short}}$:

$$V_T = n \cdot S_T + n \cdot [\max(0, K_{\text{long}} - S_T) - \max(0, K_{\text{short}} - S_T)]$$

**Cost:** $P(K_{\text{long}}) - P(K_{\text{short}})$

**Protection:** Limited to $K_{\text{long}} - K_{\text{short}}$.

#### 1.3.6.4 Covered Call

Sell call at strike $K_{\text{call}}$ against long stock:

$$V_T = n \cdot \min(S_T, K_{\text{call}})$$

**Cost:** Negative (receive premium $C(K_{\text{call}})$)

**Protection:** None. Upside capped at $K_{\text{call}}$.

### 1.3.7 Pareto Frontier Optimization

#### 1.3.7.1 Multi-Objective Formulation

For each strategy $s$, compute:

- **Cost**: $\text{Cost}(s)$ = total option premium paid
- **Protection**: $\text{Prot}(s)$ = expected downside protection at 5% quantile

A strategy $s^*$ is **Pareto efficient** if there exists no strategy $s$ such that:

$$\text{Cost}(s) \leq \text{Cost}(s^*) \quad \text{and} \quad \text{Prot}(s) \geq \text{Prot}(s^*)$$

with at least one strict inequality.

#### 1.3.7.2 Pareto Dominance

Strategy A dominates strategy B if:

$$\text{Cost}(A) \leq \text{Cost}(B) \quad \text{and} \quad \text{Prot}(A) \geq \text{Prot}(B)$$

with at least one strict inequality. Dominated strategies are never optimal.

#### 1.3.7.3 Frontier Construction

1. Generate candidate strategies by varying strikes across a grid
2. For each strategy, compute (cost, protection)
3. Remove dominated strategies
4. Plot efficient frontier: protection (x-axis) vs cost (y-axis)

The investor selects a point on the frontier based on their risk-return preferences and budget constraints.

### 1.3.8 Interpretation

The options hedging framework provides:

1. **Explicit cost-benefit trade-off**: The Pareto frontier quantifies the price of protection, enabling rational hedging decisions rather than ad-hoc choices.
2. **Volatility surface integration**: SVI calibration ensures consistent pricing across all strikes, preventing arbitrage and enabling accurate hedge valuation.
3. **American option pricing**: Longstaff-Schwartz handles early exercise optimally, critical for accurately pricing equity options (which are typically American).
4. **Strategy diversity**: From maximum protection (protective put) to income generation (covered call), the framework spans defensive to opportunistic hedging.
5. **Portfolio Greeks**: Full Greeks calculation enables dynamic risk management, position sizing, and delta-neutral adjustments.

The framework recognizes that there is no universally optimal hedge - the best strategy depends on the investor's downside tolerance, willingness to pay for protection, and market outlook. The Pareto frontier makes this trade-off explicit and quantifiable.

## 1.4 Gamma Scalping

### 1.4.1 Mathematical Framework

Gamma scalping exploits the convexity of option payoffs by dynamically delta-hedging a long options position. The strategy profits when realized volatility exceeds the implied volatility paid for the option, capturing P&L from frequent rehedging along the option's gamma profile.

#### 1.4.1.1 Black-Scholes Model

Let $S_t$ denote the underlying asset price at time $t$, with dynamics under the risk-neutral measure $\mathbb{Q}$:

$$dS_t = rS_t dt + \sigma S_t dW_t$$

where $r$ is the risk-free rate, $\sigma$ is the volatility, and $W_t$ is a standard Brownian motion.

The Black-Scholes price for a European call option at time $t$ with strike $K$ and expiry $T$ is:

$$C(S_t, t) = S_t N(d_1) - Ke^{-r(T-t)} N(d_2)$$

where:

$$\begin{aligned} d_1 &= \frac{\ln(S_t/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \\ d_2 &= d_1 - \sigma\sqrt{T-t} \end{aligned}$$

and $N(\cdot)$ is the cumulative standard normal distribution.

The European put price via put-call parity is:

$$P(S_t, t) = C(S_t, t) - S_t + Ke^{-r(T-t)}$$

#### 1.4.1.2 Greeks

The option's Greeks quantify sensitivities to market parameters:

**Delta** ($\Delta$): sensitivity to underlying price

$$\begin{aligned} \Delta_{\text{call}} &= N(d_1) \\ \Delta_{\text{put}} &= N(d_1) - 1 \end{aligned}$$

**Gamma** ($\Gamma$): rate of change of delta (convexity)

$$\Gamma = \frac{n(d_1)}{S_t \sigma \sqrt{T-t}}$$

where $n(\cdot) = N'(\cdot) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is the standard normal PDF.

**Theta** ($\Theta$): time decay (per year)

$$\begin{aligned} \Theta_{\text{call}} &= -\frac{S_t n(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)} N(d_2) \\ \Theta_{\text{put}} &= -\frac{S_t n(d_1)\sigma}{2\sqrt{T-t}} + rKe^{-r(T-t)} N(-d_2) \end{aligned}$$

**Vega** ($\mathcal{V}$): sensitivity to volatility (per 1% vol change)

$$\mathcal{V} = \frac{S_t n(d_1)\sqrt{T-t}}{100}$$

**Rho** ($\rho$): sensitivity to interest rate (per 1% rate change)

$$\begin{aligned} \rho_{\text{call}} &= \frac{K(T-t)e^{-r(T-t)} N(d_2)}{100} \\ \rho_{\text{put}} &= -\frac{K(T-t)e^{-r(T-t)} N(-d_2)}{100} \end{aligned}$$

### 1.4.2 Delta-Hedging Mechanism

#### 1.4.2.1 Portfolio Construction

Consider a portfolio consisting of:

- Long $n_{\text{opt}}$ option contracts (calls or puts)
- Short $\Delta_t$ shares of the underlying (delta hedge)
- Cash position earning/paying the risk-free rate

The portfolio value at time $t$ is:

$$V_t = n_{\text{opt}} \cdot V_{\text{opt}}(S_t, t) - \Delta_t \cdot S_t + C_t$$

where $V_{\text{opt}}(S_t, t)$ is the option value and $C_t$ is the cash balance.

#### 1.4.2.2 Delta-Neutral Hedging

To maintain delta neutrality, the hedge position is:

$$\Delta_t = n_{\text{opt}} \cdot \delta_{\text{opt}}(S_t, t)$$

where $\delta_{\text{opt}}$ is the option delta. This ensures:

$$\frac{\partial V_t}{\partial S_t} = 0$$

to first order, insulating the portfolio from small price movements.

#### 1.4.2.3 Rehedging Strategies

Four rehedging strategies are implemented:

**(1) Threshold Rehedging**

Rehedge when the absolute delta drift exceeds a threshold $\epsilon_\delta$:

$$|\Delta_{\text{current}} - \Delta_{\text{target}}| > \epsilon_\delta$$

Typical threshold: $\epsilon_\delta \in [0.05, 0.15]$.

**(2) Time-Based Rehedging**

Rehedge at fixed time intervals $\tau$ (e.g., daily, every 6 hours):

$$t - t_{\text{last rehedge}} \geq \tau$$

**(3) Hybrid Rehedging**

Combines threshold and time-based rules:

$$(|\Delta_{\text{current}} - \Delta_{\text{target}}| > \epsilon_\delta) \text{ OR } (t - t_{\text{last rehedge}} \geq \tau)$$

**(4) Volatility-Adaptive Rehedging**

Adjusts threshold based on realized volatility:

$$\epsilon_\delta(t) = \epsilon_{\text{base}} \cdot \left(1 + \alpha \cdot \frac{\sigma_{\text{realized}}(t) - \sigma_{\text{IV}}}{\sigma_{\text{IV}}}\right)$$

where $\alpha \in [0.5, 1.5]$ controls sensitivity. This widens the threshold when realized vol exceeds implied (reducing hedging frequency during profitable volatility conditions) and tightens it when realized vol is low.

### 1.4.3 P&L Attribution

The instantaneous P&L of a delta-hedged option portfolio can be decomposed via Itô's lemma.

#### 1.4.3.1 Hedged Portfolio Dynamics

For a delta-neutral portfolio $\Pi_t = V_{\text{opt}}(S_t, t) - \delta_{\text{opt}} S_t$, the instantaneous change is:

$$d\Pi_t = \frac{\partial V_{\text{opt}}}{\partial t} dt + \frac{1}{2} \frac{\partial^2 V_{\text{opt}}}{\partial S_t^2} (dS_t)^2 + \text{h.o.t.}$$

Using $dS_t = \mu S_t dt + \sigma S_t dW_t$ and $(dS_t)^2 = \sigma^2 S_t^2 dt$:

$$d\Pi_t = \Theta dt + \frac{1}{2} \Gamma \sigma^2 S_t^2 dt$$

#### 1.4.3.2 Gamma P&L

The gamma P&L accrues from the portfolio's convexity:

$$\text{Gamma P\&L} = \frac{1}{2} \Gamma \cdot (S_{t+1} - S_t)^2$$

This is always non-negative for a long option position ($\Gamma > 0$), capturing profit from large price moves in either direction.

#### 1.4.3.3 Theta Decay

Time decay erodes the option premium:

$$\text{Theta P\&L} = \Theta \cdot \Delta t$$

For long options, $\Theta < 0$, representing the daily cost of holding the position.

#### 1.4.3.4 Vega P&L

Changes in implied volatility affect option value:

$$\text{Vega P\&L} = \mathcal{V} \cdot (\sigma_{\text{IV},t+1} - \sigma_{\text{IV},t}) \cdot 100$$

For long positions, increases in implied volatility generate positive vega P&L.

#### 1.4.3.5 Total Attributed P&L

$$\text{Total P\&L} \approx \text{Gamma P\&L} + \text{Theta P\&L} + \text{Vega P\&L} + \text{Transaction Costs}$$

The residual unexplained P&L arises from higher-order Greeks (vanna, volga) and discrete rehedging errors.

### 1.4.4 Transaction Costs

Each delta rebalance incurs proportional transaction costs:

$$\text{Cost}_t = c \cdot |\Delta S_t|$$

where $c$ is the cost in basis points (typically 5–10 bps for liquid equities) and $\Delta S_t$ is the notional value traded.

Cumulative costs over the hedging period $[0, T]$:

$$\text{Total Costs} = \sum_{i=1}^{N_{\text{rehedges}}} c \cdot |\Delta_i - \Delta_{i-1}| \cdot S_{t_i}$$

Higher rehedging frequency increases transaction costs, creating a trade-off between delta accuracy and cost efficiency.

### 1.4.5 Profitability Condition

#### 1.4.5.1 Realized vs Implied Volatility

The strategy is profitable when:

$$\sigma_{\text{realized}} > \sigma_{\text{implied}}$$

**Intuition:** The gamma P&L accumulates based on realized price movements ($\sigma_{\text{realized}}$), while theta decay reflects the cost of the option premium paid ($\sigma_{\text{implied}}$).

Over the life of the option, the expected P&L (ignoring transaction costs) is approximately:

$$\mathbb{E}[\text{P\&L}] \propto \frac{1}{2}\Gamma S^2(\sigma^2_{\text{realized}} - \sigma^2_{\text{implied}})T$$

When $\sigma_{\text{realized}} > \sigma_{\text{implied}}$, gamma profits exceed theta decay.

#### 1.4.5.2 Breakeven Realized Volatility

Incorporating transaction costs $c$ and rehedging frequency $f$, the breakeven condition becomes:

$$\sigma^{\text{BE}}_{\text{realized}} = \sqrt{\sigma^2_{\text{implied}} + \frac{2cf}{\Gamma S^2 T}}$$

The strategy requires realized volatility to exceed this breakeven threshold.

### 1.4.6 Implementation

#### 1.4.6.1 Position Initialization

At $t = 0$:

- Purchase $n_{\text{opt}}$ straddles (long call + long put at strike $K$)
- Initial delta: $\Delta_0 = n_{\text{opt}} \cdot (\delta_{\text{call}} + \delta_{\text{put}})$
- Hedge by selling $\Delta_0$ shares

For at-the-money (ATM) straddles where $S_0 \approx K$:

$$\Delta_{\text{straddle}} = N(d_1) + (N(d_1) - 1) = 2N(d_1) - 1 \approx 0$$

so the initial hedge is near zero.

#### 1.4.6.2 Simulation Loop

For each time step $t$:

1. Observe new underlying price $S_t$
2. Compute current option values and Greeks
3. Determine if rehedging is triggered (based on strategy)
4. If triggered:
    - Compute new target delta $\Delta_{\text{target}} = n_{\text{opt}} \cdot \delta_{\text{opt}}(S_t, t)$
    - Trade $\Delta_{\text{target}} - \Delta_{\text{current}}$ shares
    - Incur transaction costs
    - Update cash balance
5. Attribute P&L to gamma, theta, vega
6. Update portfolio value

#### 1.4.6.3 Expiry Handling

At expiry $T$:

- Option payoff: $\max(0, S_T - K)$ for calls, $\max(0, K - S_T)$ for puts
- Close hedge position (sell/buy back shares)
- Realize final P&L

### 1.4.7 Performance Metrics

#### 1.4.7.1 Total P&L

Net profit after all costs:

$$\text{Net P\&L} = \text{Option Payoff} + \text{Hedging Gains} - \text{Option Premium Paid} - \text{Transaction Costs}$$

#### 1.4.7.2 Return on Capital

$$\text{ROC} = \frac{\text{Net P\&L}}{\text{Initial Capital}}$$

where initial capital is the option premium plus margin for short stock position.

#### 1.4.7.3 Sharpe Ratio

$$\text{Sharpe} = \frac{\mathbb{E}[\text{Daily P\&L}]}{\text{Std}[\text{Daily P\&L}]}$$

#### 1.4.7.4 Hedge Effectiveness

$$\text{Effectiveness} = 1 - \frac{\text{Var}(\Delta V_{\text{hedged}})}{\text{Var}(\Delta V_{\text{unhedged}})}$$

Measures the reduction in portfolio variance from delta-hedging.

#### 1.4.7.5 Realized Volatility

$$\sigma_{\text{realized}} = \sqrt{252} \cdot \sqrt{\frac{1}{N} \sum_{i=1}^{N} r_i^2}$$

where $r_i = \log(S_i/S_{i-1})$ are log-returns.

### 1.4.8 Interpretation

Gamma scalping is a market-neutral strategy that:

1. **Profits from volatility mispricing**: The strategy is inherently long realized volatility and short implied volatility. It succeeds when markets move more than option prices anticipated.
2. **Benefits from large moves**: Gamma P&L is quadratic in price changes, generating outsized profits during volatile periods regardless of direction.
3. **Incurs theta decay**: The cost of maintaining long gamma is the daily time decay, which must be offset by sufficient price movement.
4. **Requires active management**: Frequent rehedging is needed to maintain delta neutrality, introducing path-dependency and transaction costs.
5. **Faces execution risk**: Slippage and transaction costs erode gamma profits, particularly for high-frequency rehedging strategies.

The strategy is best suited for traders who:

- Expect realized volatility to exceed implied volatility

- Have low transaction costs (market makers, proprietary desks)
- Can monitor and rehedge positions dynamically
- Prefer market-neutral exposure with positive convexity

## 1.5 Volatility Arbitrage

### 1.5.1 Motivation

Volatility arbitrage exploits mispricings between implied volatility (IV) embedded in option prices and forecasts of realized volatility (RV). When IV systematically deviates from expected RV, delta-neutral strategies can profit from vol convergence while minimizing directional risk.

This framework addresses:

- Forecasting realized volatility using GARCH, EWMA, and HAR models
- Detecting IV–RV mispricings and arbitrage opportunities
- Identifying no-arbitrage violations (butterfly, calendar, put-call parity)
- Constructing delta-neutral positions to isolate vol exposure
- Managing vega, gamma, and theta risks in volatility trading

### 1.5.2 Realized Volatility Forecasting

#### 1.5.2.1 GARCH(1,1) Model

The Generalized Autoregressive Conditional Heteroskedasticity model captures volatility clustering:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

where $\epsilon_t = r_t - \mu$ are return innovations. Stationarity requires $\alpha + \beta < 1$. The long-run variance is:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$$

#### 1.5.2.2 Exponentially Weighted Moving Average (EWMA)

RiskMetrics-style volatility forecast:

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1 - \lambda)r_{t-1}^2$$

with decay factor $\lambda \in [0.94, 0.97]$ for daily data. Recent shocks receive exponentially declining weights.

#### 1.5.2.3 Heterogeneous Autoregressive (HAR) Model

Multi-horizon realized volatility regression:

$$\mathrm{RV}_{t\to t+h} = \beta_0 + \beta_d \mathrm{RV}_{t-1} + \beta_w \mathrm{RV}_{t-5:t-1} + \beta_m \mathrm{RV}_{t-22:t-1} + \epsilon_t$$

where $\mathrm{RV}_{t-5:t-1}$ is weekly realized vol and $\mathrm{RV}_{t-22:t-1}$ is monthly. This captures heterogeneous market participants with different rebalancing horizons.

### 1.5.3 IV–RV Mispricing Detection

#### 1.5.3.1 Arbitrage Signal

At time $t$, for an option with maturity $T$:

$$\mathrm{Signal}_t = \mathrm{IV}_t - \mathbb{E}_t[\mathrm{RV}_{t\to T}]$$

where $\mathbb{E}_t[\mathrm{RV}_{t\to T}]$ is the forecast from GARCH, EWMA, or HAR. Positive signal suggests selling volatility (options overpriced); negative signal suggests buying volatility.

#### 1.5.3.2 Threshold Strategy

Enter trade when $|\text{Signal}_t| > \tau \cdot \sigma_{\text{signal}}$, where $\tau \in [1.5, 2.5]$ and $\sigma_{\text{signal}}$ is the historical standard deviation of forecast errors. This filters noise and targets statistically significant mispricings.

### 1.5.4 No-Arbitrage Constraint Violations

#### 1.5.4.1 Butterfly Arbitrage

For strikes $K_1 < K_2 < K_3$ with equal spacing $K_2 - K_1 = K_3 - K_2 = \Delta K$:

$$\text{Butterfly}(K_2) = C(K_1) - 2C(K_2) + C(K_3) \geq 0$$

Violation implies negative Arrow-Debreu state price. Arbitrage: buy butterfly spread.

#### 1.5.4.2 Calendar Arbitrage

For maturities $T_1 < T_2$ at fixed strike $K$:

$$C(K, T_2) \geq C(K, T_1)$$

Violation: buy long-dated call, sell short-dated call. As $T_1 \to 0$, the short call expires worthless while the long call retains value.

#### 1.5.4.3 Put-Call Parity Violation

For European options:

$$C - P = Se^{-qT} - Ke^{-rT}$$

If $C - P > Se^{-qT} - Ke^{-rT}$: sell call, buy put, buy stock, borrow $Ke^{-rT}$. If $C - P < Se^{-qT} - Ke^{-rT}$: buy call, sell put, short stock, lend $Ke^{-rT}$.

### 1.5.5 Delta-Neutral Volatility Trading

#### 1.5.5.1 Straddle with Delta Hedging

At-the-money straddle: $V = C_{\text{ATM}} + P_{\text{ATM}}$. Initial delta:

$$\Delta_V = \Delta_C + \Delta_P \approx e^{-qT} N(d_1) + e^{-qT}(N(d_1) - 1) \approx 0$$

Hedge with $n_S = -\Delta_V / \Delta_S$ shares. Rehedge when $\Delta$ drifts beyond threshold $\delta \in [0.05, 0.15]$.

#### 1.5.5.2 P&L Attribution

Daily P&L from delta-hedged straddle:

$$\text{P\&L}_t = \underbrace{\frac{1}{2}\Gamma \cdot (\Delta S)^2}_{\text{Gamma P\&L}} - \underbrace{\Theta \cdot \Delta t}_{\text{Theta decay}} + \underbrace{\mathcal{V} \cdot \Delta\sigma}_{\text{Vega P\&L}}$$

Profitable if realized gamma profits exceed theta decay and vega losses.

### 1.5.6 Risk Management

#### 1.5.6.1 Greeks Exposure

Monitor portfolio-level Greeks:

- **Delta**: Keep $|\Delta_{\text{port}}| < 0.05 \times S$ to maintain neutrality
- **Gamma**: Positive for long vol, negative for short vol
- **Vega**: Sensitivity to IV changes; long vol has positive vega
- **Theta**: Time decay; short vol earns theta, long vol pays theta

#### 1.5.6.2 VaR and Stress Testing

Historical simulation VaR for vol portfolio:

$$\text{VaR}_{95\%} = -\inf\{x : P(\text{P\&L} \leq x) \leq 0.05\}$$

Stress test scenarios: IV spike (+10 vol points), RV spike (2× historical), correlation breakdown.

### 1.5.7 Interpretation

1. **Forecast-dependent**: Returns hinge on RV forecasting accuracy. Misspecified models produce false signals.
2. **Transaction costs erode profits**: Frequent delta rehedging incurs slippage and commissions, particularly for illiquid underlyings.
3. **Model risk**: GARCH parameters estimated on historical data may not persist; regime changes invalidate forecasts.
4. **Path-dependent**: Even if IV converges to RV eventually, interim P&L depends on realized gamma vs theta decay dynamics.

The strategy is best suited for:

- Traders with robust RV forecasting models and low transaction costs
- Market-neutral desks seeking to harvest vol risk premium
- Options market makers identifying arbitrage violations
- Investors willing to monitor and rehedge delta dynamically

## 1.6 Variance Swaps

### 1.6.1 Motivation

Variance swaps provide pure volatility exposure without delta or gamma hedging complexity. The payoff depends solely on realized variance over the contract period, making them ideal instruments for volatility trading and harvesting the variance risk premium (VRP).

This framework addresses:

- Pricing variance swaps via Carr-Madan replication with vanilla options
- Static replication using optimized log-spaced strike grids
- Measuring realized variance with multiple high-efficiency estimators
- Forecasting realized variance via EWMA and GARCH models
- Computing VRP and generating threshold-based trading signals

### 1.6.2 Variance Swap Payoff

At maturity $T$, the variance swap pays:

$$\text{Payoff} = N_{\text{var}} \times (\text{RV}^2 - K_{\text{var}})$$

where $N_{\text{var}}$ is the variance notional, $\text{RV}^2$ is the realized variance over $[0, T]$, and $K_{\text{var}}$ is the strike variance (fair value at inception). The *vega notional* converts between variance and volatility space:

$$N_{\text{vega}} = \frac{N_{\text{var}}}{2\sqrt{K_{\text{var}}}}$$

### 1.6.3 Realized Variance Estimators

The close-to-close estimator computes realized variance from log returns:

$$\hat{\sigma}^2_{\mathrm{CC}} = \frac{252}{N} \sum_{i=1}^{N} r_i^2, \quad r_i = \ln\left(\frac{S_i}{S_{i-1}}\right)$$

where 252 annualizes daily variance. Higher-efficiency estimators use intraday range information to reduce sampling noise:

#### 1.6.3.1 Parkinson (5× efficiency)

Uses the high-low range, exploiting the fact that extremes contain more information than closing prices alone:

$$\hat{\sigma}^2_{\mathrm{P}} = \frac{252}{N} \sum_{i=1}^{N} \frac{\left(\ln H_i/L_i\right)^2}{4\ln 2}$$

#### 1.6.3.2 Garman-Klass (8× efficiency)

Combines open-high-low-close data for further efficiency gains:

$$\hat{\sigma}^2_{\mathrm{GK}} = \frac{252}{N} \sum_{i=1}^{N} \left[\frac{1}{2}\left(\ln H_i/L_i\right)^2 - (2\ln 2 - 1)\left(\ln C_i/O_i\right)^2\right]$$

#### 1.6.3.3 Rogers-Satchell (drift-independent)

Consistent even when the underlying has non-zero drift:

$$\hat{\sigma}^2_{\mathrm{RS}} = \frac{252}{N} \sum_{i=1}^{N} \left[\ln\frac{H_i}{C_i} \cdot \ln\frac{H_i}{O_i} + \ln\frac{L_i}{C_i} \cdot \ln\frac{L_i}{O_i}\right]$$

#### 1.6.3.4 Yang-Zhang (14× efficiency)

The most efficient estimator, combining overnight and intraday components:

$$\hat{\sigma}^2_{\mathrm{YZ}} = \hat{\sigma}^2_{\mathrm{overnight}} + k\,\hat{\sigma}^2_{\mathrm{OC}} + (1-k)\,\hat{\sigma}^2_{\mathrm{RS}}, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

### 1.6.4 Variance Forecasting

Realized variance estimators measure the past; trading signals require a forecast. Two models are implemented:

#### 1.6.4.1 EWMA

Exponentially weighted moving average with decay factor $\lambda \in [0.90, 0.97]$:

$$\hat{\sigma}^2_t = \lambda\,\hat{\sigma}^2_{t-1} + (1-\lambda)\,r_t^2$$

#### 1.6.4.2 GARCH(1,1)

Conditional variance with mean-reverting dynamics:

$$\hat{\sigma}^2_t = \omega + \alpha\,r_{t-1}^2 + \beta\,\hat{\sigma}^2_{t-1}$$

where $\alpha + \beta < 1$ ensures stationarity and $\omega/(1-\alpha-\beta)$ is the long-run variance.

### 1.6.5 Carr-Madan Replication

#### 1.6.5.1 Model-Free Fair Variance

Under the risk-neutral measure, the fair variance strike is obtained from a portfolio of out-of-the-money options spanning all strikes. In practice, the integral is discretized over a finite strike grid:

$$K_{\text{var}} \approx \frac{2e^{rT}}{T} \sum_i \frac{\Delta K_i}{K_i^2} \cdot V(K_i)$$

where $V(K_i) = P(K_i)$ for $K_i < F$ and $V(K_i) = C(K_i)$ for $K_i \geq F$, with $F = S_0\, e^{(r-q)T}$ the forward price. The cutoff at the forward (rather than spot) avoids double-counting near-the-money options and is consistent with the Carr-Madan (1998) derivation.

#### 1.6.5.2 Static Replication

Each option in the replication portfolio is weighted by:

$$w_i = \frac{2e^{rT}}{T} \cdot \frac{\Delta K_i}{K_i^2} \cdot N_{\text{var}}$$

where $\Delta K_i$ uses the midpoint rule: $\Delta K_i = (K_{i+1} - K_{i-1})/2$ for interior strikes. The implementation uses log-spaced strikes across $[0.70\, S_0,\ 1.30\, S_0]$ to concentrate resolution near ATM while covering the wings. Option prices at each strike are computed from an SVI volatility surface parameterized as:

$$w(k) = a + b\big(\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\big)$$

where $k = \ln(K/S)$ is log-moneyness and $(a, b, \rho, m, \sigma)$ are calibrated per expiry from the options chain.

#### 1.6.5.3 Portfolio Greeks

The replication portfolio's aggregate Greeks (delta, gamma, vega, theta, rho) are computed by summing weighted Black-Scholes Greeks across all legs. A well-constructed replication should be approximately delta-neutral; the implementation checks $|\Delta_{\text{total}}| < \epsilon$ with configurable tolerance.

### 1.6.6 Variance Risk Premium (VRP)

#### 1.6.6.1 Definition

The variance risk premium is the difference between implied and expected realized variance:

$$\text{VRP}_t = \text{IV}_t^2 - \mathbb{E}_t[\text{RV}_{t\to T}^2]$$

Empirically, VRP $> 0$ on average: investors overpay for volatility protection, creating a persistent premium that can be harvested by selling variance. The percentage VRP normalizes by the implied level:

$$\text{VRP\%} = \frac{\text{VRP}}{\text{IV}^2} \times 100$$

#### 1.6.6.2 Signal Generation

Trading signals are generated from VRP thresholds with confidence scoring:

- **Short variance**: VRP% $\geq +2\%$. Implied volatility is rich relative to forecast; sell variance to harvest the premium. Confidence: min(1, |VRP%|/10).
- **Long variance**: VRP% $\leq -1\%$. Implied volatility is cheap; buy variance for mean reversion. Confidence: min(1, |VRP%|/5).
- **Neutral**: VRP between thresholds; no position warranted.

Position size is scaled by confidence times the target vega notional. Kelly criterion provides an upper bound: $f^* = \mu/\sigma^2$, capped at half-Kelly for conservatism.

#### 1.6.6.3 VRP Significance Testing

The module tests whether the mean VRP is statistically different from zero via a $t$-test. A rolling $z$-score detector flags regime changes when the VRP deviates from its recent mean by more than two standard deviations, signaling shifts in the volatility environment.

#### 1.6.6.4 Backtesting

The rolling strategy sells 1-month variance swaps monthly and tracks cumulative P&L:

$$\text{P\&L}_t = -N_{\text{var}} \times (\text{RV}_t^2 - K_{\text{var},t})$$

with a 20-period rolling Sharpe ratio to monitor strategy health. Historical backtests on equity indices (SPY) typically show Sharpe ratios of 1.0–2.0 during calm regimes.

### 1.6.7 Jump Risk and Tail Events

Large jumps contribute disproportionately to $\text{RV}^2$. A single $-10\%$ move adds 0.01 to annual variance. Short variance positions are exposed to convex losses during crashes—March 2020 saw variance swaps lose 10–20× in days. Risk management considerations include position sizing to cap maximum loss, tail hedges via OTM puts, stop-loss triggers on MTM drawdowns, and diversification across maturities.

### 1.6.8 Interpretation

1. **Pure vol exposure**: No delta or gamma hedging required; P&L depends solely on realized vs implied variance.
2. **VRP harvesting**: Systematically profitable if implied variance exceeds realized variance on average. The signal framework provides disciplined entry and sizing.
3. **Estimator choice matters**: Yang-Zhang is 14× more efficient than close-to-close, enabling tighter confidence intervals on RV forecasts and earlier signal detection.
4. **Jump risk**: Highly vulnerable to tail events. The VRP percentage and regime detection help identify when implied vol is insufficient compensation for crash risk.

The strategy is best suited for:

- Institutional volatility traders with risk appetite for tail events
- Market makers providing variance swap liquidity
- Hedge funds seeking uncorrelated returns from vol risk premium
- Investors willing to size positions conservatively and monitor VRP regime shifts

## 1.7 Skew Trading

### 1.7.1 Motivation

Volatility skew reflects the market's preference for downside protection, manifesting as higher implied volatility for out-of-the-money puts relative to calls. Skew trading exploits mean reversion in skew metrics, profiting when skew deviates from historical norms and reverts.

This framework addresses:

- Quantifying skew via risk reversal (RR25) and butterfly (BF25) metrics
- Detecting skew extremes and mean reversion signals
- Constructing risk reversal and butterfly spread trades
- Managing vega, delta, and gamma exposures in skew positions
- Hedging against directional moves and volatility regime shifts

### 1.7.2 Skew Metrics

#### 1.7.2.1 Risk Reversal (RR25)

Risk reversal measures the difference in implied volatility between 25-delta call and 25-delta put:

$$\text{RR25} = \text{IV}(25\Delta_{\text{call}}) - \text{IV}(25\Delta_{\text{put}})$$

Negative RR25 indicates downside skew (put IV > call IV), typical in equity markets. Positive RR25 indicates upside skew, seen in commodities during supply shocks.

#### 1.7.2.2 Butterfly (BF25)

Butterfly measures the convexity of the volatility smile:

$$\text{BF25} = \frac{\text{IV}(25\Delta_{\text{call}}) + \text{IV}(25\Delta_{\text{put}})}{2} - \text{IV}_{\text{ATM}}$$

High BF25 means wings (OTM options) are expensive relative to ATM, indicating fat tails or demand for tail hedges.

#### 1.7.2.3 Historical Context

Compare current skew to rolling percentiles:

$$\text{Skew Score} = \frac{\text{RR25}_t - \mu_{\text{RR25}}}{\sigma_{\text{RR25}}}$$

where $\mu$ and $\sigma$ are 252-day rolling mean and standard deviation. Extreme scores ($|z| > 2$) signal potential mean reversion.

### 1.7.3 Risk Reversal Trade

#### 1.7.3.1 Structure

If RR25 is excessively negative (puts too expensive):

$$\text{Position} = \text{Buy } 25\Delta \text{ Call} + \text{Sell } 25\Delta \text{ Put}$$

This is a long risk reversal, profiting if skew flattens (put IV decreases or call IV increases).

Conversely, if RR25 is too positive:

$$\text{Position} = \text{Sell } 25\Delta \text{ Call} + \text{Buy } 25\Delta \text{ Put}$$

#### 1.7.3.2 Greeks

For long risk reversal:

- **Delta**: $\Delta_{\text{call}} - |\Delta_{\text{put}}| \approx 0.25 - 0.25 = 0$ (approximately delta-neutral)
- **Vega**: $\mathcal{V}_{\text{call}} - \mathcal{V}_{\text{put}}$ (differential vega exposure)
- **Gamma**: Slightly positive from long call, negative from short put

#### 1.7.3.3 P&L Drivers

$$\text{P\&L} = \underbrace{\mathcal{V}_{\text{call}} \cdot \Delta\text{IV}_{\text{call}}}_{\text{Call vega}} - \underbrace{\mathcal{V}_{\text{put}} \cdot \Delta\text{IV}_{\text{put}}}_{\text{Put vega}} - \underbrace{\Theta \cdot \Delta t}_{\text{Theta decay}}$$

Profitable if RR25 reverts toward mean, narrowing the IV differential.

### 1.7.4 Butterfly Spread Trade

#### 1.7.4.1 Structure

If BF25 is excessively high (wings overpriced):

$$\text{Position} = \text{Sell } 25\Delta \text{ Call} + \text{Sell } 25\Delta \text{ Put} + \text{Buy } 2 \times \text{ATM Straddle}$$

This is short butterfly, profiting if BF25 compresses.

If BF25 is too low (wings underpriced):

$$\text{Position} = \text{Buy } 25\Delta \text{ Call} + \text{Buy } 25\Delta \text{ Put} + \text{Sell } 2 \times \text{ATM Straddle}$$

#### 1.7.4.2 Greeks

For short butterfly:

- **Delta**: Approximately zero (ATM straddle delta offsets wing deltas)
- **Vega**: Negative wing vega, positive ATM vega; net vega exposure to BF25
- **Gamma**: Positive ATM gamma, negative wing gamma

#### 1.7.4.3 P&L Attribution

$$\text{P\&L} = \underbrace{(\mathcal{V}_{\text{ATM}} - \mathcal{V}_{\text{wings}}) \cdot \Delta\text{BF25}}_{\text{Skew compression}} + \underbrace{\Gamma_{\text{net}} \cdot (\Delta S)^2/2}_{\text{Gamma P\&L}} - \Theta \cdot \Delta t$$

Profitable if BF25 mean-reverts and realized volatility aligns with expectations.

### 1.7.5 Mean Reversion Signal Construction

#### 1.7.5.1 Z-Score Trigger

Enter trade when skew metric exceeds historical threshold:

$$|\text{Skew Score}_t| > \tau, \quad \tau \in [1.5, 2.5]$$

Higher $\tau$ reduces trade frequency but increases signal reliability.

#### 1.7.5.2 Bollinger Band Strategy

Alternative: enter when RR25 or BF25 breaches $\pm 2\sigma$ Bollinger bands:

$$\text{Upper Band} = \mu_{20} + 2\sigma_{20}, \quad \text{Lower Band} = \mu_{20} - 2\sigma_{20}$$

where $\mu_{20}$ and $\sigma_{20}$ are 20-day moving statistics.

#### 1.7.5.3 Exit Rules

Close position when:

1. Skew metric reverts to within $\pm 0.5\sigma$ of mean (take profit)
2. Time to expiration $< 7$ days (avoid gamma risk and pin risk)
3. Mark-to-market loss exceeds -2% of position value (stop-loss)

### 1.7.6 Risk Management

#### 1.7.6.1 Delta Hedging

Even nominally delta-neutral positions drift as spot moves. Rehedge when:

$$|\Delta_{\text{port}}| > \delta \cdot S, \quad \delta \in [0.05, 0.10]$$

Use futures or stock to adjust delta without disturbing skew exposure.

#### 1.7.6.2 Scenario Analysis

Stress test positions under:

- **Spot crash (-10%)**: Short puts in risk reversal lose; BF25 widens
- **Vol spike (+10 vol points)**: All positions gain vega; assess net exposure
- **Skew widening**: RR25 becomes more negative; butterfly wings expensive

Ensure maximum loss under worst-case scenario $< 5\%$ of capital.

#### 1.7.6.3 Correlation Risk

Skew and spot returns are correlated: market drops $\Rightarrow$ skew steepens. Hedging spot exposure may inadvertently reduce skew trade profitability. Separate directional from skew exposure using futures overlays.

### 1.7.7 Interpretation

1. **Mean reversion assumption**: Strategy relies on skew returning to historical norms; structural regime shifts invalidate signals.
2. **Transaction costs**: Frequent rehedging and bid-ask spreads erode profits, particularly in illiquid options markets.
3. **Theta decay**: Long options positions pay theta; must be compensated by skew reversion or realized volatility.
4. **Event risk**: Earnings, geopolitical shocks can cause skew to diverge further before reverting.

The strategy is best suited for:

- Options traders with access to tight bid-ask spreads
- Volatility arbitrageurs seeking to monetize skew mispricings
- Institutional desks with delta hedging infrastructure
- Investors with patience for mean reversion over weeks to months

## 1.8 FX Hedging & Futures Options

### 1.8.1 Motivation

Portfolios with international exposure face currency risk: fluctuations in exchange rates can dominate returns, particularly when the domestic currency (e.g., USD) depreciates. Traditional spot FX hedging requires 100% capital commitment. Futures and futures options provide capital-efficient alternatives, requiring only 3–5% margin while delivering 20–40$\times$ leverage.

This framework addresses:

- Hedging multi-currency portfolios against FX moves
- Capital-efficient implementation via CME currency futures
- Pricing and Greeks for futures options (Black-76 model)
- Dynamic hedge ratio optimization
- Margin management and roll strategies

### 1.8.2 Forward and Futures Pricing

#### 1.8.2.1 Covered Interest Rate Parity

Let $S$ denote the spot exchange rate (units of domestic currency per unit of foreign currency), $r_d$ the domestic interest rate, $r_f$ the foreign interest rate, and $T$ the time to maturity.

The no-arbitrage forward rate is:

$$F = S \cdot e^{(r_d - r_f)T}$$

**Interpretation:** If domestic rates exceed foreign rates ($r_d > r_f$), the forward trades at a premium to spot (contango). Conversely, if $r_d < r_f$, the forward is at a discount (backwardation).

#### 1.8.2.2 Forward Points

The difference between forward and spot:

$$\text{Forward Points} = F - S = S \cdot \left(e^{(r_d - r_f)T} - 1\right)$$

For small rate differentials and short maturities, the approximation holds:

$$F - S \approx S \cdot (r_d - r_f)T$$

#### 1.8.2.3 Futures Price

Under the assumption of deterministic interest rates, futures and forwards have identical prices:

$$\text{Futures Price} = F = S \cdot e^{(r_d - r_f)T}$$

In practice, marking-to-market and margin flows introduce small deviations, but these are typically negligible for liquid FX futures.

#### 1.8.2.4 Basis

The basis is the spread between futures and spot:

$$\text{Basis} = \text{Futures Price} - S$$

**Contango** (Basis $> 0$): Futures trade above spot. **Backwardation** (Basis $< 0$): Futures trade below spot.

### 1.8.3 Black-76 Model for Futures Options

#### 1.8.3.1 Model Setup

The Black-76 model prices European options on futures. Unlike Black-Scholes (which models spot price dynamics), Black-76 assumes the futures price $F_t$ follows:

$$dF_t = \sigma F_t dW_t$$

under the risk-neutral measure, with no drift term (since futures are martingales).

#### 1.8.3.2 Option Pricing Formulas

**Call Option:**

$$C = e^{-rT} \left[F \cdot N(d_1) - K \cdot N(d_2)\right]$$

**Put Option:**

$$P = e^{-rT} \left[K \cdot N(-d_2) - F \cdot N(-d_1)\right]$$

where:

$$d_1 = \frac{\ln(F/K) + (\sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

and $r$ is the risk-free discount rate.

#### 1.8.3.3 Key Differences from Black-Scholes

- Uses futures price $F$ instead of spot $S$
- No dividend yield term (futures already incorporate the cost of carry)
- Simpler structure: only two inputs ($F$ and $\sigma$) beyond standard parameters

#### 1.8.3.4 Greeks for Futures Options

**Delta:**

$$\Delta_{\text{call}} = e^{-rT} N(d_1)$$
$$\Delta_{\text{put}} = -e^{-rT} N(-d_1)$$

**Gamma:**

$$\Gamma = e^{-rT} \frac{n(d_1)}{F\sigma\sqrt{T}}$$

**Theta (per day):**

$$\Theta_{\text{call}} = -\frac{e^{-rT} F n(d_1)\sigma}{2\sqrt{T} \cdot 365} + \frac{rC}{365}$$
$$\Theta_{\text{put}} = -\frac{e^{-rT} F n(d_1)\sigma}{2\sqrt{T} \cdot 365} + \frac{rP}{365}$$

**Vega (per 1% vol):**

$$\mathcal{V} = \frac{e^{-rT} F n(d_1)\sqrt{T}}{100}$$

**Rho (per 1% rate):**

$$\rho_{\text{call}} = -\frac{T \cdot C}{100}$$
$$\rho_{\text{put}} = -\frac{T \cdot P}{100}$$

Note: Rho for futures options reflects the discounting effect, not the carry cost (which is embedded in $F$).

### 1.8.4 Margin and Capital Efficiency

#### 1.8.4.1 Initial Margin

CME futures require initial margin (typically 3–5% of contract notional) to open a position. For the EUR/USD futures (6E):

- Contract size: EUR 125,000
- Notional (at $F = 1.10$): \$137,500
- Typical initial margin: \$2,500
- Leverage: $137{,}500/2{,}500 = 55\times$

#### 1.8.4.2 Maintenance Margin

The maintenance margin (typically 80% of initial margin) is the minimum account balance to avoid a margin call:

$$\text{Maintenance Margin} \approx 0.8 \times \text{Initial Margin}$$

If the margin account falls below this level due to adverse price moves, additional funds must be deposited.

#### 1.8.4.3 Variation Margin

Futures are marked-to-market daily. The daily P&L is settled in cash:

$$\text{Variation Margin}_t = (F_t - F_{t-1}) \times \text{Contract Size} \times \text{Position}$$

This settlement is credited or debited from the margin account.

#### 1.8.4.4 SPAN Methodology

CME uses Standard Portfolio Analysis of Risk (SPAN) to compute risk-based margin requirements. SPAN evaluates portfolio risk across 16 scenarios (price moves of $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$ and volatility changes of $\pm 10\%$, $\pm 20\%$, $\pm 30\%$), taking the worst-case loss as the margin requirement.

A simplified SPAN estimate is:

$$\text{SPAN Margin} \approx 0.04 \times \text{Contract Value}$$

for a $\pm 10\%$ price move scenario.

### 1.8.5 Hedge Ratio Optimization

#### 1.8.5.1 Exposure Calculation

For a portfolio with $n$ positions in different currencies, the net FX exposure in USD for currency $i$ is:

$$E_i = \sum_{j \in \text{positions}} \text{Quantity}_j \times \text{Price}_j \times \text{FX Rate}_j \times \mathbb{I}[\text{Currency}_j = i]$$

#### 1.8.5.2 Futures Contracts Needed

Given an exposure $E$ in currency units and a hedge ratio $h \in [-1, 1]$:

$$N_{\text{contracts}} = \left\lfloor \frac{h \cdot E \cdot S}{\text{Contract Size}} \right\rfloor$$

where $S$ is the current spot rate and Contract Size is the notional per futures contract.

For a full hedge ($h = -1$), this formula gives the number of short futures contracts needed to offset long currency exposure.

#### 1.8.5.3 Minimum Variance Hedge Ratio

The optimal hedge ratio minimizing portfolio variance is:

$$h^* = \frac{\text{Cov}(\Delta S, \Delta F)}{\text{Var}(\Delta F)} = \rho_{S,F} \cdot \frac{\sigma_S}{\sigma_F}$$

where:

- $\rho_{S,F}$ is the correlation between spot and futures returns
- $\sigma_S$ is the volatility of spot returns
- $\sigma_F$ is the volatility of futures returns

In practice, for highly liquid FX futures, $\rho_{S,F} \approx 1$ and $\sigma_S \approx \sigma_F$, so $h^* \approx -1$ (full hedge).

#### 1.8.5.4 Hedge Effectiveness

The effectiveness of the hedge is measured by variance reduction:

$$\text{Effectiveness} = 1 - \frac{\text{Var(Hedged P\&L)}}{\text{Var(Unhedged P\&L)}}$$

Values above 80% indicate a highly effective hedge. Lower effectiveness suggests basis risk or imperfect correlation.

#### 1.8.6 Roll Management

Futures contracts have finite expiry. To maintain a hedge beyond the contract's maturity, positions must be rolled to the next contract month.

##### 1.8.6.1 Roll Decision

Typically, rolls occur 5–10 days before expiry to avoid illiquidity and delivery risk:

$$\text{Days to Expiry} \leq \text{Roll Threshold} \quad \Rightarrow \quad \text{Execute Roll}$$

##### 1.8.6.2 Roll P&L

The P&L from rolling depends on the basis:

$$\text{Roll P\&L} = (F_{\text{new}} - F_{\text{old}}) \times \text{Contract Size} \times \text{Position}$$

**Positive roll yield:** If $F_{\text{new}} < F_{\text{old}}$ for a short hedge, the roll generates profit. **Negative roll yield:** If $F_{\text{new}} > F_{\text{old}}$, the roll incurs a cost.

Over time, cumulative roll costs can be significant for persistent contango or backwardation.

#### 1.8.7 Hedge Strategies

##### 1.8.7.1 Static Hedge

A fixed hedge ratio $h$ maintained throughout the backtest:

$$N_{\text{contracts}} = \text{constant}$$

**Advantages:** Low transaction costs, simple to implement. **Disadvantages:** Does not adapt to changing correlations or exposures.

##### 1.8.7.2 Dynamic Hedge (Time-Based)

Rebalance periodically (e.g., weekly, monthly):

$$t - t_{\text{last rebalance}} \geq \tau \quad \Rightarrow \quad \text{Rehedge}$$

**Advantages:** Adapts to price changes and exposure drift. **Disadvantages:** Higher transaction costs.

##### 1.8.7.3 Minimum Variance Hedge

Optimize the hedge ratio based on rolling-window correlation and volatility estimates:

$$h^*(t) = \rho_{S,F}(t) \cdot \frac{\sigma_S(t)}{\sigma_F(t)}$$

Lookback window typically 60–90 days.

**Advantages:** Maximizes hedge effectiveness. **Disadvantages:** Sensitive to estimation error in correlation.

##### 1.8.7.4 Optimal Cost Hedge

Rebalance only when delta drift exceeds a threshold:

$$|\Delta_{\text{current}} - \Delta_{\text{target}}| > \epsilon \quad \Rightarrow \quad \text{Rehedge}$$

**Advantages:** Balances effectiveness and transaction costs. **Disadvantages:** May allow temporary exposure drift.

### 1.8.8 Performance Metrics

#### 1.8.8.1 Unhedged P&L

Cumulative exposure P&L without hedging:

$$\text{Unhedged P\&L}_T = E \cdot (S_T - S_0)$$

#### 1.8.8.2 Hedged P&L

Combined exposure and futures hedge P&L:

$$\text{Hedged P\&L}_T = E \cdot (S_T - S_0) + N \cdot (F_T - F_0) \cdot \text{Contract Size} - \text{Costs}$$

#### 1.8.8.3 Sharpe Ratio

$$\text{Sharpe}_{\text{hedged}} = \frac{\mathbb{E}[\text{Daily Hedged P\&L}]}{\text{Std}[\text{Daily Hedged P\&L}]}$$

Compare to the unhedged Sharpe ratio to assess risk-adjusted improvement.

#### 1.8.8.4 Maximum Drawdown

$$\text{MDD} = \max_{t \in [0,T]} \left( \max_{s \in [0,t]} \text{P\&L}_s - \text{P\&L}_t \right)$$

Hedging should reduce maximum drawdown significantly relative to the unhedged position.

### 1.8.9 Interpretation

The FX hedging framework enables:

1. **Capital efficiency**: Futures require 3–5% margin vs 100% for spot FX, freeing capital for other investments.
2. **Leverage**: 20–40× leverage allows hedging large exposures with minimal upfront cost, critical for portfolios with substantial international holdings.
3. **Flexibility**: Multiple hedge strategies (static, dynamic, minimum variance) adapt to different market conditions and cost constraints.
4. **Tail risk management**: Futures options provide convexity, allowing non-linear hedges that protect against extreme moves while preserving upside.
5. **Transparency**: Futures are exchange-traded with standardized specifications, daily settlement, and centralized clearing, reducing counterparty risk.

The Black-76 framework for futures options extends the toolkit beyond linear hedging:

- **Collars**: Buy put, sell call to limit downside and cap upside
- **Ratio spreads**: Sell multiple out-of-the-money options to finance at-the-money protection
- **Volatility hedges**: Use vega exposure to hedge against implied volatility spikes

Optimal hedge design depends on:

- Risk tolerance and downside limits
- Transaction cost environment
- Correlation stability between spot and futures
- Expected holding period and roll frequency
- Capital constraints and margin availability

The framework rejects naive full hedging in favor of risk-return optimization, recognizing that currency exposure may offer diversification benefits or expected returns that justify partial hedging or selective hedging by currency.

## 1.9 Dispersion Trading

### 1.9.1 Motivation

Dispersion trading exploits the difference between index implied volatility and the weighted average implied volatility of its constituents. When index options are expensive relative to single-name options (or vice versa), the implied correlation embedded in index volatility deviates from realized correlation, creating arbitrage opportunities.

This framework addresses:

- Extracting implied correlation from volatility surfaces
- Detecting dispersion mispricings via z-score analysis
- Constructing delta-neutral dispersion trades
- Managing portfolio Greeks across index and constituent legs
- Attributing P&L to correlation convergence

### 1.9.2 Implied Correlation from Volatility

#### 1.9.2.1 Volatility Decomposition

For an index composed of $n$ constituents with weights $w_i$ and implied volatilities $\sigma_i$, the index variance under perfect correlation would be:

$$\sigma_{\text{index}}^2 = \left( \sum_{i=1}^{n} w_i \sigma_i \right)^2$$

Under imperfect correlation, the actual index variance is:

$$\sigma_{\text{index}}^2 = \sum_{i=1}^{n} w_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{j>i}^{n} w_i w_j \sigma_i \sigma_j \rho_{ij}$$

where $\rho_{ij}$ is the pairwise correlation between constituents $i$ and $j$.

#### 1.9.2.2 Implied Correlation

Assuming a single average correlation $\bar{\rho}$ across all pairs, we can solve for the implied correlation:

$$\bar{\rho}_{\text{impl}} = \frac{\sigma_{\text{index}}^2 - \sum_{i=1}^{n} w_i^2 \sigma_i^2}{2 \sum_{i=1}^{n} \sum_{j>i}^{n} w_i w_j \sigma_i \sigma_j}$$

This implied correlation reflects the market's expectation of how correlated constituent returns will be over the option's life.

### 1.9.3 Dispersion Metrics

#### 1.9.3.1 Dispersion Level

The dispersion level measures the spread between weighted average constituent IV and index IV:

$$D = \sum_{i=1}^{n} w_i \sigma_i - \sigma_{\text{index}}$$

Positive dispersion ($D > 0$) indicates single-name options are expensive relative to the index; negative dispersion indicates the reverse.

#### 1.9.3.2 Dispersion Z-Score

To normalize dispersion across time and market regimes:

$$Z_{\text{disp}} = \frac{D - \mu_D}{\sigma_D}$$

where $\mu_D$ and $\sigma_D$ are the rolling mean and standard deviation of historical dispersion levels (typically 60–252 day lookback).

Extreme z-scores ($|Z| > 2$) signal potential mean reversion opportunities.

### 1.9.4 Trade Construction

#### 1.9.4.1 Long Dispersion

When implied correlation is high relative to expected realized correlation ($Z_{\text{disp}} < -2$):

$$\text{Position} = \text{Long Single-Name Straddles} + \text{Short Index Straddle}$$

This profits if realized correlation is lower than implied, causing constituent volatilities to outperform index volatility.

#### 1.9.4.2 Short Dispersion

When implied correlation is low ($Z_{\text{disp}} > 2$):

$$\text{Position} = \text{Short Single-Name Straddles} + \text{Long Index Straddle}$$

This profits if realized correlation exceeds implied correlation.

#### 1.9.4.3 Delta Neutrality

Aggregate portfolio delta is maintained near zero:

$$\Delta_{\text{port}} = \sum_{i=1}^{n} w_i \Delta_i - \Delta_{\text{index}} \approx 0$$

Rehedge when $|\Delta_{\text{port}}| > 0.05 \times S_{\text{index}}$.

### 1.9.5 P&L Attribution

The dispersion trade P&L decomposes into:

$$\text{P\&L} = \underbrace{\sum_{i=1}^{n} w_i \text{P\&L}_i}_{\text{Single Names}} - \underbrace{\text{P\&L}_{\text{index}}}_{\text{Index}}$$

Further attribution:

- **Gamma P&L**: From realized volatility vs implied
- **Vega P&L**: From IV changes (correlation convergence)
- **Theta decay**: Time value erosion
- **Correlation P&L**: Residual from correlation mean reversion

### 1.9.6 Interpretation

1. **Correlation risk premium**: Index options embed a correlation risk premium; short dispersion harvests this premium but faces tail risk during market stress when correlations spike.
2. **Regime sensitivity**: During crises, correlations approach 1.0, causing dispersion trades to suffer. Long dispersion provides convexity during stress.
3. **Transaction costs**: Managing multiple single-name positions incurs higher costs than index-only strategies.
4. **Liquidity constraints**: Single-name option liquidity varies; illiquid names widen bid-ask spreads and execution risk.

The strategy is best suited for:

- Volatility arbitrageurs with access to tight spreads across index and single-name options
- Institutional desks capable of managing multi-leg positions
- Investors seeking correlation exposure independent of market direction

## 1.10 Earnings Volatility Scanner

### 1.10.1 Motivation

Earnings announcements create predictable volatility events. Implied volatility rises into earnings as uncertainty increases, then collapses post-announcement ("IV crush"). This framework identifies opportunities where the IV term structure signals profitable calendar spreads or straddles around earnings.

This framework addresses:

- Analyzing IV term structure slope and curvature
- Comparing implied volatility to realized volatility
- Sizing positions using Kelly criterion from historical backtest statistics
- Filtering opportunities by liquidity, IV/RV ratio, and term structure slope

### 1.10.2 Term Structure Analysis

#### 1.10.2.1 Term Structure Slope

The IV term structure slope measures the rate of IV change across expirations:

$$\text{Slope} = \frac{\sigma_{\text{back}} - \sigma_{\text{front}}}{\text{DTE}_{\text{back}} - \text{DTE}_{\text{front}}}$$

where $\sigma_{\text{front}}$ and $\sigma_{\text{back}}$ are the front-month and back-month ATM implied volatilities, and DTE is days to expiration.

Negative slope (backwardation) indicates front-month IV exceeds back-month, typical before earnings.

#### 1.10.2.2 Term Structure Ratio

The ratio provides a scale-invariant measure:

$$\text{Ratio} = \frac{\sigma_{\text{back}}}{\sigma_{\text{front}}}$$

Ratios below 1.0 indicate backwardation; ratios significantly below 0.9 suggest extreme front-month IV elevation.

#### 1.10.2.3 IV/RV Comparison

The implied-to-realized volatility ratio:

$$\text{IV/RV} = \frac{\sigma_{\text{implied,30d}}}{\sigma_{\text{realized,30d}}}$$

Ratios above 1.1 suggest options are expensive relative to recent realized volatility, favoring short volatility strategies.

### 1.10.3 Expected Move Calculation

The annualized ATM implied volatility converts to an expected stock move:

$$E[\text{Move}] = \sigma_{\text{ATM}} \sqrt{\frac{\text{DTE}}{365}}$$

This represents the one-standard-deviation expected price change by expiration, used to assess whether straddle pricing is fair relative to historical earnings moves.

### 1.10.4 Kelly Criterion Sizing

#### 1.10.4.1 Kelly Fraction

From backtest statistics (win rate $p$, average win $W$, average loss $L$):

$$f^* = \frac{p \cdot W - (1-p) \cdot L}{W \cdot L} = \frac{\text{Edge}}{\text{Avg Win}}$$

Simplified for equal win/loss magnitudes:

$$f^* = \frac{\mu}{\sigma^2}$$

where $\mu$ is mean return and $\sigma$ is standard deviation.

#### 1.10.4.2 Fractional Kelly

Full Kelly is aggressive; conservative sizing uses fractions:

- Half Kelly: $f = 0.5 \times f^*$
- Quarter Kelly: $f = 0.25 \times f^*$

Half Kelly is typically recommended, balancing growth rate against drawdown risk.

### 1.10.5 Trade Structures

#### 1.10.5.1 Calendar Spread

Sell front-month straddle (captures IV crush), buy back-month straddle (hedge):

$$\text{P\&L} = (\text{Front Premium Decay}) - (\text{Back Premium Change})$$

Profits if IV crush in front month exceeds back month decay and stock stays near strike.

#### 1.10.5.2 Short Straddle

Sell ATM call and put:

$$\text{Max Profit} = \text{Call Premium} + \text{Put Premium}$$

$$\text{Breakeven} = K \pm \text{Total Premium}$$

Profits if realized move is less than implied move.

### 1.10.6 Filter Criteria

Opportunities are filtered by:

- Term structure slope $\leq -5\%$ (significant backwardation)
- Average volume $\geq$ 1M shares (liquidity)
- IV/RV ratio $\geq 1.1$ (options expensive)
- Days to earnings $\leq 14$ (imminent catalyst)

### 1.10.7 Interpretation

1. **IV crush monetization**: The strategy profits from the predictable collapse of implied volatility post-earnings.
2. **Gap risk**: Large earnings surprises can cause moves exceeding the expected range, resulting in significant losses for short straddles.
3. **Term structure persistence**: Backwardation typically persists until earnings; early entry captures more theta decay.
4. **Sector effects**: Tech and growth stocks exhibit larger earnings moves; adjust expected move assumptions by sector.

The strategy is best suited for:

- Options traders comfortable with event-driven volatility
- Investors with systematic earnings calendars and backtested statistics
- Traders willing to size positions conservatively via Kelly fractions

## 1.11 Forward Factor Scanner

### 1.11.1 Motivation

The forward factor measures backwardation in the volatility term structure by comparing near-term implied volatility to forward-starting volatility. Extreme forward factors signal calendar spread opportunities where front-month options are overpriced relative to deferred expirations.

This framework addresses:

- Computing forward volatility from the variance term structure
- Calculating the forward factor as a normalized backwardation signal
- Constructing ATM and double calendar spreads
- Sizing positions based on forward factor magnitude

### 1.11.2 Forward Volatility Calculation

#### 1.11.2.1 Variance Decomposition

Total variance to time $T$ is additive:

$$\sigma_T^2 \cdot T = \sigma_1^2 \cdot T_1 + V_{\text{fwd}} \cdot (T - T_1)$$

where $V_{\text{fwd}}$ is the forward variance from $T_1$ to $T$.

#### 1.11.2.2 Forward Variance

Solving for forward variance:

$$V_{\text{fwd}} = \frac{\sigma_2^2 \cdot T_2 - \sigma_1^2 \cdot T_1}{T_2 - T_1}$$

where $\sigma_1$ and $\sigma_2$ are the implied volatilities at expirations $T_1$ and $T_2$.

#### 1.11.2.3 Forward Volatility

The annualized forward volatility:

$$\sigma_{\text{fwd}} = \sqrt{V_{\text{fwd}}}$$

This represents the market's implied volatility for the period between $T_1$ and $T_2$.

### 1.11.3 Forward Factor

#### 1.11.3.1 Definition

The forward factor measures backwardation strength:

$$\text{FF} = \frac{\sigma_1 - \sigma_{\text{fwd}}}{\sigma_{\text{fwd}}} = \frac{\sigma_1}{\sigma_{\text{fwd}}} - 1$$

#### 1.11.3.2 Interpretation

- FF $> 0$: Backwardation (front IV $>$ forward IV); calendar spreads are attractive
- FF $= 0$: Flat term structure
- FF $< 0$: Contango (front IV $<$ forward IV); avoid calendar spreads

Threshold: FF $\geq 0.20$ (20% backwardation) signals actionable opportunity.

### 1.11.4 Calendar Spread Structures

#### 1.11.4.1 ATM Calendar

Sell front-month ATM call, buy back-month ATM call at same strike:

$$\text{Debit} = C_{\text{back}}(K) - C_{\text{front}}(K)$$

$$\text{Max Profit} \approx 0.75 \times \text{Debit} \quad \text{(conservative estimate)}$$

Profits if stock stays near strike and front-month IV decays faster than back-month.

#### 1.11.4.2 Double Calendar

Sell front-month 35$\Delta$ call and put, buy back-month at same strikes:

$$\text{Debit} = [C_{\text{back}}(K_C) - C_{\text{front}}(K_C)] + [P_{\text{back}}(K_P) - P_{\text{front}}(K_P)]$$

$$\text{Max Profit} \approx 1.0 \times \text{Debit}$$

Wider profit zone than ATM calendar; benefits from both call and put IV convergence.

### 1.11.5 Position Sizing

#### 1.11.5.1 Kelly Fraction by Forward Factor

Expected return scales with forward factor magnitude:

- FF $\geq 1.0$: Expected return $\approx 80\%$, Kelly $\approx 8\%$
- FF $\geq 0.5$: Expected return $\approx 50\%$, Kelly $\approx 5\%$
- FF $\geq 0.2$: Expected return $\approx 30\%$, Kelly $\approx 3\%$

From backtest: win rate $p \approx 0.66$, win/loss ratio $b \approx 1.5$.

### 1.11.6 Interpretation

1. **Term structure mean reversion**: Extreme backwardation tends to normalize as front-month IV decays faster than back-month.
2. **Event sensitivity**: Earnings or other catalysts in the front month cause backwardation; post-event, term structure flattens.
3. **Gamma risk**: Large moves can cause losses even if IV converges; double calendars provide wider breakevens.
4. **Theta dynamics**: Front-month theta exceeds back-month; time decay favors the position.

The strategy is best suited for:

- Volatility traders seeking term structure arbitrage
- Systematic scanners monitoring multiple tickers and DTE combinations
- Investors comfortable with defined-risk spreads

## 1.12 Pairs Trading

### 1.12.1 Motivation

Pairs trading exploits mean reversion in the price spread between two cointegrated assets. When two stocks share a common stochastic trend, their spread is stationary and reverts to a long-run equilibrium, enabling market-neutral profits.

This framework addresses:

- Testing for cointegration using Engle-Granger and Johansen methods
- Estimating hedge ratios via OLS and Total Least Squares (TLS) regression
- Computing spread z-scores for entry/exit signals
- Calculating mean reversion half-life for holding period estimation
- Monitoring half-life stability to detect weakening mean reversion

### 1.12.2 Cointegration Testing

#### 1.12.2.1 Engle-Granger Two-Step Method

**Step 1:** Regress one price series on the other:

$$Y_t = \alpha + \beta X_t + \epsilon_t$$

where $\beta$ is the hedge ratio and $\epsilon_t$ are the residuals.

**Step 2:** Test residuals for stationarity using the Augmented Dickey-Fuller (ADF) test.

#### 1.12.2.2 Augmented Dickey-Fuller Test

The ADF regression with $p$ lags:

$$\Delta\epsilon_t = \alpha + \rho\epsilon_{t-1} + \sum_{i=1}^{p} \gamma_i \Delta\epsilon_{t-i} + u_t$$

Hypothesis test:

- $H_0$: $\rho = 0$ (unit root, non-stationary)
- $H_1$: $\rho < 0$ (stationary)

Test statistic:

$$t = \frac{\hat{\rho}}{\mathrm{SE}(\hat{\rho})}$$

Reject $H_0$ if $t <$ critical value (e.g., $-2.86$ at 5% for $n = 100$).

#### 1.12.2.3 Lag Selection

Optimal lag $p$ minimizes the Bayesian Information Criterion (BIC):

$$\mathrm{BIC}(p) = n \ln(\mathrm{RSS}/n) + p \ln(n)$$

Maximum lag from Schwert's rule:

$$p_{\max} = \left\lfloor 12 \cdot \left(\frac{n}{100}\right)^{0.25} \right\rfloor$$

#### 1.12.2.4 Johansen Cointegration Test

The Johansen test is inherently symmetric and does not require choosing a dependent variable. For two price series $Z_t = [X_t, Y_t]'$, the Vector Error Correction Model (VECM) with one lag is:

$$\Delta Z_t = \Pi Z_{t-1} + \mu + \varepsilon_t$$

where $\Pi = \alpha\beta'$ is the error correction matrix with $\beta$ the cointegrating vector and $\alpha$ the adjustment speeds.

The procedure computes residuals $R_0$ and $R_1$ from regressing $\Delta Z_t$ and $Z_{t-1}$ on a constant, then forms the moment matrices:

$$S_{ij} = \frac{1}{T} R_i' R_j, \quad i, j \in \{0, 1\}$$

The eigenvalues $\hat{\lambda}_1 \geq \hat{\lambda}_2$ of the matrix $S_{11}^{-1} S_{10} S_{00}^{-1} S_{01}$ yield the trace statistic for testing rank $r = 0$ (no cointegration):

$$\mathrm{Trace}(r = 0) = -T \sum_{i=1}^{2} \ln(1 - \hat{\lambda}_i)$$

For two variables at 5% significance: reject $H_0$ (no cointegration) if Trace $> 15.41$.

The cointegrating vector is extracted from the eigenvector of $\hat{\lambda}_1$, normalized as $[1, -\beta]$, giving the Johansen hedge ratio directly. For the $2 \times 2$ case, all matrix operations (inversion, eigenvalue decomposition) have closed-form analytical solutions.

### 1.12.3 Spread Construction

#### 1.12.3.1 OLS Hedge Ratio

The OLS hedge ratio $\beta$ determines the relative position sizes:

$$\mathrm{Spread}_t = Y_t - \beta X_t - \alpha$$

For each \$1 long in $Y$, short $\beta$ dollars in $X$.

#### 1.12.3.2 Total Least Squares Regression

OLS is asymmetric: regressing $Y$ on $X$ gives a different hedge ratio than $X$ on $Y$. Total Least Squares (TLS) resolves this by minimizing the perpendicular distance to the regression line rather than the vertical distance.

For centered data $(\tilde{x}, \tilde{y})$, form the $2 \times 2$ scatter matrix:

$$S = \begin{bmatrix} \sum \tilde{x}_i^2 & \sum \tilde{x}_i \tilde{y}_i \\ \sum \tilde{x}_i \tilde{y}_i & \sum \tilde{y}_i^2 \end{bmatrix}$$

The eigenvalues are:

$$\lambda = \frac{\mathrm{tr}(S) \pm \sqrt{\mathrm{tr}(S)^2 - 4\det(S)}}{2}$$

The TLS slope comes from the eigenvector $[a, b]$ corresponding to the smallest eigenvalue:

$$\beta_{\mathrm{TLS}} = -\frac{a}{b} = \frac{S_{xy}}{S_{xx} - \lambda_{\min}}$$

The key property is symmetry: if $\beta_{\mathrm{TLS}}(X \to Y) = \hat{\beta}$, then $\beta_{\mathrm{TLS}}(Y \to X) \approx 1/\hat{\beta}$, making the cointegration test invariant to the choice of dependent variable.

#### 1.12.3.3 Spread Z-Score

Normalize the spread for signal generation:

$$Z_t = \frac{\mathrm{Spread}_t - \mu_{\mathrm{spread}}}{\sigma_{\mathrm{spread}}}$$

where $\mu$ and $\sigma$ are rolling statistics (typically 20–60 day lookback).

### 1.12.4 Mean Reversion Half-Life

#### 1.12.4.1 AR(1) Model

Fit an autoregressive model to the spread:

$$\Delta S_t = \alpha + \phi S_{t-1} + \epsilon_t$$

where $\phi = \rho - 1$ and $\rho$ is the AR(1) coefficient.

#### 1.12.4.2 Half-Life Calculation

The half-life is the expected time for the spread to revert halfway to equilibrium:

$$t_{1/2} = -\frac{\ln 2}{\ln \rho} = -\frac{\ln 2}{\ln(1 + \phi)}$$

Typical tradeable pairs have half-lives of 5–20 days.

#### 1.12.4.3 Dynamic Half-Life Monitoring

The half-life is not constant — it drifts as the cointegration relationship strengthens or weakens. A rolling window estimate tracks this:

$$t_{1/2}^{(\mathrm{rolling})} = -\frac{\ln 2}{\ln(1 + \hat{\phi}_w)}$$

where $\hat{\phi}_w$ is the AR(1) coefficient estimated from the most recent $w$ observations (typically $w = 60$).

Comparing the rolling half-life to the full-sample baseline:

$$r = \frac{t_{1/2}^{(\mathrm{rolling})}}{t_{1/2}^{(\mathrm{baseline})}}$$

- $r < 2.0$: Mean reversion stable, position is safe to hold
- $r \geq 2.0$: Mean reversion weakening, consider reducing exposure or exiting

This provides an early warning signal that the cointegration relationship may be breaking down, complementing the stop-loss rules based on z-score expansion.

### 1.12.5 Trading Signals

#### 1.12.5.1 Entry Rules

- **Long spread** (buy $Y$, sell $X$): $Z_t < -2.0$
- **Short spread** (sell $Y$, buy $X$): $Z_t > 2.0$

#### 1.12.5.2 Exit Rules

- **Take profit**: $|Z_t| < 0.5$ (spread reverted)
- **Stop loss**: $|Z_t| > 4.0$ (divergence, possible regime change)
- **Time stop**: Position held $> 2 \times t_{1/2}$ (mean reversion failed)

### 1.12.6 Position Sizing

#### 1.12.6.1 Dollar Neutrality

For notional $N$:

- Long $Y$: $N/(1+\beta)$ dollars
- Short $X$: $N \cdot \beta/(1+\beta)$ dollars

This ensures the portfolio is approximately dollar-neutral.

#### 1.12.6.2 Beta Neutrality

Adjust for market beta:

$$\text{Shares}_X = \text{Shares}_Y \times \frac{\beta_Y}{\beta_X} \times \beta_{\text{hedge}}$$

### 1.12.7 Performance Metrics

- **Win rate**: Fraction of trades closed at profit
- **Profit factor**: Gross profits / Gross losses
- **Sharpe ratio**: $\mu_{\text{return}}/\sigma_{\text{return}}$
- **Maximum drawdown**: Largest peak-to-trough decline
- **Average holding period**: Mean trade duration

### 1.12.8 Interpretation

1. **Cointegration vs. correlation**: High correlation does not imply cointegration; pairs must share a common stochastic trend.
2. **Regime changes**: Cointegration relationships can break down due to fundamental changes (M&A, sector rotation).
3. **Execution costs**: Frequent rebalancing and short borrowing costs erode profits; target pairs with low transaction costs.
4. **Capacity constraints**: Pairs trading alpha decays with AUM; strategies are capacity-limited.

The strategy is best suited for:

- Quantitative traders with robust cointegration testing infrastructure
- Market-neutral funds seeking low-beta returns
- Systematic investors with access to short selling and low borrowing costs

## 1.13 Pre-Earnings Straddle Scanner

### 1.13.1 Motivation

Pre-earnings straddles profit from stock moves exceeding the implied move priced into options. A machine learning model predicts expected returns based on IV term structure signals, enabling systematic selection of high-probability earnings trades.

This framework addresses:

- Extracting predictive signals from IV and realized volatility relationships
- Applying a linear regression model to forecast straddle returns
- Sizing positions using Kelly criterion from historical trade statistics
- Ranking opportunities for portfolio construction

### 1.13.2 Signal Construction

#### 1.13.2.1 Four Predictive Signals

The model uses four signals derived from implied and realized volatility:

**Signal 1:** Current implied vs. last implied ratio

$$S_1 = \frac{\sigma_{\text{curr}}}{\sigma_{\text{last}}}$$

**Signal 2:** Current implied minus last realized gap

$$S_2 = \sigma_{\text{curr}} - \text{RV}_{\text{last}}$$

**Signal 3:** Current implied vs. average implied ratio

$$S_3 = \frac{\sigma_{\text{curr}}}{\bar{\sigma}}$$

**Signal 4:** Current implied minus average realized gap

$$S_4 = \sigma_{\text{curr}} - \overline{\text{RV}}$$

where "last" refers to the previous earnings cycle and "average" refers to the trailing 4–8 earnings cycles.

### 1.13.3 Linear Regression Model

#### 1.13.3.1 Prediction Equation

Expected straddle return:

$$\hat{r} = \beta_0 + \beta_1 S_1 + \beta_2 S_2 + \beta_3 S_3 + \beta_4 S_4$$

Coefficients are estimated via OLS on historical earnings trades.

#### 1.13.3.2 Typical Coefficients

From backtest calibration:

- $\beta_0 \approx 0.033$ (intercept)
- $\beta_1 \approx -0.050$ (negative: high relative IV is bearish)
- $\beta_2 \approx -0.040$ (negative: large IV-RV gap is bearish)
- $\beta_3 \approx -0.060$ (negative: elevated vs. history is bearish)
- $\beta_4 \approx -0.050$ (negative: IV above avg RV is bearish)

### 1.13.4 Implied Move Calculation

The ATM straddle prices an expected move:

$$E[\text{Move}] = \sigma_{\text{ATM}} \sqrt{\frac{\text{DTE}}{365}}$$

The straddle is profitable if the realized move exceeds this implied move.

### 1.13.5 Kelly Sizing

#### 1.13.5.1 Historical Statistics

From backtest (typical values):

- Win rate: $p \approx 0.42$
- Average win: $W \approx 15\%$
- Average loss: $L \approx 50\%$

#### 1.13.5.2 Kelly Fraction

$$f^* = \frac{p \cdot W - (1-p) \cdot L}{W} = \frac{0.42 \times 0.15 - 0.58 \times 0.50}{0.15} \approx -1.5$$

Negative Kelly indicates the strategy is unprofitable on average; only trade when predicted return $\hat{r}$ exceeds a threshold.

#### 1.13.5.3 Conditional Sizing

When $\hat{r} > 0$:

$$f = \min\left(\frac{\hat{r}}{\sigma_{\hat{r}}^2}, 0.10\right)$$

Cap at 10% of account to limit single-event risk.

### 1.13.6 Recommendation Thresholds

- **Strong Buy**: $\hat{r} \geq 5\%$
- **Buy**: $\hat{r} \geq 2\%$
- **Pass**: $\hat{r} < 2\%$

### 1.13.7 Interpretation

1. **Signal interpretation**: Negative coefficients indicate that high implied volatility (relative to history or realized) predicts lower straddle returns—options are overpriced.
2. **Model limitations**: Linear regression assumes stable relationships; coefficient drift requires periodic recalibration.
3. **Event clustering**: Multiple earnings in the same week can concentrate risk; diversify across dates.
4. **Tail risk**: Earnings surprises can cause moves far exceeding expectations; position sizing limits catastrophic losses.

The strategy is best suited for:

- Systematic earnings traders with historical databases
- Quantitative investors comfortable with machine learning signals
- Traders willing to accept low win rates with positive expected value

## 1.14 Skew Verticals Scanner

### 1.14.1 Motivation

Volatility skew—the tendency for OTM put IV to exceed ATM IV—reflects demand for downside protection. When skew is extreme relative to history, vertical spreads can profit from skew normalization while providing defined risk.

This framework addresses:

- Quantifying call and put skew relative to ATM volatility
- Computing skew z-scores for mean reversion signals
- Constructing bull call and bear put vertical spreads
- Scoring opportunities via multi-factor edge analysis
- Estimating probability of profit and expected value

### 1.14.2 Skew Metrics

#### 1.14.2.1 Normalized Skew

Call skew (typically negative in equity markets):

$$\text{Skew}_{\text{call}} = \frac{\sigma_{\text{ATM}} - \sigma_{25\Delta\ \text{call}}}{\sigma_{\text{ATM}}}$$

Put skew (typically positive):

$$\text{Skew}_{\text{put}} = \frac{\sigma_{25\Delta\ \text{put}} - \sigma_{\text{ATM}}}{\sigma_{\text{ATM}}}$$

#### 1.14.2.2 Skew Z-Score

Normalize skew against historical distribution:

$$Z_{\text{skew}} = \frac{\text{Skew}_{\text{curr}} - \mu_{\text{skew}}}{\sigma_{\text{skew}}}$$

Extreme z-scores ($|Z| > 2$) signal mean reversion opportunities.

#### 1.14.2.3 Variance Risk Premium

The VRP measures richness of options:

$$\text{VRP} = \sigma_{\text{ATM}} - \text{RV}_{30d}$$

Positive VRP indicates options are expensive relative to realized volatility.

### 1.14.3 Vertical Spread Construction

#### 1.14.3.1 Bull Call Spread

Buy lower-strike call, sell higher-strike call:

$$\text{Debit} = C(K_{\text{long}}) - C(K_{\text{short}})$$

$$\text{Max Profit} = (K_{\text{short}} - K_{\text{long}}) - \text{Debit}$$

$$\text{Max Loss} = \text{Debit}$$

$$\text{Breakeven} = K_{\text{long}} + \text{Debit}$$

#### 1.14.3.2 Bear Put Spread

Buy higher-strike put, sell lower-strike put:

$$\text{Debit} = P(K_{\text{long}}) - P(K_{\text{short}})$$

$$\text{Max Profit} = (K_{\text{long}} - K_{\text{short}}) - \text{Debit}$$

$$\text{Max Loss} = \text{Debit}$$

$$\text{Breakeven} = K_{\text{long}} - \text{Debit}$$

#### 1.14.3.3 Strike Selection

Valid strikes must satisfy sanity filters:

- ATM candidates: Strike within 5% of spot, IV $< 200\%$, mid price $\geq \$0.10$
- Short leg candidates: Strike 5–15% from long strike, IV $< 200\%$, IV $< 2.5 \times \sigma_{\text{ATM}}$
- Minimum reward/risk: $\geq 1.5 : 1$ for debit spreads, $\geq 0.1 : 1$ for credit spreads

The relaxed IV ceiling of 200% (vs. 100%) accommodates high-volatility names where OTM options routinely exceed 100% IV.

### 1.14.4 Probability of Profit

#### 1.14.4.1 Lognormal Approximation

Assuming lognormal price distribution:

$$\text{POP} = N\left(\frac{\ln(\text{Breakeven}/S)}{\sigma\sqrt{T}}\right)$$

where $N(\cdot)$ is the standard normal CDF.

For bull call spreads, POP increases as the breakeven approaches the current price.

#### 1.14.4.2 Expected Value

$$E[V] = \text{POP} \times \text{Max Profit} - (1 - \text{POP}) \times \text{Max Loss}$$

$$\text{Expected Return} = \frac{E[V]}{\text{Debit}} \times 100\%$$

### 1.14.5 Multi-Factor Edge Score

A composite score (0–100) combines multiple factors:

- **Skew component** (40 points): $\min(|Z_{\text{skew}}| \times 10, 40)$
- **Momentum component** (20 points): Directional alignment with price trend
- **Reward/risk component** (30 points): Tiered by ratio
  - $\geq 10 : 1 \rightarrow 30$ points
  - $\geq 7 : 1 \rightarrow 25$ points
  - $\geq 5 : 1 \rightarrow 20$ points
  - $\geq 3 : 1 \rightarrow 10$ points
- **Expected value component** (10 points): Tiered by return
  - $\geq 50\% \rightarrow 10$ points
  - $\geq 20\% \rightarrow 7$ points
  - $\geq 0\% \rightarrow 0$ points

### 1.14.6 Filter Criteria

Opportunities must pass:

- Skew filter: $Z_{\text{skew}} < -2.0$ (extreme skew)
- IV/RV filter: VRP $> 0$ (positive variance risk premium) and OTM IV $>$ RV
- Momentum filter: Score $> 0.3$ for bull calls, score $< -0.3$ for bear puts

The IV/RV filter ensures options are overpriced relative to realized volatility. Positive VRP indicates the market systematically pays more for volatility protection than historical moves justify, creating edge for premium sellers.

### 1.14.7 Recommendation Thresholds

- **Strong Buy**: Edge score $\geq 60$ and all filters pass
- **Buy**: Edge score $\geq 40$
- **Pass**: Edge score $< 40$ or filter failure

### 1.14.8 Interpretation

1. **Skew mean reversion**: Extreme skew tends to normalize; selling expensive OTM options via spreads captures this convergence.
2. **Defined risk**: Vertical spreads cap maximum loss at the debit paid, unlike naked option sales.
3. **Momentum alignment**: Trading with the trend increases probability of profit; counter-trend trades require stronger skew signals.
4. **Liquidity considerations**: Wide bid-ask spreads on OTM options reduce edge; prefer liquid underlyings.

The strategy is best suited for:

- Options traders seeking defined-risk directional exposure
- Systematic scanners monitoring skew across multiple underlyings
- Investors comfortable with multi-factor signal combination

## 1.15 Perpetual Futures Pricing

Perpetual futures are derivatives contracts without expiration dates, dominant in cryptocurrency markets. The anchoring of futures price to spot is achieved via periodic funding payments. This section implements the no-arbitrage pricing framework.

### 1.15.1 Contract Types

#### 1.15.1.1 Linear Perpetual Futures

A linear perpetual provides exposure to one unit of base currency $b$ from the perspective of an investor in quote currency $a$. The futures price $f_t$ is quoted in units of $a$ and margining is in currency $a$.

#### 1.15.1.2 Inverse Perpetual Futures

An inverse perpetual uses the base currency $b$ for margining and settlement. The contract size is expressed in units of quote currency $a$. This allows speculation on the exchange rate without holding fiat currency.

#### 1.15.1.3 Quanto Perpetual Futures

A quanto perpetual uses a third currency $c$ (different from $a$ and $b$) for margining and settlement. The payoff requires a convexity adjustment for the correlation between the two exchange rates.

### 1.15.2 Funding Mechanism

The funding payment from long to short consists of two components:

1. **Premium term**: $\kappa_t(f_t - x_t)$ where $\kappa_t > 0$ controls anchoring intensity
2. **Interest term**: $\iota_t x_t$ reflecting interest rate differential

The cash flow at time $t+1$ from a long position is:

$$(f_{t+1} - f_t) - \kappa_t(f_t - x_t) - \iota_t x_t$$

### 1.15.3 No-Arbitrage Pricing

#### 1.15.3.1 Linear Perpetual Futures Price

Under constant interest rates $r_a$, $r_b$ and funding parameters $\kappa$, $\iota$:

**Discrete-time** (Proposition 1):

$$f_t = \frac{(\kappa - \iota)(1 + r_b)}{r_b - r_a + \kappa(1 + r_b)} \cdot x_t$$

**Continuous-time** (Proposition 3):

$$f_t = \frac{\kappa - \iota}{\kappa + r_b - r_a} \cdot x_t$$

subject to the condition $\kappa + r_b - r_a > 0$.

#### 1.15.3.2 Inverse Perpetual Futures Price

Let $x_t^* = 1/x_t$ denote the $a/b$ exchange rate.

**Continuous-time** (Proposition 4):

$$f_{I,t} = \frac{\kappa_I + r_a - r_b}{\kappa_I - \iota_I} \cdot x_t$$

#### 1.15.3.3 Quanto Perpetual Futures Price

For a contract on the $c/a$ exchange rate $z_t$, margined in currency $b$:

$$f_{q,t} = \frac{\kappa - \iota}{r_c - \sigma_x^* \sigma_z - r_a + \kappa} \cdot z_t$$

where $\sigma_x^* \sigma_z$ is the covariance between the two exchange rates (convexity adjustment).

### 1.15.4 Perfect Anchoring

The interest factor that guarantees $f_t = x_t$ at all times:

**Linear** (Corollary 1):

$$\iota_t = \frac{r_{a,t} - r_{b,t}}{1 + r_{b,t}} < \kappa_t \quad \text{(discrete)}$$

$$\iota_t = r_{a,t} - r_{b,t} < \kappa_t \quad \text{(continuous)}$$

**Inverse** (Corollary 2):

$$\iota_{I,t} = \frac{r_{b,t} - r_{a,t}}{1 + r_{a,t}} < \kappa_{I,t} \quad \text{(discrete)}$$

### 1.15.5 Random Maturity Representation

A key result is that the perpetual futures price equals the risk-neutral expectation of the spot at a random time:

$$f_t = E_t^{Q_a}[x_{\theta_t}]$$

where $\theta_t$ is distributed according to:

- **Discrete**: $\theta \sim \text{Geometric}(\kappa)$ with mean $1/\kappa$
- **Continuous**: $\theta \sim \text{Exponential}(\kappa)$ with mean $1/\kappa$

### 1.15.6 Basis Analysis

The basis (futures minus spot) under constant parameters:

$$\text{Basis} = f_t - x_t = \left(\frac{\kappa - \iota}{\kappa + r_b - r_a} - 1\right) x_t$$

- **Contango** ($f_t > x_t$): When $r_a > r_b$ (quote currency has higher interest)
- **Backwardation** ($f_t < x_t$): When $r_b > r_a$ (base currency has higher interest)

The magnitude of basis is:

1. Increasing in $|r_a - r_b|$ (interest rate spread)
2. Decreasing in $\kappa$ (anchoring intensity)
3. Zero when $\iota = r_a - r_b$ (perfect anchoring)

### 1.15.7 Everlasting Options

Everlasting options track a function $\varphi(x_t)$ of the spot price instead of the spot itself. The price is:

$$f_{o,t} = E_t^{Q_a}\left[\varphi(x_{\theta_{o,t}})\right]$$

where $\theta_{o,t} \sim \text{Exponential}(\kappa_o)$.

#### 1.15.7.1 Black-Scholes Everlasting Options

Assume the exchange rate follows geometric Brownian motion:

$$dx_t/x_t = (r_a - r_b)dt + \sigma_x^* dZ_t^a$$

**Everlasting Call** (Proposition 6):

$$c(x) = \begin{cases} x^{\Theta} K^{1-\Theta} \cdot \frac{\Pi(r_a - r_b) - \kappa}{(\Pi - \Theta)(\kappa - r_a + r_b)} & x \leq K \\ x^{\Pi} K^{1-\Pi} \cdot \frac{\Theta(r_a - r_b) - \kappa}{(\Pi - \Theta)(\kappa - r_a + r_b)} + f(x) - K & x > K \end{cases}$$

where $\Pi < 0$ and $\Theta > 1$ are roots of:

$$(r_a - r_b)\xi + \frac{1}{2}\xi(\xi - 1)\|\sigma_x\|^2 - \kappa = 0$$

and $f(x) = \kappa x/(\kappa - r_a + r_b)$ is the perpetual futures price with $\iota = 0$.

**Put-Call Parity**:

$$c_t - p_t = f(x_t) - K$$

#### 1.15.7.2 Greeks

**Call Delta**:

$$\Delta_c(x) = \begin{cases} \Theta \cdot x^{\Theta-1} K^{1-\Theta} \cdot \text{coef}_1 & x \leq K \\ \Pi \cdot x^{\Pi-1} K^{1-\Pi} \cdot \text{coef}_2 + f'(x) & x > K \end{cases}$$

where $f'(x) = \kappa/(\kappa - r_a + r_b)$.

### 1.15.8 Implementation

The module provides:

1. **Perpetual futures pricing**: Linear, inverse, and quanto contracts
2. **Funding analysis**: Perfect anchoring interest factors
3. **Basis computation**: Theoretical vs market basis
4. **Everlasting options**: Black-Scholes pricing with Greeks
5. **Market data**: Real-time funding rates from exchanges

### 1.15.9 Practical Considerations

#### 1.15.9.1 Funding Rate Annualization

Most exchanges use 8-hour funding periods (3 per day):

$$\kappa_{\text{annual}} = \kappa_{8h} \times 3 \times 365 = 1095 \times \kappa_{8h}$$

#### 1.15.9.2 Interest Rate Proxy

For cryptocurrency perpetuals:

- $r_a$ (USD): Use Fed Funds rate or SOFR ($\approx$ 4–5% as of 2025)
- $r_b$ (BTC/ETH): Typically assumed $\approx 0$ or use staking yield

#### 1.15.9.3 Arbitrage Opportunities

When market futures price deviates from theoretical price:

- **Market** $>$ **Theoretical**: Short futures, long spot (cash-and-carry)
- **Market** $<$ **Theoretical**: Long futures, short spot (reverse cash-and-carry)

## 1.16 Liquidity Analysis

Liquidity analysis evaluates a security's tradability via price impact, volume characteristics, and spread metrics. Two complementary frameworks are implemented: a structural liquidity scoring system and a predictive volume signal engine.

### 1.16.1 Structural Liquidity Metrics

#### 1.16.1.1 Amihud Illiquidity Ratio

The Amihud (2002) illiquidity ratio measures price impact per unit of trading volume:

$$\text{ILLIQ} = \frac{1}{D} \sum_{d=1}^{D} \frac{|R_d|}{V_d^{\$}} \times 10^6$$

$R_d$: daily return on day $d$

$V_d^{\$} = P_d \times \text{Volume}_d$: dollar volume

$D$: lookback window (default 20 days)

Lower values indicate higher liquidity; ILLIQ $< 0.01$ signals highly liquid large-caps, while ILLIQ $> 10$ suggests severe illiquidity.

#### 1.16.1.2 Turnover Ratio

Turnover measures the fraction of shares outstanding traded per day:

$$\text{Turnover} = \frac{\bar{V}_D}{\text{Shares Outstanding}}$$

where $\bar{V}_D$ is the average daily volume over the lookback window. Turnover $> 5\%$ indicates active trading interest; turnover $< 0.1\%$ suggests a thinly traded name.

#### 1.16.1.3 Spread Proxy

In the absence of tick-level bid-ask data, the high-low range serves as a spread proxy:

$$\text{Spread Proxy}(\%) = \frac{1}{D} \sum_{d=1}^{D} \frac{H_d - L_d}{C_d} \times 100$$

where $H_d$, $L_d$, $C_d$ are the daily high, low, and close. Spread $< 0.5\%$ is tight; spread $> 3\%$ indicates significant friction.

#### 1.16.1.4 Volume Volatility

The coefficient of variation of volume measures trading consistency:

$$\text{VV} = \frac{\sigma(\text{Volume})}{\mu(\text{Volume})}$$

Low volume volatility (VV $< 0.3$) indicates stable, predictable liquidity; high values (VV $> 1.0$) signal erratic trading patterns.

#### 1.16.1.5 Relative Volume

Relative volume compares the latest session's volume to its recent average:

$$\text{RVOL} = \frac{V_{\text{today}}}{\bar{V}_D}$$

RVOL $\geq 2.0\times$ constitutes a volume surge; RVOL $\ll 1.0$ indicates unusual quietness.

### 1.16.2 Composite Liquidity Score

The composite score aggregates structural metrics on a 0–100 scale, starting from a base of 50 and adjusting by tier:

| Metric | Condition | Adjustment |
|---|---|---|
| Amihud | $< 0.01$ | $+15$ |
| Amihud | $> 10$ | $-15$ |
| Turnover | $> 5\%$ | $+15$ |
| Turnover | $< 0.1\%$ | $-10$ |
| Vol Volatility | $< 0.3$ | $+10$ |
| Vol Volatility | $> 1.0$ | $-10$ |
| Spread Proxy | $< 0.5\%$ | $+10$ |
| Spread Proxy | $> 3.0\%$ | $-10$ |

$$\text{Liquidity Score} = \text{clamp}\left(50 + \sum \text{adjustments},\ 0,\ 100\right)$$

Tiers: Excellent ($\geq 80$), Good ($\geq 65$), Fair ($\geq 50$), Poor ($\geq 35$), Very Poor ($< 35$).

### 1.16.3 Volume-Based Predictive Signals

#### 1.16.3.1 On-Balance Volume (OBV)

OBV is a cumulative indicator that links volume to price direction:

$$\text{OBV}_t = \text{OBV}_{t-1} + \text{sign}(R_t) \cdot V_t$$

where $\text{sign}(R_t) = +1$ if $C_t > C_{t-1}$, $-1$ if $C_t < C_{t-1}$, and 0 otherwise.

**Divergence detection**: Compare the linear regression slopes of price and OBV over a window $w$. Bullish divergence occurs when price slopes down but OBV slopes up; bearish divergence is the reverse.

$$\text{OBV Strength} = \min\left(100,\ \frac{|\text{slope}(\text{OBV})|}{\bar{V}_w} \times 100\right)$$

Divergence signals are scaled by a factor of 0.7 relative to confirmation signals.

#### 1.16.3.2 Volume Surge Detection

A surge is flagged when the current volume exceeds the prior-window average by a threshold multiple:

$$\text{Surge} = \mathbb{I}\left[\frac{V_{\text{today}}}{\bar{V}_w} \geq \tau\right], \quad \tau = 2.0$$

Surges often precede breakouts or breakdowns and warrant attention regardless of direction.

#### 1.16.3.3 Smart Money Flow

Volume-weighted return estimates net institutional participation:

$$\text{SMF} = \frac{\sum_{d=1}^{w} R_d \cdot V_d}{\sum_{d=1}^{w} V_d} \times 100$$

SMF $> 1.0$ suggests accumulation (net buying on volume); SMF $< -1.0$ suggests distribution.

### 1.16.4 Composite Signal Score

The predictive signal score combines five indicators via weighted sum:

$$S = 0.30 \cdot \text{OBV}_{\text{str}} + 0.20 \cdot (\text{RVOL} - 1) \cdot 20 + 0.20 \cdot \text{Vol Slope} + 0.15 \cdot \rho_{|R|,V} \cdot 30 + 0.15 \cdot \text{SMF} \cdot 5$$

$\text{OBV}_{\text{str}}$: OBV strength (directional)

Vol Slope: volume trend slope (%/day)

$\rho_{|R|,V}$: correlation between absolute returns and volume changes

SMF: smart money flow

Classification: Strong Bullish ($S > 30$), Bullish ($S > 10$), Neutral, Bearish ($S < -10$), Strong Bearish ($S < -30$).

### 1.16.5 Interpretation

1. **Structural vs predictive**: The liquidity score measures current tradability; the signal score forecasts directional pressure from volume patterns.
2. **Volume precedes price**: OBV divergences and smart money flow often lead price by several sessions, providing early warning of trend changes.
3. **Liquidity and volatility**: Illiquid names exhibit higher volatility and wider spreads during stress; the Amihud ratio captures this nonlinearly.
4. **Position sizing**: Low liquidity scores should translate to smaller position sizes to avoid excessive market impact.

# 2 Valuation

The tools of valuation empower the investor to judge stocks in a manner that is detached from behavioral biases and therefore from market sentiment. Valuation emphasizes companies' fundamentals and financial health.

By relying on valuation – regardless of an active or passive investing style – we reject the Efficient Market Hypothesis (EMH) in at least its strong and semi-strong forms.

## 2.1 DCF (Deterministic)

### 2.1.1 Discounted Cash Flow Foundation

Let us derive a stock's value per share based on discounted free cash flow and regard this value as intrinsic.

Interest, e.g., on investment streams, compounds to form the future value:

$$\text{FutureValue}_n = \sum_{t=0}^{n} \text{InvestedCash}_t \cdot (1 + \text{InterestRate})^{n-t}$$

To calculate the present value, e.g. of a business, we undo the compounding effect. The intrinsic value of a financial asset is the discounted value of its future cash flows:

$$\text{PresentValue} = \sum_{t=1}^{n} \frac{\text{CashFlow}_t}{(1 + \text{DiscountRate})^t}$$

One can go the free-cash-flow-to-equity (FCFE) and/or the free-cash-flow-to-firm (FCFF) route. It will depend on the company's circumstances to which to give modeling precedence. FCFE should lead to a more insightful intrinsic value per share (IVPS), if the debt policy is known and reliable, and FCFF should do so, if the debt structure is uncertain or complex.

### 2.1.2 Cash Flow Mechanics

#### 2.1.2.1 Free Cash Flow to Equity

$$\text{FCFE} = \text{NI} - (1 - \text{TDR}) \cdot (\text{CAPX} - \text{D} - \Delta\text{WC})$$

FCFE: free cash flow to equity

NI: net income

TDR: target debt ratio

CAPX: capital expenditure

D: depreciation & amortization

$\Delta\text{WC} = \Delta\text{CA} - \Delta\text{CL}$: change in working capital

CA: current assets

CL: current liabilities

#### 2.1.2.2 Free Cash Flow to Firm

$$\text{FCFF} = \text{EBIT} \cdot (1 - \text{CTR}) + \text{D} - \text{CAPX} - \Delta\text{WC}$$

FCFF: free cash flow to firm

EBIT: earnings before interest and taxes

CTR: corporate tax rate

### 2.1.3 Cost of Capital

#### 2.1.3.1 Levered Beta

$$\beta_{\mathrm{L}} = \beta_{\mathrm{U}} \cdot \left(1 + (1 - \mathrm{CTR}) \cdot \frac{\mathrm{MVB}}{\mathrm{MVE}}\right)$$

$\beta_{\mathrm{L}}$: levered beta

$\beta_{\mathrm{U}}$: unlevered beta

MVB: market value of debt

MVE: market value of equity

#### 2.1.3.2 Cost of Equity (CAPM)

$$\mathrm{CE} = \mathrm{RFR} + \beta_{\mathrm{L}} \cdot \mathrm{ERP}$$

CE: cost of equity

RFR: risk-free rate

ERP: equity risk premium

#### 2.1.3.3 Currency-Adjusted Equity Risk Premium

$$\mathrm{ERPT} = (1 + \mathrm{ERPS}) \cdot \left(\frac{1 + \mathrm{IT}}{1 + \mathrm{IS}}\right) - 1$$

ERPT: equity risk premium of target currency

ERPS: equity risk premium of source currency

IT: inflation of target

IS: inflation of source

#### 2.1.3.4 Weighted Average Cost of Capital

$$\mathrm{WACC} = \frac{\mathrm{MVE}}{\mathrm{MVE} + \mathrm{MVB}} \cdot \mathrm{CE} + \frac{\mathrm{MVB}}{\mathrm{MVE} + \mathrm{MVB}} \cdot \mathrm{CB} \cdot (1 - \mathrm{CTR})$$

WACC: weighted average cost of capital

CB: pre-tax cost of debt

### 2.1.4 Growth Rate Estimation

#### 2.1.4.1 FCFE Growth Rate

$$\begin{aligned}\mathrm{FCFEGR} &= \mathrm{ROE} \cdot \mathrm{FCFERR} \\ &= \left(\frac{\mathrm{NI}}{\mathrm{BVE}}\right) \cdot \left(1 - \frac{\mathrm{DP}}{\mathrm{NI}}\right)\end{aligned}$$

FCFEGR: FCFE growth rate estimate

ROE: return on equity

FCFERR: FCFE retention ratio

BVE: book value of equity

DP: dividends paid

#### 2.1.4.2 FCFF Growth Rate

$$\begin{aligned}
\text{FCFFGR} &= \text{ROIC} \cdot \text{FCFFRR} \\
&= \left(\frac{\text{NOPAT}}{\text{IC}}\right) \cdot \left(\frac{\text{CAPX} - \text{D} + \Delta\text{WC}}{\text{NOPAT}}\right) \\
&= \left(\frac{\text{EBIT} \cdot (1 - \text{ETR})}{\text{IC}}\right) \cdot \left(\frac{\text{CAPX} - \text{D} + \Delta\text{WC}}{\text{EBIT} \cdot (1 - \text{ETR})}\right)
\end{aligned}$$

FCFFGR: FCFF growth rate estimate

ROIC: return on invested capital

FCFFRR: FCFF reinvestment rate

NOPAT: net operating profit after taxes

IC: invested capital

ETR: effective tax rate

### 2.1.5 Present Value Calculation

#### 2.1.5.1 FCFE-Based Valuation

$$\begin{aligned}
\text{PVE} &= \text{PVFCFE} + \text{PVTVFCFE} \\
&= \left(\sum_{t=1}^{h} \frac{\text{FCFE}_t}{(1 + \text{CE})^t}\right) + \text{PVTVFCFE} \\
&= \left(\sum_{t=1}^{h} \frac{\text{FCFE}_0 \cdot (1 + \text{FCFEGR})^t}{(1 + \text{CE})^t}\right) + \text{PVTVFCFE} \\
&= \text{PVFCFE} + \left(\sum_{t=h+1}^{\infty} \frac{\text{FCFE}_t}{(1 + \text{CE})^t}\right) \\
&= \text{PVFCFE} + \left(\sum_{t=h+1}^{\infty} \frac{\text{FCFE}_h \cdot (1 + \text{TGR})^{t-h}}{(1 + \text{CE})^t}\right) \\
&= \text{PVFCFE} + \left(\sum_{t=h+1}^{\infty} \frac{\text{FCFE}_h}{(1 + \text{CE})^h} \cdot \left(\frac{1 + \text{TGR}}{1 + \text{CE}}\right)^{t-h}\right) \\
&= \text{PVFCFE} + \left(\frac{\text{FCFE}_h}{(1 + \text{CE})^h} \sum_{k=1}^{\infty} \left(\frac{1 + \text{TGR}}{1 + \text{CE}}\right)^{k}\right) \\
&\overset{\text{GGM}}{=} \text{PVFCFE} + \left(\frac{\text{FCFE}_h}{(1 + \text{CE})^h} \cdot \frac{\frac{1+\text{TGR}}{1+\text{CE}}}{1 - \frac{1+\text{TGR}}{1+\text{CE}}}\right) \\
&= \text{PVFCFE} + \left(\frac{\text{FCFE}_h}{(1 + \text{CE})^h} \cdot \frac{1 + \text{TGR}}{\text{CE} - \text{TGR}}\right) \\
&= \text{PVFCFE} + \left(\frac{\text{FCFE}_h \cdot (1 + \text{TGR})}{(\text{CE} - \text{TGR})} \cdot \frac{1}{(1 + \text{CE})^h}\right) \\
&= \text{PVFCFE} + \left(\frac{\text{FCFE}_{h+1}}{(\text{CE} - \text{TGR})} \cdot \frac{1}{(1 + \text{CE})^h}\right) \\
&= \text{PVFCFE} + \left(\frac{\text{TVFCFE}}{(1 + \text{CE})^h}\right) \\
&= \left(\sum_{t=1}^{h} \frac{\text{FCFE}_0 \cdot (1 + \text{FCFEGR})^t}{(1 + \text{CE})^t}\right) + \left(\frac{\text{FCFE}_h \cdot (1 + \text{TGR})}{(\text{CE} - \text{TGR}) \cdot (1 + \text{CE})^h}\right)
\end{aligned}$$

PVE: present value of equity

PVFCFE: present value of free cash flow to equity

PVTVFCFE: present value of terminal value of FCFE

$h$: growth forecast horizon (before terminal growth rate into perpetuity)

GGM: Gordon growth model (closed-form solution of infinite geometric series)

TGR: terminal growth rate

TVFCFE: terminal value of FCFE

#### 2.1.5.2 FCFF-Based Valuation

$$
\begin{aligned}
\mathrm{PVF} &= \mathrm{PVFCFF} + \mathrm{PVTVFCFF} \\
&= \left(\sum_{t=1}^{h} \frac{\mathrm{FCFF}_t}{(1+\mathrm{WACC})^t}\right) + \mathrm{PVTVFCFF} \\
&= \left(\sum_{t=1}^{h} \frac{\mathrm{FCFF}_0 \cdot (1+\mathrm{FCFFGR})^t}{(1+\mathrm{WACC})^t}\right) + \mathrm{PVTVFCFF} \\
&= \mathrm{PVFCFF} + \left(\sum_{t=h+1}^{\infty} \frac{\mathrm{FCFF}_t}{(1+\mathrm{WACC})^t}\right) \\
&= \mathrm{PVFCFF} + \left(\sum_{t=h+1}^{\infty} \frac{\mathrm{FCFF}_h \cdot (1+\mathrm{TGR})^{t-h}}{(1+\mathrm{WACC})^t}\right) \\
&= \mathrm{PVFCFF} + \left(\sum_{t=h+1}^{\infty} \frac{\mathrm{FCFF}_h}{(1+\mathrm{WACC})^h} \cdot \left(\frac{1+\mathrm{TGR}}{1+\mathrm{WACC}}\right)^{t-h}\right) \\
&= \mathrm{PVFCFF} + \left(\frac{\mathrm{FCFF}_h}{(1+\mathrm{WACC})^h} \sum_{k=1}^{\infty} \left(\frac{1+\mathrm{TGR}}{1+\mathrm{WACC}}\right)^{k}\right) \\
&\overset{\mathrm{GGM}}{=} \mathrm{PVFCFF} + \left(\frac{\mathrm{FCFF}_h}{(1+\mathrm{WACC})^h} \cdot \frac{\frac{1+\mathrm{TGR}}{1+\mathrm{WACC}}}{1-\frac{1+\mathrm{TGR}}{1+\mathrm{WACC}}}\right) \\
&= \mathrm{PVFCFF} + \left(\frac{\mathrm{FCFF}_h}{(1+\mathrm{WACC})^h} \cdot \frac{1+\mathrm{TGR}}{\mathrm{WACC}-\mathrm{TGR}}\right) \\
&= \mathrm{PVFCFF} + \left(\frac{\mathrm{FCFF}_h \cdot (1+\mathrm{TGR})}{(\mathrm{WACC}-\mathrm{TGR})} \cdot \frac{1}{(1+\mathrm{WACC})^h}\right) \\
&= \mathrm{PVFCFF} + \left(\frac{\mathrm{FCFF}_{h+1}}{(\mathrm{WACC}-\mathrm{TGR})} \cdot \frac{1}{(1+\mathrm{WACC})^h}\right) \\
&= \mathrm{PVFCFF} + \left(\frac{\mathrm{TVFCFF}}{(1+\mathrm{WACC})^h}\right) \\
&= \left(\sum_{t=1}^{h} \frac{\mathrm{FCFF}_0 \cdot (1+\mathrm{FCFFGR})^t}{(1+\mathrm{WACC})^t}\right) + \left(\frac{\mathrm{FCFF}_h \cdot (1+\mathrm{TGR})}{(\mathrm{WACC}-\mathrm{TGR}) \cdot (1+\mathrm{WACC})^h}\right)
\end{aligned}
$$

PVF: present value of the firm

PVFCFF: present value of free cash flow to firm

PVTVFCFF: present value of terminal value of FCFF

TVFCFF: terminal value of FCFF

#### 2.1.5.3 Reconciliation

$$\mathrm{PVE} \approx \mathrm{PVF} - \mathrm{MVB} + \mathrm{CNOA}$$

CNOA: cash and non-operating assets

#### 2.1.5.4 Intrinsic Value Per Share

$$\mathrm{IVPS} = \frac{\mathrm{PVE}}{\mathrm{SO}}$$

IVPS: intrinsic value per share

SO: shares outstanding

### 2.1.6 Interpretation

Having derived an IVPS, it may serve as a reference point when judging the market price. Supposing that, ideally, both (rather than just one of) the FCFE and FCFF based IVPS have been modeled well and are therefore reliable indicators, we may interpret the nine resulting constellations as follows.

| | **FCFE > Price** | **FCFE ≈ Price** | **FCFE < Price** |
|---|---|---|---|
| **FCFF > Price** | **strong buy:** The firm's assets and operations generate more value than what is priced in by the market, and equity holders retain it — low leverage or efficient debt structure. | **buy:** The firm's assets and operations are underpriced, but excess value is absorbed by debt or reinvestment, leaving equity fairly valued. | **caution:** The firm's assets and operations are underpriced, but debt or reinvestment absorbs most cash flows — equity claims more than it economically receives. |
| **FCFF ≈ Price** | **buy:** The firm's assets and operations are fairly priced, but equity captures a disproportionately large share — market underprices the equity upside. | **hold:** The present value of free cash flows — to the firm (before payments to debt holders) and to equity (after them) — is consistent with market prices; no mispricing is evident. | **speculative:** The business generates enough pre-financing cash flow to justify its market price, but equity holders retain too little after payments to debt holders. |
| **FCFF < Price** | **caution:** The business is overvalued, but equity appears cheap due to temporarily favorable debt terms — value may be unstable under a leveraged structure. | **speculative:** Equity is fairly priced, but depends on cash flows from a business generating less than what its market price would suggest — any decline in operations could undermine equity value. | **avoid:** There isn't sufficient cash flow to the business or the equity for the fundamentals to justify the high market price. |

### 2.1.7 Sector-Specific Models

The standard FCFE/FCFF framework breaks down for three categories of companies whose economics fundamentally differ from the general corporate model. For each, we derive a purpose-built valuation and sensitivity analysis.

#### 2.1.7.1 Bank Valuation (Excess Return Model)

For banks, interest expense is an operating cost (not a financing cost), and deposits – the primary source of debt – are the raw material of the business. FCFE and FCFF are therefore not meaningful. Instead, we value banks through the excess return model:

$$\mathrm{FV}_{\mathrm{bank}} = \mathrm{BVE} + \sum_{t=1}^{h} \frac{(\mathrm{ROE}_t - \mathrm{CE}) \cdot \mathrm{BVE}_t}{(1+\mathrm{CE})^t} + \frac{(\mathrm{ROE}_\infty - \mathrm{CE}) \cdot \mathrm{BVE}_h}{(\mathrm{CE} - \mathrm{TGR}) \cdot (1+\mathrm{CE})^h}$$

where ROE mean-reverts toward the cost of equity at rate $\lambda$:

$$\mathrm{ROE}_{t+1} = \mathrm{ROE}_t + \lambda \cdot (\mathrm{CE} - \mathrm{ROE}_t)$$

BVE: book value of equity

$\mathrm{ROE}_t$: return on equity in year $t$

$\mathrm{ROE}_\infty$: terminal ROE (cost of equity plus sustainable spread)

$\lambda$: mean-reversion speed (typically $\approx 0.15$)

The model's implied P/BV follows a Gordon-growth-style identity:

$$\frac{P}{B} = \frac{\mathrm{ROE} - g}{\mathrm{CE} - g}$$

When ROE = CE, the stock should trade at book value. The sensitivity analysis sweeps ROE (5–25%), cost of equity (6–12%), and sustainable growth (0–6%).

#### 2.1.7.2 Insurance Valuation (Float-Based Model)

Insurers collect premiums before paying claims, creating a pool of investable capital called *float*. If underwriting is profitable (combined ratio < 100%), this float is obtained at negative cost – policyholders effectively pay the insurer to hold their money.

$$\mathrm{FV}_{\mathrm{ins}} = \mathrm{BVE} + \mathrm{UW}_{\mathrm{PV}} + \mathrm{Float}_{\mathrm{PV}}$$

where the present value of underwriting profits is:

$$\mathrm{UW}_{\mathrm{PV}} = \sum_{t=1}^{h} \frac{P_t \cdot (1 - \mathrm{CR}_t)}{(1 + \mathrm{CE})^t} + \frac{P_h \cdot (1 - \mathrm{CR}_\infty)}{(\mathrm{CE} - \mathrm{TGR}) \cdot (1 + \mathrm{CE})^h}$$

and the float value is a perpetuity of net investment income:

$$\mathrm{Float}_{\mathrm{PV}} = \frac{F \cdot (\mathrm{IY} - \mathrm{CoF})}{\mathrm{CE} - \mathrm{TGR}}$$

$P_t$: premiums earned in year $t$

$\mathrm{CR}_t$: combined ratio (loss ratio + expense ratio)

$F$: float (unearned premiums + loss reserves)

IY: investment yield on float

CoF: cost of float ($= \mathrm{CR} - 1$ if $\mathrm{CR} > 1$; 0 otherwise)

The combined ratio mean-reverts toward the industry average ($\approx 98\%$). The sensitivity analysis sweeps combined ratio (85–110%), investment yield (1–6%), and cost of equity (6–12%).

#### 2.1.7.3 Oil & Gas E&P Valuation (NAV Model)

Exploration and production companies are valued on the net asset value of their proven reserves, projected using a decline-curve model:

$$\mathrm{FV}_{\mathrm{og}} = \frac{1}{2}\left(\mathrm{NAV} + \mathrm{PV}_{10}\right)$$

where

$$\mathrm{NAV} = \sum_{t=1}^{T_{\mathrm{res}}} \frac{Q_t \cdot (\mathrm{NB}_t) \cdot (1 - \tau)}{(1 + r)^t} - D$$

$Q_t = Q_0 \cdot (1 - d)^t$: production in year $t$ with decline rate $d$

$\mathrm{NB} = p_{\mathrm{oil}} \cdot \alpha + p_{\mathrm{gas}} \cdot 6 \cdot (1 - \alpha) - c_{\mathrm{lift}}$: netback per BOE

$\alpha$: oil production fraction

$c_{\mathrm{lift}}$: lifting cost per BOE

$T_{\text{res}}$: reserve life (capped at 30 years)

$\text{PV}_{10}$: SEC-standard valuation at $r = 10\%$

$D$: net debt

The blended fair value averages NAV (company-specific discount rate) and PV-10 (standard 10% rate). The sensitivity analysis sweeps oil price ($40–$120/bbl), lifting cost ($5–$25/BOE), and discount rate (6–14%).

### 2.1.8 Sensitivity Analysis

Each valuation model produces a four-panel sensitivity plot: three line charts showing fair value as a function of one swept parameter (with the market price as a reference line and the implied parameter value marked), and a tornado diagram ranking parameters by their impact on fair value.

| Model | Parameter Axes | Output |
|---|---|---|
| Standard DCF | Growth rate, Discount rate, Terminal growth | FCFE & FCFF IVPS |
| Bank | ROE, Cost of equity, Sustainable growth | Fair value |
| Insurance | Combined ratio, Investment yield, Cost of equity | Fair value |
| Oil & Gas | Oil price, Lifting cost, Discount rate | Blended NAV |

The implied parameter value at which fair value equals the market price reveals what the market is pricing in. For banks, this is the implied ROE; for insurers, the implied combined ratio; for O&G companies, the implied oil price. Deviation between the actual and implied parameter indicates the degree to which the market under- or over-prices the asset.

## 2.2 DCF (Probabilistic)

### 2.2.1 Motivation

The deterministic DCF model relies on point estimates for key inputs (growth rates, discount rates, tax rates). However, these parameters are inherently uncertain. Probabilistic DCF addresses this by:

1. Treating financial statement line items as random variables
2. Propagating uncertainty through the cash flow projection
3. Generating a distribution of intrinsic values rather than a single point estimate
4. Quantifying valuation confidence via probability metrics

### 2.2.2 Input Sampling

#### 2.2.2.1 Historical Distributions

The following variables are sampled independently in each simulation iteration $i = 1, \ldots, N$ (typically $N = 5000$):

- Net income: $\mathrm{NI}^{(i)}$
- Earnings before interest and taxes: $\mathrm{EBIT}^{(i)}$
- Capital expenditures: $\mathrm{CAPX}^{(i)}$
- Depreciation & amortization: $\mathrm{D}^{(i)}$
- Current assets: $\mathrm{CA}^{(i)}$
- Current liabilities: $\mathrm{CL}^{(i)}$

For each variable $X$, we compute empirical mean $\mu_X$ and standard deviation $\sigma_X$ from the historical time series (typically 4–8 quarters).

#### 2.2.2.2 Sampling Distributions

We choose the sampling distribution based on the sign and scale of the variable:

- If $\mu_X > 0$, the variable is sampled from a log-normal distribution to ensure positivity:

$$\log\text{-mean} = \log(\max(\mu_X, 10^{-3})), \quad \log\text{-std} = \frac{\sigma_X}{\mu_X}$$

$$X^{(i)} \sim \mathrm{LogNormal}(\log\text{-mean}, \log\text{-std})$$

- If $\mu_X \leq 0$, the variable falls back to a normal distribution:

$$X^{(i)} \sim \mathcal{N}(\mu_X, \sigma_X^2)$$

No additional prior smoothing is applied to these inputs. Their randomness enters directly into the cash flow projection models.

#### 2.2.2.3 Outlier Squashing

To reduce the impact of extreme outliers, selected variables like NI and EBIT are post-processed using a squashing function:

$$x \mapsto \begin{cases} x, & x \leq \tau \\ \tau + \log(1 + (x - \tau)), & x > \tau \end{cases}$$

where the threshold $\tau$ is:

$$\tau = \min\{3 \cdot \max(\text{historical values}),\ 0.3 \cdot \text{MarketCap}\}$$

This logarithmic damping prevents unrealistic extrapolation while preserving the shape of the distribution.

### 2.2.3 Growth Rate Sampling

#### 2.2.3.1 Fundamental Growth Identity

Two growth rates are sampled in each simulation:

- FCFE growth rate: $g_{\text{FCFE}}$
- FCFF growth rate: $g_{\text{FCFF}}$

Both follow the fundamental identity:

$$g = \text{return} \times \text{reinvestment rate}$$

For FCFE:

$$\begin{aligned} g_{\text{FCFE}} &= \text{ROE} \times (1 - \text{payout ratio}) \\ &= \frac{\text{NI}}{\text{BVE}} \times \left(1 - \frac{\text{DP}}{\text{NI}}\right) \end{aligned}$$

For FCFF:

$$\begin{aligned} g_{\text{FCFF}} &= \text{ROIC} \times \text{reinvestment rate} \\ &= \frac{\text{NOPAT}}{\text{IC}} \times \frac{\text{CAPX} - \text{D} + \Delta\text{WC}}{\text{NOPAT}} \end{aligned}$$

where $\text{NOPAT} = 0.75 \cdot \text{EBIT}$ (approximating $(1 - \text{CTR}) \cdot \text{EBIT}$ with a fixed 25% tax rate).

#### 2.2.3.2 Bayesian Smoothing with Sector Priors

Each return and reinvestment term is blended as:

$$\text{final} = 0.5 \cdot \text{empirical} + 0.5 \cdot \text{prior}$$

Sector-specific priors are sampled from:

- ROE and ROIC: $\text{Beta}(2, 5)$ scaled to $[0, 0.4]$
- Retention and reinvestment rates: $\text{Beta}(2, 2)$ scaled to $[0, 1]$

This regularization prevents overfitting to recent historical data while incorporating industry-level information.

#### 2.2.3.3 Growth Rate Distribution

The blended product gives a per-year estimate. Growth rates are then sampled from a normal distribution using the historical mean and standard deviation:

$$g^{(i)} \sim \mathcal{N}(\mu_g, \sigma_g^2)$$

and clipped to user-defined bounds:

$$g^{(i)} \in [g_{\min}, g_{\max}]$$

Typical bounds are $g_{\min} = -0.05$ (allowing slight contraction) and $g_{\max} = 0.15$ (capping growth at 15%).

This ensures simulated growth is consistent with both financial fundamentals and prior regularization.

### 2.2.4 Stochastic Discount Rates

In addition to stochastic growth rates, the discount rate components (risk-free rate, beta, and equity risk premium) may also be sampled to capture uncertainty in the cost of capital.

#### 2.2.4.1 Discount Rate Component Sampling

For each simulation $i$, the following components are sampled from normal distributions:

- Risk-free rate:
$$\mathrm{RFR}^{(i)} \sim \mathcal{N}(\mathrm{RFR}_{\mathrm{base}}, \sigma^2_{\mathrm{RFR}})$$
with typical volatility $\sigma_{\mathrm{RFR}} = 0.005$ (50 basis points).
- Leveraged beta:
$$\beta_{\mathrm{L}}^{(i)} \sim \mathcal{N}(\beta_{\mathrm{L,base}}, \sigma^2_{\beta})$$
with typical volatility $\sigma_{\beta} = 0.10$.
- Equity risk premium:
$$\mathrm{ERP}^{(i)} \sim \mathcal{N}(\mathrm{ERP}_{\mathrm{base}}, \sigma^2_{\mathrm{ERP}})$$
with typical volatility $\sigma_{\mathrm{ERP}} = 0.01$ (100 basis points).

#### 2.2.4.2 Cost of Capital Recomputation

Using the sampled components, the cost of equity for simulation $i$ is:

$$\mathrm{CE}^{(i)} = \mathrm{RFR}^{(i)} + \beta_{\mathrm{L}}^{(i)} \cdot \mathrm{ERP}^{(i)}$$

Similarly, the weighted average cost of capital is recomputed:

$$\mathrm{WACC}^{(i)} = \frac{\mathrm{MVE}}{\mathrm{MVE} + \mathrm{MVB}} \cdot \mathrm{CE}^{(i)} + \frac{\mathrm{MVB}}{\mathrm{MVE} + \mathrm{MVB}} \cdot \mathrm{CB} \cdot (1 - \mathrm{CTR})$$

This stochastic treatment of discount rates significantly increases the dispersion of the intrinsic value distribution, providing a more realistic quantification of valuation uncertainty. The feature is controlled by the `use_stochastic_discount_rates` configuration flag.

### 2.2.5 Cash Flow Projection

#### 2.2.5.1 Explicit Forecast Period

For each simulation $i$, cash flows are projected over $h$ years (typically $h = 5$) using the sampled inputs and growth rate.

#### 2.2.5.2 Time-Varying Growth Rates

Rather than applying a constant growth rate throughout the projection period, the model implements exponential mean reversion toward the terminal growth rate. This prevents unrealistic perpetual high-growth assumptions.

For each simulation $i$ and forecast year $t \in \{1, \ldots, h\}$, the growth rate decays according to:

$$g_t^{(i)} = g_{\mathrm{terminal}} + \left(g_0^{(i)} - g_{\mathrm{terminal}}\right) \cdot e^{-\lambda t}$$

where:

- $g_0^{(i)}$ is the initial sampled growth rate (FCFE or FCFF growth from the fundamental identity)
- $g_{\mathrm{terminal}}$ is the long-run terminal growth rate (typically 2–3%)
- $\lambda \in [0, 1]$ is the mean reversion speed parameter (typical value: 0.3)
- $t$ is the year index within the projection period

**Interpretation of $\lambda$:**

- $\lambda = 0$: No reversion, constant growth $g_t^{(i)} = g_0^{(i)}$ for all $t$
- $\lambda = 0.3$: Moderate reversion (recommended), growth decays smoothly over 5–7 years

- $\lambda = 1.0$: Fast reversion, growth approaches terminal rate within 3–4 years

This time-varying formulation yields more conservative valuations than constant-growth models, as high initial growth rates gradually moderate to sustainable long-term levels. The feature is controlled by the `use_time_varying_growth` configuration flag.

#### 2.2.5.3 Cash Flow Computation

Using the time-varying growth rates $g_t^{(i)}$, the projected cash flows are:

For FCFE:

$$\mathrm{FCFE}_t^{(i)} = \mathrm{NI}_t^{(i)} + \mathrm{D}_t^{(i)} - \mathrm{CAPX}_t^{(i)} - \Delta\mathrm{WC}_t^{(i)} + \mathrm{NetBorrowing}_t^{(i)}$$

where $\mathrm{NetBorrowing}_t^{(i)} = \mathrm{TDR} \cdot \mathrm{Reinvestment}_t^{(i)}$

For FCFF:

$$\mathrm{FCFF}_t^{(i)} = \mathrm{NOPAT}_t^{(i)} + \mathrm{D}_t^{(i)} - \mathrm{CAPX}_t^{(i)} - \Delta\mathrm{WC}_t^{(i)}$$

with $\mathrm{NOPAT}_t^{(i)} = (1 - \mathrm{CTR}) \cdot \mathrm{EBIT}_t^{(i)}$

#### 2.2.5.4 Terminal Value

The terminal value uses a Gordon growth model:

$$\mathrm{TV}^{(i)} = \frac{\mathrm{CashFlow}_h^{(i)} \cdot (1 + g_{\mathrm{terminal}})}{r - g_{\mathrm{terminal}}}$$

where:

- $r = \mathrm{CE}$ for FCFE or $r = \mathrm{WACC}$ for FCFF
- $g_{\mathrm{terminal}}$ is the terminal growth rate (typically 2–3%)

To avoid division by zero or unstable extrapolation, the denominator is lower-bounded:

$$r - g_{\mathrm{terminal}} \geq 0.01$$

The terminal value is then discounted to present:

$$\mathrm{PV}_{\mathrm{terminal}}^{(i)} = \frac{\mathrm{TV}^{(i)}}{(1 + r)^h}$$

#### 2.2.5.5 Present Value

The present value of cash flows is:

$$\mathrm{PV}_{\mathrm{years}}^{(i)} = \sum_{t=1}^{h} \frac{\mathrm{CashFlow}_t^{(i)}}{(1 + r)^t}$$

Final present value of the firm:

$$\mathrm{PV}^{(i)} = \mathrm{PV}_{\mathrm{years}}^{(i)} + \mathrm{PV}_{\mathrm{terminal}}^{(i)}$$

For FCFF, market value of debt per share is subtracted to get the per-share equity value:

$$\mathrm{IVPS}_{\mathrm{FCFF}}^{(i)} = \frac{\mathrm{PV}_{\mathrm{FCFF}}^{(i)} - \mathrm{MVB} + \mathrm{CNOA}}{\mathrm{SO}}$$

### 2.2.6 Statistical Analysis

#### 2.2.6.1 Simulation Matrix

After $N$ simulations, we obtain an intrinsic value distribution for each asset and method (FCFE, FCFF):

$$\{V^{(1)}, V^{(2)}, \ldots, V^{(N)}\}$$

#### 2.2.6.2 Kernel Density Estimation

To estimate the probability density of intrinsic value, we apply a Gaussian kernel density estimator (KDE):

$$\hat{f}(v) = \frac{1}{Nh} \sum_{i=1}^{N} K\left(\frac{v - V^{(i)}}{h}\right)$$

where $K(\cdot)$ is the Gaussian kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$$

and $h$ is the bandwidth selected using Scott's rule:

$$h = N^{-1/5} \cdot \hat{\sigma}$$

where $\hat{\sigma}$ is the sample standard deviation of $\{V^{(i)}\}$.

#### 2.2.6.3 Value Surplus View

Each simulation sample is transformed into a percentage surplus over market price:

$$\text{Surplus}^{(i)} = \frac{V^{(i)} - P}{P}$$

We then apply KDE on the transformed values to estimate the distribution of relative mispricing. This highlights the extent to which the simulated valuations exceed or fall short of the current price.

From the KDE and simulation samples, we extract:

- $\mathbb{P}[\text{Surplus} > 0]$: probability the asset is undervalued
- $\mathbb{P}[\text{Surplus} < 0]$: probability it is overvalued

These quantities are calculated empirically via:

$$\frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[V^{(i)} > P] \quad \text{and} \quad \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[V^{(i)} < P]$$

### 2.2.7 Multi-Asset Portfolio Frontiers

To explore portfolio-level valuation outcomes, we simulate many random portfolios over the set of assets.

#### 2.2.7.1 Simulation Matrix

Let $n$ be the number of assets and $S_i^{(j)}$ the $j$-th simulation of value surplus (as a percentage of market price) for asset $i$. The simulation matrix is:

$$\mathbf{S} \in \mathbb{R}^{N \times n}, \quad \text{with } S_{ji} = \frac{V_i^{(j)} - P_i}{P_i}$$

#### 2.2.7.2 Random Portfolio Generation

We generate $M$ random portfolio weight vectors $\mathbf{w}^{(k)} \in \mathbb{R}^n$ (typically $M = 5000$) using a Dirichlet distribution over the simplex:

$$\mathbf{w}^{(k)} \sim \text{Dir}(\alpha \mathbf{1}), \quad \alpha = 1$$

This ensures:

$$\sum_{i=1}^{n} w_i^{(k)} = 1, \quad w_i^{(k)} \geq 0 \quad \forall i$$

and provides uniform sampling over the weight simplex.

#### 2.2.7.3 Portfolio Return Distribution

For each portfolio $k$ and simulation $j$, the portfolio surplus is:

$$R_j^{(k)} = \sum_{i=1}^{n} w_i^{(k)} S_{ji} = (\mathbf{w}^{(k)})^\top \mathbf{S}_j$$

where $\mathbf{S}_j$ is the $j$-th row of the simulation matrix.

#### 2.2.7.4 Portfolio Statistics

For each portfolio $k$:

- **Expected return** (mean surplus):

$$\mu^{(k)} = \frac{1}{N} \sum_{j=1}^{N} R_j^{(k)}$$

- **Standard deviation** (valuation risk):

$$\sigma^{(k)} = \sqrt{\frac{1}{N-1} \sum_{j=1}^{N} (R_j^{(k)} - \mu^{(k)})^2}$$

  Alternatively, using the covariance matrix:

$$\sigma^{(k)} = \sqrt{(\mathbf{w}^{(k)})^\top \mathbf{\Sigma} \mathbf{w}^{(k)}}$$

  where $\mathbf{\Sigma}$ is the empirical covariance of $\mathbf{S}$.

- **Probability of loss**:

$$p_{\text{loss}}^{(k)} = \frac{1}{N} \sum_{j=1}^{N} \mathbb{I}[R_j^{(k)} < 0]$$

- **Downside deviation** (Sortino denominator):

$$\sigma_{\text{down}}^{(k)} = \sqrt{\frac{1}{N} \sum_{j=1}^{N} \min(0, R_j^{(k)})^2}$$

- **Conditional Value-at-Risk** (CVaR at 95%): Let $\alpha = 0.05$. Sort the portfolio returns: $R_{(1)}^{(k)} \leq R_{(2)}^{(k)} \leq \cdots \leq R_{(N)}^{(k)}$.

  The VaR is:

$$\text{VaR}_{95}^{(k)} = R_{(\lfloor \alpha N \rfloor)}^{(k)}$$

  The CVaR is:

$$\text{CVaR}_{95}^{(k)} = \frac{1}{\lfloor \alpha N \rfloor} \sum_{j=1}^{\lfloor \alpha N \rfloor} R_{(j)}^{(k)}$$

- **Maximum drawdown**: Compute cumulative returns over the simulation series and measure the largest peak-to-trough decline.

#### 2.2.7.5 Efficient Frontier Plots

Six distinct frontier views are produced, each optimizing a different risk-return trade-off:

1. **Mean vs Standard Deviation** ($\mu$–$\sigma$): Mean-variance analysis. Identifies:
   - Minimum variance portfolio: $\mathbf{w}_{\text{minvar}} = \arg\min_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$
   - Maximum Sharpe portfolio: $\mathbf{w}_{\text{maxSharpe}} = \arg\max_{\mathbf{w}} \frac{\mu(\mathbf{w})}{\sigma(\mathbf{w})}$
2. **Mean vs Probability of Loss** ($\mu$–$p_{\text{loss}}$): Focuses on minimizing the chance of overvaluation. Identifies the portfolio with minimum $p_{\text{loss}}$ for a given expected return.
3. **Mean vs Downside Deviation** ($\mu$–$\sigma_{\text{down}}$): Sortino-style frontier. Identifies the portfolio with maximum Sortino ratio:
$$\text{Sortino}(\mathbf{w}) = \frac{\mu(\mathbf{w})}{\sigma_{\text{down}}(\mathbf{w})}$$
4. **Mean vs CVaR** ($\mu$–CVaR): Tail risk frontier. Identifies the portfolio minimizing expected tail loss.
5. **Mean vs VaR** ($\mu$–VaR): Similar to CVaR but focuses on the quantile rather than the conditional expectation.
6. **Mean vs Maximum Drawdown** ($\mu$–MDD): Calmar-style frontier. Identifies the portfolio with maximum Calmar ratio:
$$\text{Calmar}(\mathbf{w}) = \frac{\mu(\mathbf{w})}{\text{MDD}(\mathbf{w})}$$

In each plot, the efficient frontier curve and key labeled portfolios (e.g., maximum return, minimum risk, optimal ratio) are annotated. Portfolio composition legends show the top 15 holdings for each optimal portfolio.

### 2.2.8 Interpretation

The probabilistic DCF framework yields a summary of valuation statistics and qualitative signals:

- **Strong buy**: if both models (FCFE and FCFF) yield undervaluation ($\mathbb{P}[\text{Surplus} > 0] > 0.6$).
- **Buy**: if only one model suggests undervaluation.
- **Avoid**: if both suggest overvaluation ($\mathbb{P}[\text{Surplus} < 0] > 0.6$).
- **Hold**: if both suggest fair valuation.

The probabilistic approach provides several advantages over deterministic DCF:

1. **Uncertainty quantification**: Rather than a single IVPS point estimate, we obtain a full distribution $\hat{f}(v)$, allowing probabilistic statements about valuation.
2. **Confidence calibration**: The probability of undervaluation/overvaluation provides a confidence measure for investment decisions.
3. **Portfolio-level analysis**: By combining simulation matrices across assets, we can construct efficient portfolios based on intrinsic value rather than historical prices.
4. **Regime-invariant correlations**: The low inter-asset correlations ( 0.0) observed in DCF-based returns reflect fundamental independence, unlike market-based returns which exhibit regime-dependent correlation structures.
5. **Tail risk visibility**: The CVaR and maximum drawdown frontiers explicitly address worst-case scenarios, aligning with investor risk preferences during stress.

The framework enables the construction of valuation-based portfolios that systematically exploit mispricing while managing fundamental uncertainty.

## 2.3 GARP / PEG Analysis

Growth at a Reasonable Price (GARP) bridges value and growth investing by seeking companies with solid growth prospects trading at reasonable valuations. The PEG ratio is the primary metric.

### 2.3.1 PEG Ratio

The Price/Earnings-to-Growth ratio normalizes valuation by growth rate:

$$\text{PEG} = \frac{\text{P/E}}{\text{Growth Rate} \times 100}$$

where Growth Rate is expressed as a decimal (e.g., 15% = 0.15). A PEG of 1.0 implies fair value; below 1.0 suggests undervaluation relative to growth.

#### 2.3.1.1 Growth Rate Selection

The model selects growth rate with the following priority:

1. 1-year EPS growth (most recent)
2. Revenue growth (fallback)
3. Earnings growth (secondary fallback)
4. 5-year analyst estimate (last resort)

All growth rates are capped at 50% to prevent distortion from outliers.

#### 2.3.1.2 PEGY Ratio

For dividend-paying stocks, the dividend-adjusted PEG accounts for yield:

$$\text{PEGY} = \frac{\text{P/E}}{\text{Growth Rate}(\%) + \text{Dividend Yield}(\%)}$$

#### 2.3.1.3 Fair Value Estimation

Assuming a fair PEG of 1.0:

$$\text{Fair P/E} = \text{Growth Rate} \times 100$$

$$\text{Fair Price} = \text{Fair P/E} \times \text{EPS}$$

### 2.3.2 GARP Scoring

The composite GARP score (0–100) aggregates five components:

| Component | Weight | Criteria |
|---|---|---|
| PEG Score | 30 | PEG $< 0.5$: 30; $< 1.0$: 25; $< 1.5$: 15; $< 2.0$: 5 |
| Growth Score | 25 | $> 25\%$: 25; $> 15\%$: 20; $> 10\%$: 15; $> 5\%$: 10 |
| FCF Conversion | 20 | FCF/NI $> 1.0$: 20; $> 0.8$: 15; $> 0.5$: 10 |
| Balance Sheet | 15 | D/E $< 0.3$: 15; $< 0.5$: 10; $< 1.0$: 5 |
| ROE Score | 10 | $> 20\%$: 10; $> 15\%$: 7; $> 10\%$: 5 |

### 2.3.3 PEG Assessment

PEG $< 0.5$: Very Undervalued

PEG 0.5–1.0: Undervalued

PEG 1.0–1.5: Fairly Valued

PEG 1.5–2.0: Moderately Expensive

PEG $> 2.0$: Expensive

## 2.4 Growth Analysis

Growth analysis evaluates companies from a pure growth investing perspective, focusing on revenue acceleration, margin trajectory, and capital efficiency.

### 2.4.1 Rule of 40

The Rule of 40 is a benchmark for growth companies, particularly in technology and SaaS:

$$\text{Rule of 40} = \text{Revenue Growth}(\%) + \text{Operating Margin}(\%)$$

Companies exceeding 40 demonstrate efficient growth—either growing rapidly with acceptable margins or growing moderately with strong profitability.

$\geq 40$: Excellent

30–40: Good

20–30: Moderate

$< 20$: Concerning

### 2.4.2 Operating Leverage

Operating leverage measures how earnings growth amplifies revenue growth:

$$\text{Operating Leverage} = \frac{\text{Earnings Growth}}{\text{Revenue Growth}}$$

Leverage $> 1$ indicates positive operating leverage (margins expanding with scale).

### 2.4.3 EV/Revenue Efficiency

Normalized valuation per unit of growth:

$$\text{EV/Rev per Growth} = \frac{\text{EV/Revenue}}{\text{Revenue Growth}(\%)}$$

Lower values indicate more attractive valuations relative to growth rate.

### 2.4.4 Implied Growth Rate

The market-implied growth rate from current valuation:

$$g_{\text{implied}} = r - \frac{1}{\text{Forward P/E}}$$

where $r$ is the required return (typically 10%). This reveals what growth rate the market is pricing in.

### 2.4.5 Growth Classification

$> 40\%$: Hypergrowth

20–40%: High Growth

10–20%: Moderate Growth

5–10%: Slow Growth

$< 5\%$: No Growth / Declining

### 2.4.6 Growth Scoring

The composite growth score (0–100) weights five factors:

| Component | Weight | Key Thresholds |
|---|---|---|
| Revenue Growth | 25 | $> 40\%$: 25; $> 20\%$: 18; $> 10\%$: 10 |
| Earnings Growth | 20 | $> 50\%$: 20; $> 30\%$: 16; $> 20\%$: 12 |
| Margin Quality | 20 | Gross $> 60\%$, Op $> 25\%$, FCF $> 20\%$ |
| Efficiency | 20 | Rule of 40, Operating Leverage |
| Quality (ROE/ROIC) | 15 | ROE $> 25\%$, ROIC $> 20\%$ |

## 2.5 Relative Valuation

Relative valuation compares a target company against peer companies using standardized multiples, deriving fair value from comparable company analysis.

### 2.5.1 Peer Similarity Scoring

Peers are weighted by similarity across four dimensions (0–100 total):

| Dimension | Weight | Scoring |
|---|---|---|
| Industry | 30 | Same industry: 30; Same sector: 15 |
| Size | 25 | Market cap ratio 0.5–2.0×: 25 |
| Growth | 25 | Growth differential $< 5\%$: 25 |
| Profitability | 20 | EBITDA margin diff $< 5$pp: 20 |

### 2.5.2 Multiple Analysis

For each valuation multiple $M$, we compute:

#### 2.5.2.1 Premium/Discount

$$\text{Premium}(\%) = \frac{M_{\text{target}} - M_{\text{peer median}}}{M_{\text{peer median}}} \times 100$$

#### 2.5.2.2 Implied Valuation

For equity multiples (P/E, P/B, P/S, P/FCF):

$$\text{Implied Price} = \text{Metric} \times M_{\text{peer median}}$$

For enterprise value multiples (EV/EBITDA, EV/Revenue):

$$\text{Implied EV} = \text{Metric} \times M_{\text{peer median}}$$

$$\text{Implied Price} = \frac{\text{Implied EV} - \text{Debt} + \text{Cash}}{\text{Shares}}$$

### 2.5.3 Fair Premium Adjustment

Companies with superior fundamentals deserve a premium:

$$\text{Fair Premium} = \text{Growth Premium} + \text{Margin Premium}$$

where:

Growth Premium: Each 5pp growth advantage $\rightarrow$ 10% premium

Margin Premium: Each 5pp margin advantage $\rightarrow$ 5% premium

Capped at: $-30\%$ to $+50\%$

### 2.5.4 Relative Score

The relative score (0–100) measures undervaluation:

$$\text{Score} = 50 - \text{Average Premium}(\%)$$

Higher scores indicate greater undervaluation relative to peers.

### 2.5.5 Multiples Analyzed

- **Equity multiples**: P/E (trailing, forward), P/B, P/S, P/FCF
- **EV multiples**: EV/EBITDA, EV/EBIT, EV/Revenue

## 2.6 Normalized Multiples

Normalized multiples analysis compares a company's valuation multiples against sector benchmarks using percentile-based scoring with quality adjustments. Unlike relative valuation, which operates on a small peer group, this framework benchmarks against the full sector distribution.

### 2.6.1 Multiple Extraction

Ten valuation multiples are computed:

#### 2.6.1.1 Price Multiples

- P/E (TTM, NTM): Price / Earnings per share
- P/S (TTM): Price / Revenue per share
- P/B (TTM): Price / Book value per share
- P/FCF (TTM): Price / Free cash flow per share
- PEG: P/E / Earnings growth rate

#### 2.6.1.2 Enterprise Value Multiples

- EV/EBITDA (TTM): Enterprise value / EBITDA
- EV/EBIT (TTM): Enterprise value / EBIT
- EV/Sales (TTM): Enterprise value / Revenue
- EV/FCF (TTM): Enterprise value / Free cash flow

A multiple is considered valid when both the numerator and denominator are positive.

### 2.6.2 Percentile Ranking

Each multiple is ranked against the sector distribution using piecewise linear interpolation through three known quantiles ($Q_{25}$, $Q_{50}$, $Q_{75}$):

$$\mathrm{Pct}(x) = \begin{cases} 25 \cdot \dfrac{x}{Q_{25}} & \text{if } x \leq Q_{25} \\ 25 + 25 \cdot \dfrac{x - Q_{25}}{Q_{50} - Q_{25}} & \text{if } Q_{25} < x \leq Q_{50} \\ 50 + 25 \cdot \dfrac{x - Q_{50}}{Q_{75} - Q_{50}} & \text{if } Q_{50} < x \leq Q_{75} \\ \min\left(100,\ 75 + 25 \cdot \dfrac{x - Q_{75}}{Q_{75} - Q_{50}}\right) & \text{if } x > Q_{75} \end{cases}$$

A percentile of 50 means the company trades at the sector median; lower percentiles indicate cheaper valuation.

### 2.6.3 Premium/Discount and Implied Price

#### 2.6.3.1 Premium/Discount

$$\text{Premium}(\%) = \frac{M_{\text{company}} - M_{\text{sector median}}}{M_{\text{sector median}}} \times 100$$

#### 2.6.3.2 Implied Price

For equity multiples (P/E, P/B, P/S, P/FCF):

$$\text{Implied Price} = \text{Underlying Metric Per Share} \times M_{\text{sector median}}$$

For enterprise value multiples (EV/EBITDA, EV/EBIT, EV/Sales, EV/FCF):

$$\text{Implied Price} = \frac{(\text{Underlying Metric} \times M_{\text{sector median}}) - \text{Net Debt}}{\text{Shares Outstanding}} \times \frac{\text{Current Price}}{\text{Market Cap / Shares Outstanding}}$$

### 2.6.4 Quality Adjustment

Companies with superior fundamentals deserve a premium; quality adjustment shifts the composite percentile accordingly:

#### 2.6.4.1 Growth Premium

$$\text{Growth Premium}(\%) = \frac{g_{\text{company}} - g_{\text{sector median}}}{g_{\text{sector median}}} \times 10$$

where $g$ is revenue growth (TTM). Clamped to $[-20\%, +20\%]$.

#### 2.6.4.2 Margin Premium

$$\text{Margin Premium}(\%) = \frac{m_{\text{company}} - m_{\text{sector median}}}{m_{\text{sector median}}} \times 10$$

where $m$ is the EBITDA margin. Clamped to $[-20\%, +20\%]$.

#### 2.6.4.3 Return Premium

$$\text{Return Premium}(\%) = \frac{\text{ROE}_{\text{company}} - \text{ROE}_{\text{sector}}}{\text{ROE}_{\text{sector}}} \times 5$$

Clamped to $[-20\%, +20\%]$.

#### 2.6.4.4 Total Fair Premium

$$\text{Fair Premium}(\%) = \text{Growth} + \text{Margin} + \text{Return}$$

### 2.6.5 Composite Scoring

#### 2.6.5.1 Raw Composite Percentile

The arithmetic mean of all valid multiples' percentile ranks:

$$\overline{\text{Pct}} = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \text{Pct}(m)$$

where $\mathcal{M}$ is the set of valid multiples.

#### 2.6.5.2 Quality-Adjusted Percentile

$$\overline{\text{Pct}}_{\text{adj}} = \text{clamp}\big(\overline{\text{Pct}} - \text{Fair Premium}(\%),\ 0,\ 100\big)$$

A company deserving a 15% quality premium has its percentile reduced by 15 points, making it appear cheaper on a quality-adjusted basis.

#### 2.6.5.3 Confidence Score

Confidence (0–100) combines data coverage and agreement:

$$\text{Confidence} = \underbrace{\min(50,\ |\mathcal{M}| \times 5)}_{\text{Data score}} + \underbrace{\max(0,\ 50 - \sigma_{\text{Pct}})}_{\text{Agreement score}}$$

where $\sigma_{\text{Pct}}$ is the standard deviation of percentile ranks across multiples. Low dispersion means multiples agree, yielding higher confidence.

#### 2.6.5.4 Value Score

The value score inverts the percentile:

$$\text{Value Score} = 100 - \overline{\text{Pct}}_{\text{adj}}$$

Higher value scores indicate deeper undervaluation after quality adjustment.

### 2.6.6 Interpretation

1. **Sector context**: All comparisons are sector-relative; a "cheap" P/E in tech would be expensive in utilities. Sector benchmarks provide the appropriate reference frame.
2. **Quality matters**: A company at the 80th percentile of P/E might be fairly valued if its growth and margins are also at the 80th percentile; the quality adjustment captures this.
3. **Multiple agreement**: When P/E says cheap but EV/EBITDA says expensive, the confidence score drops, signaling caution.
4. **Cyclicality**: Trailing multiples are distorted near cycle peaks and troughs. PEG and forward P/E provide partial correction.

## 2.7 Dividend Income Analysis

Dividend income analysis evaluates stocks from an income investor's perspective, emphasizing yield sustainability, dividend growth, and payout safety.

### 2.7.1 Dividend Discount Models

#### 2.7.1.1 Gordon Growth Model (Constant Growth DDM)

For stocks with stable, perpetual dividend growth:

$$\text{Fair Value} = \frac{D_1}{r - g} = \frac{D_0(1+g)}{r - g}$$

where $D_0$ is the current dividend, $r$ is the required return, and $g$ is the dividend growth rate. Validity requires $r > g$.

#### 2.7.1.2 Two-Stage DDM

For companies transitioning from high growth to stable growth:

$$\text{Fair Value} = \sum_{t=1}^{n} \frac{D_t}{(1+r)^t} + \frac{D_{n+1}}{(r - g_{\text{terminal}})} \cdot \frac{1}{(1+r)^n}$$

#### 2.7.1.3 H-Model (Gradual Growth Decline)

$$\text{Fair Value} = \frac{D_0(1+g_L)}{r - g_L} + \frac{D_0 \cdot H \cdot (g_S - g_L)}{r - g_L}$$

where $H$ is the half-life of the high-growth period, $g_S$ is short-term growth, and $g_L$ is long-term sustainable growth.

### 2.7.2 Dividend Coverage

$$\text{EPS Coverage} = \frac{\text{EPS}}{\text{DPS}}$$

$$\text{FCF Coverage} = \frac{\text{FCF per Share}}{\text{DPS}}$$

Coverage $> 1.5\times$ is considered safe; $< 1.0\times$ indicates the dividend exceeds earnings.

### 2.7.3 Chowder Number

A metric for income-focused investors:

$$\text{Chowder} = \text{Dividend Yield}(\%) + \text{Avg Dividend Growth}(\%)$$

Values $>$ 12–14% are attractive for total return from dividends.

### 2.7.4 Dividend Aristocrat Classification

**Dividend King**: 50+ years of consecutive increases

**Dividend Aristocrat**: 25+ years

**Dividend Achiever**: 10–24 years

**Dividend Contender**: 5–9 years

**Dividend Challenger**: 1–4 years

### 2.7.5 Payout Ratio Assessment

$< 50\%$: Very Safe

50–60%: Safe

60–75%: Moderate

75–90%: Elevated

$> 90\%$: Unsustainable

$> 100\%$: Paying from reserves

### 2.7.6 Dividend Safety Scoring

The dividend safety score (0–100) aggregates:

| Component | Weight | Key Thresholds |
|---|---|---|
| Payout Ratio | 25 | $< 50\%$: 25; $< 60\%$: 20; $< 75\%$: 15 |
| Coverage | 25 | EPS $> 1.5\times$: 25; $> 1.3\times$: 18 |
| Streak | 25 | Based on consecutive years |
| Balance Sheet | 15 | D/E $< 0.5$: 15; $< 1.0$: 10 |
| Stability | 10 | ROE/ROA quality metrics |

## 2.8 REIT Valuation

Real Estate Investment Trusts require specialized valuation metrics due to significant non-cash depreciation charges and the asset-intensive nature of real estate.

### 2.8.1 Funds From Operations (FFO)

FFO is the REIT-standard earnings metric:

$$\text{FFO} = \text{Net Income} + \text{Depreciation} + \text{Amortization} + \text{Impairments} - \text{Gains on Sales}$$

FFO adds back non-cash real estate depreciation, which often exceeds economic depreciation for well-maintained properties.

### 2.8.2 Adjusted FFO (AFFO)

AFFO provides a more conservative view of recurring cash flow:

$$\text{AFFO} = \text{FFO} - \text{Maintenance CapEx} - \text{Straight-Line Rent Adj} - \text{Stock Comp}$$

AFFO better approximates true free cash flow available for distribution.

### 2.8.3 Net Asset Value (NAV)

NAV values the underlying real estate portfolio:

$$\text{Property Value} = \frac{\text{NOI}}{\text{Cap Rate}}$$

$$\text{NAV} = \text{Property Value} + \text{Cash} + \text{Other Assets} - \text{Total Debt}$$

$$\text{NAV per Share} = \frac{\text{NAV}}{\text{Shares Outstanding}}$$

#### 2.8.3.1 Sector Cap Rates

| Sector | Cap Rate | Sector | Cap Rate |
|---|---|---|---|
| Industrial | 5.0% | Healthcare | 6.5% |
| Data Center | 5.5% | Office | 7.0% |
| Self Storage | 5.5% | Retail | 7.0% |
| Residential | 5.0% | Hotel | 8.0% |

### 2.8.4 Valuation Multiples

$$\text{P/FFO} = \frac{\text{Price}}{\text{FFO per Share}}$$

$$\text{P/AFFO} = \frac{\text{Price}}{\text{AFFO per Share}}$$

$$\text{Premium/Discount to NAV} = \frac{\text{Price} - \text{NAV per Share}}{\text{NAV per Share}}$$

### 2.8.5 REIT Payout Analysis

REITs must distribute $\geq 90\%$ of taxable income. The AFFO payout ratio indicates sustainability:

$$\text{AFFO Payout Ratio} = \frac{\text{Annual Dividend}}{\text{AFFO per Share}}$$

$< 75\%$: Very Safe (room for growth)

75–85%: Healthy

85–95%: Tight

$> 95\%$: At Risk

### 2.8.6 Mortgage REIT Metrics

Mortgage REITs (mREITs) invest in mortgages rather than properties:

$$\text{Net Interest Income} = \text{Interest Income} - \text{Interest Expense}$$

$$\text{Net Interest Margin} = \frac{\text{NII}}{\text{Average Earning Assets}}$$

$$\text{Price/Book} = \frac{\text{Price}}{\text{Book Value per Share}}$$

For mREITs, P/B near 1.0 is typical; significant discounts may indicate credit concerns.

## 2.9 ETF Analysis

ETF analysis evaluates exchange-traded funds on cost efficiency, tracking quality, liquidity, and structural characteristics.

### 2.9.1 Premium/Discount to NAV

$$\text{Premium/Discount} = \frac{\text{Price} - \text{NAV}}{\text{NAV}} \times 100$$

ETFs should trade near NAV; persistent premiums or discounts indicate inefficiencies or structural issues.

### 2.9.2 Tracking Error

Tracking error measures deviation from the benchmark:

$$\text{Tracking Error} = \sqrt{\text{Var}(R_{\text{ETF}} - R_{\text{Benchmark}})}$$

$< 0.10\%$: Excellent

0.10–0.25%: Good

0.25–0.50%: Acceptable

$> 0.50\%$: Poor

### 2.9.3 Total Cost of Ownership

Beyond expense ratio, total cost includes:

$$\text{TCO} = \text{Expense Ratio} + \text{Bid-Ask Spread} + \text{Tracking Drag}$$

### 2.9.4 ETF Quality Scoring

The ETF quality score (0–100) aggregates:

| Component | Weight | Key Thresholds |
|---|---|---|
| Cost | 25 | ER $< 0.05\%$: 25; $< 0.10\%$: 20; $< 0.20\%$: 15 |
| NAV Gap | 10 | Gap $< 0.05\%$: 10; $< 0.10\%$: 8; $< 0.25\%$: 6 |
| Tracking | 25 | TE $< 0.10\%$: 25; $< 0.25\%$: 20; $< 0.50\%$: 15 |
| Liquidity | 20 | Spread, volume, AUM |
| Size (AUM) | 20 | $>$ \$100B: 20; $>$ \$10B: 15; $>$ \$1B: 10 |

### 2.9.5 Derivatives ETF Analysis

Specialized metrics for options-based and leveraged ETFs:

#### 2.9.5.1 Capture Ratios

Capture ratios measure how much of the benchmark's return is captured in up and down markets:

$$\text{Upside Capture} = \frac{\sum_{t \in U} R_{\text{ETF},t}}{\sum_{t \in U} R_{\text{bench},t}} \times 100, \qquad \text{Downside Capture} = \frac{\sum_{t \in D} R_{\text{ETF},t}}{\sum_{t \in D} R_{\text{bench},t}} \times 100$$

where $U = \{t : R_{\text{bench},t} > 0\}$ and $D = \{t : R_{\text{bench},t} < 0\}$.

Capture efficiency isolates the asymmetry:

$$\text{Capture Efficiency} = \text{Upside Capture} - \text{Downside Capture}$$

Positive efficiency means the ETF captures more upside than downside—desirable for covered call strategies.

#### 2.9.5.2 Covered Call ETFs

Covered call ETFs (e.g., JEPI, QYLD) sacrifice upside for income. Key metrics:

- Distribution Yield: annualized income as % of NAV
- Yield vs Benchmark: $\text{Yield}_{\text{ETF}}/\text{Yield}_{\text{benchmark}}$
- Scoring sweet spot: 7–10% yield with upside capture $\geq 60\%$

#### 2.9.5.3 Volatility ETF Decay

Volatility ETFs that hold VIX futures suffer from contango roll yield:

$$\text{Roll Yield (monthly)} \approx \frac{R_{1\text{Y}}}{12}$$

where $R_{1\text{Y}}$ is the 1-year return. Persistent contango (front futures < back futures) causes compounding erosion. An ETF with $R_{1\text{Y}} < -30\%$ is flagged as severely decaying and unsuitable for buy-and-hold.

#### 2.9.5.4 Leveraged ETF Volatility Drag

Daily-rebalanced leveraged ETFs suffer from volatility drag:

$$R_{\text{leveraged}} \approx L \cdot R_{\text{index}} - \frac{L(L-1)}{2}\sigma^2$$

where $L$ is the leverage factor and $\sigma$ is realized volatility. Higher volatility accelerates decay; ETFs with $\sigma_{1\text{Y}} > 50\%$ score lowest.

#### 2.9.5.5 Buffer ETFs

Buffer (defined-outcome) ETFs use options collars to provide downside protection up to a buffer level in exchange for a capped upside:

- Buffer Level: starting point of downside protection
- Cap Level: maximum upside participation
- Remaining Buffer / Cap: current effective levels within the outcome period
- Days to Outcome Period Reset

Mid-period entry alters effective buffer and cap levels; best entered near period start.

## 2.10 Crypto Treasury Valuation

Crypto treasury valuation applies to companies holding significant cryptocurrency assets on their balance sheet, using a multiple of Net Asset Value (mNAV) framework.

### 2.10.1 Net Asset Value

$$\text{NAV} = (\text{BTC Holdings} \times P_{\text{BTC}}) + (\text{ETH Holdings} \times P_{\text{ETH}})$$

$$\text{NAV per Share} = \frac{\text{NAV}}{\text{Shares Outstanding}}$$

### 2.10.2 Multiple of NAV (mNAV)

The key valuation metric:

$$\text{mNAV} = \frac{\text{Market Cap}}{\text{NAV}}$$

mNAV $< 0.8$: Strong Buy (20%+ discount to holdings)

mNAV 0.8–1.0: Buy (discount)

mNAV 1.0–1.2: Hold (fair to slight premium)

mNAV 1.2–1.5: Caution (meaningful premium)

mNAV $> 1.5$: Overvalued (speculative premium)

### 2.10.3 Premium/Discount

$$\text{Premium}(\%) = (\text{mNAV} - 1) \times 100$$

Negative values indicate the stock trades at a discount to its crypto holdings.

### 2.10.4 Per-Share Exposure

$$\text{BTC per Share} = \frac{\text{BTC Holdings}}{\text{Shares Outstanding}}$$

This metric shows the crypto exposure obtained per share purchased.

### 2.10.5 Implied Crypto Prices

The market-implied crypto price embedded in the stock valuation:

$$\text{Implied BTC Price} = P_{\text{BTC}} \times \text{mNAV}$$

If mNAV $> 1$, the market implicitly values the company's BTC holdings above spot price.

### 2.10.6 Unrealized Gains

$$\text{Unrealized Gain} = (P_{\text{current}} - P_{\text{avg cost}}) \times \text{Holdings}$$

Significant unrealized gains may create tax overhang if liquidated.

## 2.11 NLP Sentiment Analysis

Natural language processing analyzes corporate communications to detect narrative shifts before they manifest in financial results.

### 2.11.1 Document Sources

- **SEC Filings**: MD&A (Item 7/Item 5), Risk Factors (Item 1A/Item 3D)
- **Earnings Calls**: Prepared remarks, Q&A transcripts
- **Comparison**: Current vs prior quarter, current vs historical centroid

### 2.11.2 Embedding-Based Change Detection

Documents are converted to dense vector embeddings using transformer models. Change is measured via cosine distance:

$$\Delta = 1 - \cos(\mathbf{e}_{\text{current}}, \mathbf{e}_{\text{prior}})$$

Higher $\Delta$ indicates greater semantic shift in language.

### 2.11.3 Historical Centroid

A weighted average of past embeddings with exponential decay:

$$\mathbf{c} = \frac{\sum_t w_t \cdot \mathbf{e}_t}{\sum_t w_t}, \quad w_t = e^{-\lambda(T-t)}$$

where $\lambda$ controls decay speed and $T$ is the current period. Recent quarters receive higher weight.

### 2.11.4 Hedging Language Detection

Count of uncertainty-indicating words and phrases:

- **Hedging words**: "may", "might", "could", "uncertain", "risk"
- **Evasive phrases**: "as I mentioned", "too early to say", "time will tell"

Increasing hedging language signals growing management uncertainty.

### 2.11.5 Commitment Strength Scoring

Management commitment is scored on a ladder (0–1):

| Word | Score | Word | Score |
|---|---|---|---|
| "will" | 1.0 | "intend" | 0.5 |
| "expect" | 0.8 | "target" | 0.4 |
| "anticipate" | 0.7 | "hope" | 0.3 |
| "believe" | 0.6 | "monitor" | 0.2 |

Declining commitment scores indicate weakening conviction.

### 2.11.6 Signal Ontology

Fixed categories for narrative signals (each with UP/DOWN direction):

| Category | Interpretation |
|---|---|
| Pricing Power | Ability to raise/maintain prices |
| Margin Pressure | Cost/competitive pressure on margins |
| Demand Visibility | Clarity on demand outlook |
| Competitive Threat | New or intensifying competition |
| Guidance Confidence | Management confidence in outlook |
| Management Tone | Overall tone shift |
| Regulatory Overhang | Regulatory risk exposure |
| Execution Risk | Ability to execute initiatives |

#### 2.11.7 Composite Scoring

Snippets are ranked by composite score:

$$\text{Score} = w_\Delta \cdot \Delta_{\text{embedding}} + w_h \cdot \Delta_{\text{hedging}} + w_c \cdot \Delta_{\text{commitment}}$$

where typical weights are $w_\Delta = 0.5$, $w_h = 0.25$, $w_c = 0.25$.

#### 2.11.8 Interpretation

- **High $\Delta$ + Decreasing Commitment**: Deteriorating narrative
- **Increasing Hedging**: Growing uncertainty/caution
- **Risk Factor Changes**: New risks emerging
- **Q&A Tone Shift**: Management becoming defensive

## 2.12 Tail Risk Forecast

Tail risk forecasting estimates Value-at-Risk (VaR) and Expected Shortfall (ES) using realized variance computed from high-frequency intraday returns.

### 2.12.1 Realized Variance

Daily realized variance is computed from intraday returns:

$$RV_t = \sum_{i=1}^{n} r_{t,i}^2$$

where $r_{t,i}$ are intraday log returns (typically 5-minute intervals). This provides a more accurate volatility measure than close-to-close returns by capturing intraday price movements.

### 2.12.2 HAR-RV Model

The Heterogeneous Autoregressive Realized Volatility (HAR-RV) model captures volatility persistence at multiple horizons:

$$RV_{t+1} = c + \beta_d \cdot RV_t + \beta_w \cdot RV_t^{(w)} + \beta_m \cdot RV_t^{(m)} + \varepsilon_t$$

where:

- $RV_t^{(w)} = \frac{1}{5}\sum_{i=0}^{4} RV_{t-i}$ is the weekly (5-day) average
- $RV_t^{(m)} = \frac{1}{22}\sum_{i=0}^{21} RV_{t-i}$ is the monthly (22-day) average

The HAR structure captures heterogeneous market participants operating at different frequencies (daily traders, weekly rebalancers, monthly allocators).

### 2.12.3 Jump Detection

Variance jumps are identified using a threshold approach:

$$J_t = \mathbf{1}\left\{RV_t > \mu_{RV} + k \cdot \sigma_{RV}\right\}$$

where $\mu_{RV}$ and $\sigma_{RV}$ are the rolling mean and standard deviation of realized variance, and $k = 2.5$ is the threshold multiplier. When recent jumps are detected, the forecast includes a risk premium adjustment.

### 2.12.4 Value-at-Risk

VaR at confidence level $\alpha$ estimates the maximum loss that will not be exceeded with probability $\alpha$:

$$\text{VaR}_\alpha = t_\alpha(\nu) \cdot \sigma_{\text{forecast}}$$

where $t_\alpha(\nu)$ is the quantile of the Student-t distribution with $\nu = 6$ degrees of freedom, accounting for fat tails in return distributions.

| Confidence | t(6) Quantile | Interpretation |
|---|---|---|
| 95% | 1.943 | 5% chance of exceeding |
| 99% | 3.143 | 1% chance of exceeding |

### 2.12.5 Expected Shortfall

Expected Shortfall (Conditional VaR) estimates the average loss in the worst scenarios:

$$\text{ES}_\alpha = \text{VaR}_\alpha \times m_\alpha$$

where the multiplier $m_\alpha$ accounts for the tail weight:

- $m_{0.95} = 1.25$ for 95% confidence
- $m_{0.99} = 1.35$ for 99% confidence

ES is more conservative than VaR and better captures tail risk for non-normal distributions.

#### 2.12.6 Interpretation

| Metric | Meaning |
|---|---|
| VaR 95% = 2% | 5% chance of losing more than 2% tomorrow |
| VaR 99% = 3.5% | 1% chance of losing more than 3.5% tomorrow |
| ES 95% = 2.5% | Average loss in worst 5% of days is 2.5% |
| Jump-adjusted | Recent variance jumps detected; forecast conservative |

Higher volatility stocks (e.g., NVDA: VaR 99% $\approx$ 7.5%) require larger position sizing buffers than lower volatility names (e.g., SPY: VaR 99% $\approx$ 2.3%).

# 3 Monitoring

Monitoring tools detect shifts in market microstructure before they manifest as realized losses. Rather than reacting to drawdowns, we track leading indicators embedded in the covariance structure and network topology of asset returns.

## 3.1 Systematic Risk Early-Warning Signals

### 3.1.1 Covariance Decomposition

#### 3.1.1.1 Sample Covariance Matrix

Let $\mathcal{A} = \{1, \ldots, N\}$ denote a universe of $N$ assets with return series $\{r_{i,t}\}_{t=1}^{T}$. The sample covariance matrix $\Sigma \in \mathbb{R}^{N \times N}$ has entries:

$$\Sigma_{ij} = \frac{1}{T-1} \sum_{t=1}^{T} (r_{i,t} - \bar{r}_i)(r_{j,t} - \bar{r}_j)$$

where $\bar{r}_i = \frac{1}{T} \sum_{t=1}^{T} r_{i,t}$ is the sample mean return of asset $i$.

#### 3.1.1.2 Eigenvalue Decomposition

The eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0$ of $\Sigma$ are extracted via power iteration with Hotelling deflation. Given an initial random vector $\mathbf{v}^{(0)}$, the $k$-th eigenvalue–eigenvector pair $(\lambda_k, \mathbf{v}_k)$ is found by iterating:

$$\mathbf{v}^{(m+1)} = \frac{\Sigma^{(k)} \mathbf{v}^{(m)}}{\|\Sigma^{(k)} \mathbf{v}^{(m)}\|}$$

until convergence, where $\Sigma^{(1)} = \Sigma$ and subsequent matrices are deflated:

$$\Sigma^{(k+1)} = \Sigma^{(k)} - \lambda_k \mathbf{v}_k \mathbf{v}_k^{\top}$$

The eigenvalue is recovered as $\lambda_k = (\mathbf{v}_k)^{\top} \Sigma^{(k)} \mathbf{v}_k$. Deflation removes each found component from the matrix, exposing the next largest eigenvalue in successive iterations.

#### 3.1.1.3 Signal 1: Variance Explained by First Eigenvalue

The fraction of total variance captured by the dominant eigenvalue is:

$$\text{VEF} = \frac{\lambda_1}{\sum_{j=1}^{N} \lambda_j}$$

A high VEF indicates that a single factor drives the cross-section of returns. During crises, assets move in lockstep and VEF spikes as diversification breaks down.

#### 3.1.1.4 Signal 2: Variance Explained by Eigenvalues 2–5

$$\text{VE}_{2\text{–}5} = \frac{\sum_{j=2}^{\min(5,N)} \lambda_j}{\sum_{j=1}^{N} \lambda_j}$$

When secondary eigenvalues also concentrate, multiple systemic risk channels are simultaneously active (e.g., rates, credit, and equity stress co-occurring).

### 3.1.2 Distance and Graph Construction

#### 3.1.2.1 Correlation-Based Distance

The sample correlation matrix $\rho_{ij} = \Sigma_{ij}/\sqrt{\Sigma_{ii}\Sigma_{jj}}$ is transformed into a distance metric:

$$d_{ij} = \sqrt{1 - \rho_{ij}^2}$$

This metric satisfies $d_{ij} = 0$ when assets are perfectly correlated (or anti-correlated) and $d_{ij} = 1$ when uncorrelated, capturing the strength of linear dependence regardless of sign.

#### 3.1.2.2 Minimum Spanning Tree

The minimum spanning tree (MST) of the complete graph with edge weights $\{d_{ij}\}$ is constructed via Kruskal's algorithm:

1. Sort all $\binom{N}{2}$ edges by weight in ascending order
2. For each edge $(i, j)$ in sorted order: if $i$ and $j$ belong to different connected components, add the edge to the MST (cycle detection via Union-Find with path compression and union by rank)
3. Terminate after $N - 1$ edges

The MST retains the $N-1$ strongest pairwise connections while eliminating redundancy, revealing the backbone of the correlation network.

### 3.1.3 Eigenvector Centrality

#### 3.1.3.1 MST Adjacency Matrix

The MST is represented as a weighted adjacency matrix $A \in \mathbb{R}^{N\times N}$ with inverse-distance edge weights:

$$A_{ij} = \begin{cases} 1/d_{ij} & \text{if } (i,j) \in \text{MST} \\ 0 & \text{otherwise} \end{cases}$$

Inverse-distance weighting assigns higher strength to edges between more correlated assets.

#### 3.1.3.2 Centrality Computation

Eigenvector centrality $\mathbf{c} = (c_1, \ldots, c_N)^\top$ satisfies the eigenvalue equation:

$$A\mathbf{c} = \lambda_{\max}\mathbf{c}$$

where $\lambda_{\max}$ is the largest eigenvalue of $A$. The centrality is computed via power iteration on $A$ until the iterate converges within tolerance $\varepsilon = 10^{-8}$, then normalized to unit length. A node with high centrality is connected to other high-centrality nodes, identifying systemically important assets.

#### 3.1.3.3 Signal 3: Mean Eigenvector Centrality

$$\bar{c} = \frac{1}{N}\sum_{i=1}^{N} c_i$$

High mean centrality indicates a tightly interconnected MST – contagion pathways are dense and shocks propagate rapidly across the network.

#### 3.1.3.4 Signal 4: Standard Deviation of Centrality

$$\sigma_c = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(c_i - \bar{c})^2}$$

High dispersion reveals a hub-and-spoke topology where a few nodes dominate – the "too-connected-to-fail" configuration where stress in a hub propagates to all spokes.

### 3.1.4 Regime Classification

#### 3.1.4.1 Composite Scoring

Each signal is scored on a scale of 0 to 4 based on empirical thresholds:

| Score | VEF | $\text{VE}_{2\text{–}5}$ | $\bar{c}$ | $\sigma_c$ |
|---|---|---|---|---|
| 4 | $> 0.60$ | $> 0.25$ | $> 0.25$ | $> 0.15$ |
| 3 | $> 0.50$ | $> 0.20$ | $> 0.20$ | $> 0.10$ |
| 2 | $> 0.40$ | $> 0.15$ | $> 0.15$ | $> 0.07$ |
| 1 | $> 0.30$ | $> 0.10$ | $> 0.10$ | $> 0.04$ |
| 0 | $\leq 0.30$ | $\leq 0.10$ | $\leq 0.10$ | $\leq 0.04$ |

The total score $S = s_1 + s_2 + s_3 + s_4 \in \{0, \ldots, 16\}$ determines the risk regime:

$$\text{Regime}(S) = \begin{cases} \textit{Crisis} & \text{if } S \geq 12 \\ \textit{High Risk} & \text{if } 9 \leq S < 12 \\ \textit{Elevated} & \text{if } 6 \leq S < 9 \\ \textit{Normal} & \text{if } 3 \leq S < 6 \\ \textit{Low Risk} & \text{if } S < 3 \end{cases}$$

The composite scoring approach avoids false alarms from any single indicator by requiring corroborating evidence across structurally different signals (spectral and topological).

### 3.1.5 Transition Probability

#### 3.1.5.1 Signal Normalization

Each signal is normalized to $[0, 1]$ by dividing by an empirical upper bound:

$$\begin{aligned} x_1 &= \min(1,\ \text{VEF} \,/\, 0.7) \\ x_2 &= \min(1,\ \text{VE}_{2\text{–}5} \,/\, 0.35) \\ x_3 &= \min(1,\ \bar{c} \,/\, 0.3) \\ x_4 &= \min(1,\ \sigma_c \,/\, 0.2) \end{aligned}$$

#### 3.1.5.2 Logistic Combination

The probability of transitioning to a higher-risk regime is:

$$P(\text{transition}) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

where:

$$z = -2.0 + 2.0\, x_1 + 1.5\, x_2 + 2.5\, x_3 + 1.5\, x_4$$

$x_1$: normalized eigenvalue concentration (weight 2.0)

$x_2$: normalized secondary eigenvalue share (weight 1.5)

$x_3$: normalized mean centrality (weight 2.5)

$x_4$: normalized centrality dispersion (weight 1.5)

The intercept of $-2.0$ centers the function such that $P \approx 0.5$ when signals are at moderate levels. Higher weights on VEF and mean centrality reflect their stronger empirical predictive power for systemic events.